

Nutr 101 Intro to Nutrition

Numbered Transcript

NUTR 101 • Max Lugavere

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Lecture 1: Nutritional Fundamentals

At a glance — Lugavere introduces nutritional fundamentals and what the course covers.

- 1.1 Welcome, everybody. Thanks for being here. The title of this course—I was trying to decide how I was going to frame it. Rather than do a deep dive into the weeds of nutrition science, I thought a much better use of your time and my life focus and expertise, I would say, is to really focus on the practical aspects of nutrition. How to harness the power of nutrition and various lifestyle variables for optimal performance, better health, longevity, and the like.
- 1.2 We're going to focus as much as we can from a high level on the more actionable side of nutrition, which I think is very valuable. We're going to talk about the definition of nutrition. We're going to define it. We're going to talk about the purpose of food. We're going to talk about the various components of nutrition, macronutrients. There's this misconception that to understand nutrition fully, you have to be a PhD or have some kind of advanced degree. But I think nutrition is something—health literacy is something—that I strongly believe we've been disadvantaged because of the way that we've outsourced it, like so many other aspects of life today.
- 1.3 We outsource culinary literacy, we outsource financial literacy. When our ancestors were generalists, now we live in the era of specialization, and I think we've lost something in terms of our health, in terms of our appreciation for many of these kinds of topics. I want each of you to walk away from this lecture really having a solid grasp on nutrition, and again, how to harness it for the betterment of your health and the health of your loved ones.
- 1.4 Just to share a little bit about my background. I didn't take the academic path. I went to the University of Miami, and then I became a journalist, where I worked for six years on television in the United States as a journalist, as a field producer. About six years into that role, my mother got sick. My mom developed a rare form of dementia. I've always been passionate about nutrition science and health to the point that when I started college, I actually was on a pre-med track. I ended up pivoting out of that. But when my mom got sick, I became obsessed with trying to understand everything that I possibly could about the risk factors that we all are vulnerable to with regard to chronic disease.
- 1.5 And so I became obsessed. I dove into the medical literature, and I exploited my title as a quasi-media persona and journalist to reach out to scientists, experts from around the world, so that I could learn from them. Subsequently, I've been able to write a number of books. My first book came out in 2018. It's called "Genius Foods." It's now published around the world. It's a New York Times best seller.
- 1.6 Though I'm not primarily a researcher in the technical sense, I have actually been able to publish scientific literature. You can see in the middle section of the screen, on the lower portion of the screen, I actually got to co-author a chapter in a clinician's

textbook on the clinical practice of the prevention of cognitive decline with a focus on conditions like Alzheimer's disease and other forms of dementia. This is a textbook that actually gets into the hands of clinicians, of physicians, and so I've been able to actually influence the clinical practice of dementia prevention, which to me is super exciting.

- 1.7 Dementia prevention, those two words, it's only really recently that we've been able to mention both of those words within the same sentence and be taken seriously. Ninety percent of what we know about Alzheimer's disease alone has been discovered only in the past fifteen or so years. So, this is a rapidly evolving field of science, but we now know that there are steps that you can take in your day-to-day life that can optimize the way that your brain works and batten down the hatches so that you reduce your risk for these kinds of conditions decades down the line.
- 1.8 Everything I do, my why, is motivated by the sickness that I saw in my mom. For me, being honest, being transparent, and sharing the best available evidence with you guys has become my life's purpose. Psyched to have you guys here.
- 1.9 With regard to dementia prevention, just to touch on this. I talk a lot about brain health. That's my passion. But underneath the umbrella of brain health, you have many different domains that influence brain health. You have metabolic health, body composition plays a role, cardiovascular health, fitness—e everything ultimately influences the brain, every aspect of our physiology, of our biology. It really is an interesting vantage point to take and lens through which to look at all of the other different facets that comprise human health.
- 1.10 This is just to show you that this is not pseudoscience. This was actually published in The Lancet in 2020. The Lancet is one of the highest impact factor journals that we have in the field of medicine. The potential for prevention is high. We've identified, at this point, twelve modifiable risk factors, suggesting that at least 40% of dementia cases worldwide are preventable. I would say probably way higher than that, but this is where the consensus currently stands.
- 1.11 So, generally speaking, nutrition is the process of providing or obtaining the food necessary for health and growth. Fairly basic. I think everybody generally has a sense of what nutrition really means. Nutrition is really important, and it can be as simple or as complex as we want to make it.
- 1.12 We're going to scale that hierarchy over the course of this seminar, of course. But yeah, it's a super empowering science. I happen to love it. I think it's a fun science to discuss because it's relevant to all of us. There's not one of us for whom nutrition is not relevant. It continues to be relevant across the age spectrum, and it changes also across the age spectrum. It changes depending on what gender you are to some degree. It changes depending on what your goals are.

- 1.13** As you might hear me repeat a few times, there's no one-size-fits-all nutrition plan for everybody. Nutrition is not a hat. It's something that, again, it's not one-size-fits-all; it's bespoke. We have to think of it that way.
- 1.14** I think a very common mistake that you see on social media, in diet books, is that people tend to present nutrition as if it is this one-size-fits-all science, and it's not. It's incredibly complicated. For that reason, many people actually consider nutrition to be a 'soft science.' It's not something that we can measure in each person—the effects of—with pristine accuracy the way that we can the laws of physics, for example, or geology, or other such sciences.
- 1.15** It's very difficult to study. It's actually much harder to study than drug research, and it's much less well-funded. In my view, there are many aspects of nutrition science that actually we have to think outside of the realm of science for answers. We have to use logic. We have to use common sense. We have to use reason. I think that's a crucially important way to think about this. Then, of course, there's food—the purpose of food.
- 1.16** The role that food plays in our lives is a lot different than the role that nutrition plays. Nutrition has a very simple definition. Nutrition is all about making us feel good, setting us up for the best possible health that we can have. But food plays many other roles.
- 1.17** Food is how we celebrate. It's how we bond with one another. Cooking a meal for loved ones—it's an incredible way to show love, to communicate that we love the person that we're sitting across from. We experience joy with our food. We have rituals associated with food. When we are stressed out, we reach for our favorite foods. So, food, in a way, is like our friend when we're stressed or going through uncomfortable life events.
- 1.18** So, food is something that we can't necessarily break down to nutrition and distill to its component parts because it plays such an oftentimes ineffable role in our lives, a role that can't properly be described with words. There's a culture associated with food, personal preference, there are so many aspects of it that are, again, personalized, that are individual.
- 1.19** The problem is that the food—our food supply has become super industrialized. I think this is one of the major reasons. This is not just opinion, but our industrialized food supply is one of the major reasons, one of the foundational reasons, why so many of us are unwell today, struggling with our health. Our food supply has become overrun with what are referred to as ultra-processed foods.
- 1.20** Now, it's really important to make the distinction between the different types of food processing. Most people, when they think about food, they think it's either processed or unprocessed. But there's actually a continuum of processing.
- 1.21** When you buy an apple in a supermarket, that's an unprocessed food. The second you slice the apple with a knife, you are processing it to some degree. You take that apple

slice, and you squeeze it, you create apple juice from it; you're processing it even further. You use that apple slice, and you create apple sauce; you're processing it even further. You take the apple, and you make apple-flavored gummy treats with it; you're processing it even further. So, food processing is a continuum. At one end, we have ultra-processed foods, which is a term that I'm going to come back to again and again. And at the lower end of the spectrum, you have unprocessed foods. And then, just above that, you have minimally processed foods.

- 1.22 One of the primary issues today—and ultimately, one of the defining characteristics of the Standard American Diet, which is no longer just the Standard American Diet but is now, in many ways, the Westernized diet is how it's often referred to because we've essentially exported it to other nations around the world—is that our food supply has become ultra-processed.
- 1.23 Typically, there are a few ways to recognize ultra-processed foods. We'll get into that. But generally, where you find them in the supermarket is in the aisles. You walk down the aisles of most major supermarkets and the shelf-stable, packaged, processed foods with seemingly infinite shelf lives—those are the ultra-processed foods that I'm talking about. These are the foods that people are now over-consuming. Indeed, ultimately, basing their diets around to the tune of about 60% of your average adult's calories today come from these kinds of foods: ultra-processed, shelf-stable foods. We'll get into the consequences of that. For children, that proportion is even higher; it's about 70%.
- 1.24 There was a great photo essay that was done. I don't actually know the name of the photographer, but some photographer did a project where he went around the world and he looked at a week's worth of groceries for your average family. This is a family in the UK, and they're a week's worth of groceries for them. You have to squint to see the unprocessed or minimally processed food. You can see some fruits over there, some whole fruit on the right side of the screen. But generally, the majority of items that you see in this photo are of the packaged sort. You see the Capri-Sun, you see the cereals, you see the bags of chips. This is a UK family.
- 1.25 In the US, it's not much different. In fact, it's probably a little bit worse. Barely any fresh food. You've got pizza, which is a hyper-palatable mixed dish loaded with probably innumerable industrial chemicals. You've got soft drinks and sodas. I do see some meat over there. But generally, this is how most Americans are eating. Just an overabundance of ultra-processed foods.
- 1.26 We'll talk about some of the consequences, but just to give you guys a sense, we are now seeing the connection made between the consumption of these foods and chronic disease, which, is pretty obvious, but it's always good to have data. This is a study that came out actually very recently, last year, middle of last year, linking directly ultra-processed food consumption with risk of dementia.

- 1.27** We see that for every 10% increase in ultra-processed food consumption, there's about a 25% increased risk in the development of dementia. So, it's not bueno from the standpoint of brain health, but there are other consequences as well.
- 1.28** When it comes to nutrition, there are a few major—I guess you could call them tent poles—that we want to think about, basically. The more essential tenets of nutrition come down to nutrients. As I mentioned, that was one of the defining aspects of nutrition. When it comes to nutrients, we can generally break them apart into a small number of distinct categories.
- 1.29** The first one that we'll talk about are macronutrients. 'Macro' means big. So these are the big nutrients that we generally get from the foods that we eat. And generally speaking, the three macronutrients are carbohydrates, fat, and protein. Now, each of these can be broken down further.
- 1.30** So, carbohydrates aren't carbohydrates necessarily. Fats aren't fats. Protein isn't protein. The fats that you eat don't necessarily become the fat in your body, even though they have the same word. There are many different types of carbohydrates to consume, and protein quality differs. So, we're going to break down these into the nuanced details, but just off the top, when people say macronutrients, generally, they're referring to one of the three categories: carbohydrates, fats, and proteins.
- 1.31** Now, when it comes to essential nutrients—'essential' basically means that we need them to survive. If we don't get them, we're going to develop a deficiency disease. There's no such thing as an essential carbohydrate, which isn't to say that carbohydrate inclusion in the diet isn't optimal. Carbohydrates are great. We need carbohydrates to optimize many aspects of our biology, but you wouldn't develop a deficiency disease by not consuming carbohydrates.
- 1.32** Fats, on the other hand, are essential. Certain fats are definitely essential. You might have heard of the essential fatty acids. That is actually a very, relatively speaking, there's a very small amount of essential fats that we need on a daily basis. We need to ingest certain fats like omega-3 fatty acids you might have heard of, omega-6 fatty acids on a daily basis. Those are the only fats that we need to ingest, which isn't to say that adding more fat to the diet isn't essential either. We see that higher fat diets are associated with better hormonal profiles and the like. But again, we're talking about the difference between essential versus non-essential.
- 1.33** And then, protein is absolutely essential. Protein provides amino acids, and your body is made of protein. So, when you ingest protein, your body disassembles those proteins into its constituent amino acids. There are dozens of amino acids to list, but of note, there are nine essential ones, and we're going to talk about where to find those.
- 1.34** Essentially, protein is absolutely essential. Certain fats are essential, and there's no such thing as an essential carbohydrate. But the proportion that you want to include each of these in your diet, again, varies from person to person, depends on goals. What

is not necessarily essential isn't necessarily suboptimal. We just want to be clear to make the distinction between the way that we talk about these different nutrients.

- 1.35** When it comes to carbohydrates, we have simple carbs, complex carbs, and then fiber. Fiber is a form of carbohydrates. We used to think that there was a big difference biologically between simple carbs and complex carbs. I think most people past a certain age are pretty familiar with the fact that simple carbs are often used to describe sugars, simple sugars. Added sugar, specifically, is certainly not healthy, but what we're starting to see is that the difference biologically between consuming simple carbohydrates and complex carbohydrates isn't always as black and white as one is bad, the other is good.
- 1.36** Complex carbs can often be easily broken down by our bodies almost as quickly as simple carbs are. So, the difference is moot, particularly in the context of a mixed meal. But just so you know, simple carbs, generally sugars, things like that, you can taste them on your tongue. They taste sweet. Complex carbs don't immediately taste sweet until they start getting broken down on your tongue. They're generally starches and things like that found in fruits and vegetables, primarily. Then fiber is technically a carbohydrate, but fiber we don't break down.
- 1.37** Actually, something that should be said about both fats and carbohydrates is that they're both energy. You can describe them both as providing energy to the body in different ways, but ultimately, they're both energy. Protein isn't generally used as energy in the body. Protein is a building block.
- 1.38** Carbohydrates and fat can both generally be pretty easily stored as fat. Fat is the most easily stored as fat. Then you get carbohydrates, and then protein is the most difficult for your body to store as fat because protein isn't really an energy source, and your body wants to silo energy for later if you provide it with an excess of energy.
- 1.39** Fiber doesn't really provide energy. Fiber is generally thought of as indigestible. Instead, it plays other roles. It fills us up; it makes us feel satiated. We know that it's beneficial because of the role that it now plays in the health of our guts, gut health, and the gut microbiome biome, which we're going to talk about.
- 1.40** Fiber is generally found primarily in whole plants: fruits, vegetables, things like that. We see that fiber consumption is generally associated with longevity and lower levels of inflammation and is generally thought of as being good for you. I just want to be clear that, again, there's no such thing as a one-size-fits-all diet. People generally have different fiber tolerances, but it is generally thought to be very beneficial from the standpoint of longevity. But we can talk about why fiber is thought to be beneficial.
- 1.41** Next up, we have fat. As I mentioned, there are certainly some essential fats. It's unfortunate that fat, we refer to as fat, and we also refer to the fat on our bodies as fat because a lot of people are under the misconception that fat makes you fat. It doesn't necessarily, although it can if we overconsume it.

- 1.42 But underneath the umbrella of fat, we have different types of fats. We have saturated fats. Even under saturated fats, there are many different types of saturated fats.
- 1.43 A saturated fat, we've been told for decades, are unequivocally bad for us—the evil saturated fats. But there are many different types of saturated fats, and not all saturated fats affect our biology the same way. For example, we have saturated fats in animal products like red meat, like the steak that I have this morning for breakfast; there are saturated fats in dairy; there are saturated fats found in plants—coconut fat is predominantly saturated fat—and they all impact our biology differently.
- 1.44 So, if you ever hear anybody say that saturated fats are bad, unequivocally bad, they probably don't know what they're talking about. For example, among saturated fats that are commonly found in animal products, there are three primary saturated fats that you'll find: palmitic acid, myristic acid, and stearic acid.
- 1.45 Stearic acid is an example of a saturated fat that's commonly found in animal products that has little to no impact on biomarkers associated with cardiovascular disease. And yet you'll hear again and again that saturated fats are bad.
- 1.46 Also, dairy as a food category has the highest proportion of saturated fat of any other food category. Actually, this is pretty interesting. Wherever you find in nature saturated fat—wherever you find any natural food source of fat—is going to have some combination of all of the different types of fat. Some are going to have more saturated fat; some are going to have more unsaturated fats. But in general, they tend to arrive like an entourage, but in different proportions.
- 1.47 Dairy has the highest proportion of saturated fat. It's much higher than beef, it's much higher than pork, and yet we see consistently in the observational literature that people who consume full-fat dairy tend to have better cardiometabolic health, which is a paradox. I mean, if saturated fat is so bad for us, wouldn't you see people that regularly consume full-fat dairy just dropping off of heart disease left and right? You don't see that. In fact, you see that they have better cardiovascular health. That's thought to be due to the way the fats in dairy are packaged as part of the food matrix, which is something that we'll talk about.
- 1.48 The other thing about saturated fats is that they are the most chemically stable of any of the other fats, meaning they are the most resistant to a form of chemical degradation called oxidation. Due to their chemical nature, they're able to stack together very tightly. This is the reason why saturated fats generally are solid at room temperature. It's just the way the chemical chain is organized with saturated fats. It's why they're so chemically stable. The fatty acids are just packed together very tightly. It's one of the reasons, and it's why they're solid at room temperature.
- 1.49 From a dietary standpoint, anytime you see something that is a fat that's solid at room temperature, we call that a fat. We don't call them oils. We call them like butter is

referred to as a fat as opposed to an oil. Oils tend to be liquid at room temperature, and they are primarily comprised of unsaturated fats.

- 1.50** Unsaturated fats, generally speaking, are less chemically stable than saturated fats. What gives them that chemical stability are what are called double bonds. It just makes them more stable, less prone to this oxidative process. The number of double bonds that a fatty acid has basically indicates its degree of unsaturation or saturation.
- 1.51** A fat with no double bonds is going to be unsaturated, or I'm sorry, saturated, and a fat with a double bond or more is going to be unsaturated. Among the unsaturated fats, we have monounsaturated and polyunsaturated.
- 1.52** Now, 'mono,' as the name suggests, means one, singular. Monounsaturated fats have one double bond. Polyunsaturated fats have more than one double bond. Of the unsaturated fats, monounsaturated fats, because they are more saturated, meaning they have fewer double bonds, they only have one, they're going to be more chemically stable than polyunsaturated fats. Again, saturated fats are the most chemically stable. Unsaturated fats are going to be less chemically stable than saturated fats. Among the unsaturated fats, monounsaturated is more chemically stable because it's essentially more saturated. Then polyunsaturated is going to be the least chemically stable.
- 1.53** This is relevant to our biology. It's relevant when it comes to food because, generally speaking, foods that are higher in polyunsaturated fats—or I should say food products that are higher in polyunsaturated fats—are going to be more prone to this form of chemical degradation, which can affect your biology in profound ways.
- 1.54** When it comes to whole foods, you don't have to worry about this distinction; this is a distinction that only really becomes relevant when we're talking about processed foods because wherever you find these fats in nature, nature has done the guesswork for us. Nature has packaged these fats with the appropriate amount of antioxidants to protect them. Again, in nature, wherever you find polyunsaturated fats, which, as I mentioned, are chemically unstable, monounsaturated fats, you always will find a requisite amount of antioxidants to protect those fats.
- 1.55** For example, in walnuts, walnuts are a very rich plant-based source of polyunsaturated fats, chemically very unstable, right? So, does that mean that walnuts are unhealthy? No, because walnuts are also a very rich source of vitamin E. Wherever you see polyunsaturated fats in nature, you also are going to see a requisite amount of—a requisite concentration of—vitamin E to protect those fats.
- 1.56** The problem arises in processed foods where you have polyunsaturated fats that have been extracted from the food matrix and don't necessarily have that antioxidant protection. Then that's where you run into problems that we're going to talk about. But generally, those are the two primary sources of fats that you consume: saturated and unsaturated.

- 1.57 Then we have trans fats. Underneath the umbrella of trans fats, trans fats, we often hear, are really bad for us. And generally speaking, that is true. They are very bad for us. That's because the primary occurrence in the diet where they would appear, historically, would be in the form of partially hydrogenated vegetable oils.
- 1.58 Does anybody remember those? Show of hands. They were actually very commonly used in processed foods up until, I believe, about 10 or 15 years ago when the FDA banned them. But they were in the food supply—the Western food supply—for decades because they were thought to be a healthier alternative to saturated fats. They behave similarly to saturated fats, but they're artificially manipulated so as to behave like saturated fats. What manufacturers would typically do, they would take unsaturated fats, like polyunsaturated vegetable oils, and they would pump them full of hydrogen so that they would behave more like saturated fats.
- 1.59 For the past few decades, the tide is beginning to turn, but we were told again and again that saturated fats were not good for us. They were demonized to a large extent by our governing bodies. A food manufacturer could give processed foods the texture and the consistency of food that was otherwise rich in saturated fats but market it as being low in saturated fat. It was also—they were dirt cheap to produce, unlike butter or tallow. Partially hydrogenated fats, where they cost nothing to produce, essentially. And so very appealing from a food processing standpoint.
- 1.60 We now know that those kinds of fats are poisonous to the cardiovascular system and to our brains. And so, those have been essentially outlawed, although they still exist in the food supply. And I'll talk about that; I'll talk about where we can still presently find them later on. But an important distinction needs
- 1.61 to be made: trans fats are also naturally occurring to some degree in other forms in animal products. Certain dairy products and the fat of ruminant meat also contain naturally produced trans fats, which are actually thought to be beneficial. The data is not super strong on this, but we see that they are different than the kinds of trans fats that were man-made, artificially created.
- 1.62 One example of a naturally occurring trans fat is referred to as CLA or conjugated linoleic acid. It's thought to have some anti-cancer properties, some metabolism fortifying properties, it's been associated with fat loss, although I think that the data on that is pretty weak, but CLA is an example of a naturally occurring trans fat. Again, it's found primarily in the fat of ruminant animals and the dairy produced by those animals.
- 1.63 Then protein. Protein is one of my favorite topics. Protein is super, super important. As I mentioned, it's essential. We're going to talk about protein timing and protein dosing and all the cool, fun, relevant stuff with regard to protein.
- 1.64 But essentially, protein is what most people think of when they think of meat, eggs, dairy, things like that. We find protein in plants as well. It's actually a misconception that plants don't have all of the requisite amino acids, that plant proteins are

essentially incomplete. Animal proteins do provide all of the essential amino acids in their appropriate amounts, in their plug and play amounts. Animal proteins are the highest quality proteins that we find in nature, primarily because we're meat and we're not ruminated into animals, as I mentioned. When we ingest the meat of another animal, it helps to fortify the meat that we're made of, which is a weird way to think about it, but it's how it works.

- 1.65 Plant proteins—all plants have all of the amino acids, but they have them in varying quantities. Now, if you're consuming enough protein on a plant-based diet, that becomes moot, so long as you're consuming a diverse diet, a diverse array of plant proteins. But that's not something that you need to be concerned with when you're consuming animal protein because it provides the appropriate ratios of all the different amino acids to promote human health.
- 1.66 But again, plant proteins, you'll find certain plant-based sources of protein that are low in certain essential amino acids, and then you have to combine them. People on plant-based diets will often combine grains with legumes and things like that. Plant-based protein powders—you'll often see pea protein mixed with rice protein and things like that. You do have to cobble together different protein sources on a plant-based diet. But in the context of a diet that contains adequate protein, it's generally not something that you need to worry about.
- 1.67 Also, plant protein in the context of an omnivorous diet—if you do eat animal protein and you're consuming plant protein as well, that protein does count because, at the end of the day, all proteins are broken down into its constituent amino acids. You might be getting less of a certain amino acid when you consume a certain plant that you're getting more of in another food. Ultimately, it all comes out in the wash. The more varied and diverse your diet is, particularly if it's an omnivorous diet, you don't have to necessarily worry about protein quality so much.
- 1.68 Where protein quality does become an issue is for people that are on restrictive diets. For vegans, protein quality definitely becomes an issue, and particularly for people on low-protein diets. The less protein you consume, the more you want to make sure that the protein quality that you're consuming is sufficient. Animal protein, again, is always at the top of the list in terms of protein quality.
- 1.69 But this becomes really important for certain populations. For example, anybody who's on a lower-protein diet, older populations, so older adults, like elder adults, tend to under-consume protein. So, protein quality, making sure that they're consuming adequate protein, is a really important consideration.
- 1.70 This shouldn't be construed as medical advice, but people who are on protein-restricted diets for medical purposes, you want to make sure that you're getting high-quality protein. If you're on a low-protein diet and you're consuming plant protein and you're not doing it thoughtfully, you could run into trouble.

- 1.71** The third type of protein of note would be collagen. Collagen is actually the most abundant protein in the body. We now see it on the market in terms of numerous supplements. Collagen is not thought of as an essential protein because we produce it. But that being said, there is mounting evidence to suggest that by eating more collagen, we see benefits to our joints and to various aspects of our physiology, like wound healing.
- 1.72** My dad actually recently had a total knee replacement surgery. My dad is very supplement averse, but one of the supplements that I got him to take was collagen because there's evidence that collagen can support wound healing.
- 1.73** Collagen comprises the molecules that keep our arteries elastic. So, it's a super important protein that's not necessarily essential, but again, we see benefits when we consume more of it. Collagen is essentially a repeating loop of three amino acids, which you don't really have to remember: glycine, proline, and hydroxyproline. It also contains a few of the essential amino acids with the exception of tryptophan. So, collagen is actually unlike plant protein; you actually are lacking in tryptophan with collagen. But again, in the context of a diet with enough protein and with food and protein diversity, you would count collagen as your protein, as part of your daily protein intake.
- 1.74** We actually see that glycine, which is one of the primary parts of collagen— one-third of collagen is glycine—actually plays a number of important roles in the body independent of the role that it plays in collagen synthesis. We see that glycine is important for metabolic health. It can actually be used to improve sleep. There have been a small number of studies that have shown that glycine before bed can actually boost sleep quality in certain measures of sleep quality.
- 1.75** So, collagen is pretty great. It's found primarily in the ligaments and connective tissue of animal products, organ meats, and things like that, which we used to consume a lot more of. Today, we tend to consume more the skeletal muscle of animals. Some people believe that we're actually missing out by not ingesting more collagen on a regular basis. It's definitely, I think, a worthwhile form of protein to ingest. Yeah, I'm a fan.
- 1.76** Then, of course, we have supplements. Collagen is widely available now as a type of supplement. We see them on the market as collagen peptides, but there's also other protein supplements that are more complete in their essential amino acid profile.
- 1.77** My favorite protein is whey protein. Whey protein supplements are great. You also have plant protein supplements on the market. In general, what I will say is that whey is the highest quality protein. It's extremely high-quality protein. It is a processed food, but it's made in the cheese-making process. So, I would consider it minimally processed, and I would consider it actually one of the few exceptions where ultra-processed or where processed foods actually are—you can actually get a really healthy product with food processing. So, not all food processing is bad. There are definitely

examples of processed foods and even ultra-processed foods, where you actually yield a very healthy product, and whey protein supplements, I think, are an example of that.

- 1.78** Then, you also get plant protein supplements. Now, plant protein supplements are a lot more processed, and they tend to harbor heavy metals, which is a problem. The dose of heavy metals that you're ingesting when you ingest plant protein supplements might not be all that concerning, especially if you are not ingesting it all the time, but people on vegan diets are consuming lots and lots of plant-based protein supplements. So, for that population, it could potentially be a concern. But in general, I'm a fan of protein supplements. You don't need to be a bodybuilder to reap the benefit from protein supplements, I will add. In general, I'm not a huge supplement guy, but protein supplements are great.
- 1.79** With all of the variables that we talked about in this portion of the seminar, these are all variables that can be tweaked for better health: fat, carbohydrates—we're going to get into all of it, but just wanted to lay the groundwork and establish some definitions before we dive in. But we are made of food. The foods we eat, you are what you eat, essentially.
- 1.80** Food is a form of information. It has epigenetic capacity, meaning that it actually can change the way your genes express themselves, which is super cool to think about.
- 1.81** We see now that many of the kinds of chronic diseases that people are suffering from around the world, cardiovascular disease, cancer, dementia, are responsive to food. Many people say that food is a form of medicine. I would agree with that. I think we just have to be mindful of what food can and can't do. If you have a chronic disease, I think that the capacity of food is limited. But where its power really shines is with regard to prevention and management to some degree. So, I'm a huge believer in the power of nutrition, and we're going to dive into all of this stuff.
- 1.82** We also know that food is super important from the standpoint of body composition. Body composition matters. The ratio of muscle to fat mass that you're carrying with you, all super, super relevant when it comes to long-term health, how you feel, and food plays a major role in that. I hope to share with you guys ways of optimizing your body composition. There's a ton to talk about there, which we're going to cover.
- 1.83** And then, of course, the role that nutrition plays in exercise performance. Now, I come at this as somebody who's not an athlete. I'm not a superstar athlete or anything like that. I think that knowing how to harness the power of nutrition to augment the effort that you're putting in the gym is super important. So, we're going to talk about all that stuff. But that's it for this portion of, I guess, the introductory portion of my seminar.

Lecture 2: Satiety Superstars

At a glance — 'Satiety superstars' — the fundamentals of healthy eating and foods that keep you full.

- 2.1 Part two: "The Fundamentals of Healthy Eating." Look at that beautiful slide. Thank you to Nancy for creating that for us. All right, so we're going to talk about the key components of a healthy diet, daily needs for macronutrients and micronutrients, optimizing for satiety, and the importance of fiber.
- 2.2 The satiety one is a big one. They're all important, but how do you construct a healthy diet? Well, as I mentioned, There's no such thing as a one-size-fits-all diet. But given your own unique needs, a healthy diet is going to strike a balance between the calories, aka the energy that we ingest on a daily basis, and the energy that we expend. So, this is referred to as energy balance. You've seen it online referred to as calories in, calories out. We'll talk a little bit about that. It can be a heated issue online, but I'll break it down for you.
- 2.3 Macronutrients are important. We've already defined those, but just to recap: protein, fats, carbohydrates. What is the appropriate ratio of proteins, fats, and carbohydrates for you that's going to optimize your health?
- 2.4 Micronutrients—we haven't yet touched on micronutrients. As the name suggests, micronutrients are small. You can't necessarily see from the outside what micronutrients are lurking in your foods, but whole foods tend to be rich in micronutrients. These can include vitamins; these can include minerals. Vitamins and minerals are essential, but these can also include certain compounds like phytochemicals that are not necessarily essential but that we know—that we have evidence—play a beneficial role in human health.
- 2.5 And then, we want to minimize the consumption of ultra-processed foods. As I mentioned, not all ultra-processed foods are bad, but the vast majority of ultra-processed foods are bad. I use the term 'bad' loosely. I just want to be very clear that no single food is capable of swaying your health in any one direction, although, obviously, a single food can make you feel bad after eating it, which you don't want that. Nobody wants that.
- 2.6 But really, it comes down to your dietary pattern. Dietary pattern is the most important thing. In general, we see that dietary patterns that are associated with health and longevity are very low in ultra-processed foods. We can look to any of the world's long-lived populations. We can argue ad nauseam about the proportion of meat and plant products in their diets, but one thing is certain—they tend to all have very low proportions of ultra-processed foods.
- 2.7 First up, we talked about energy balance. So, essentially, one of the things that we need to ingest from our food, aside from the nutrients, is energy. We need energy. Fats and

carbohydrates, as I mentioned, are energy. Energy is essential for any organism. It gives us the fuel to move about the world, for all of our organs to perform their various processes—whether you're a liver cell, whether you are an ovary cell, whether you are a neuron—cells require energy to create ATP, the energetic currency of cells.

- 2.8** We need calories. Calories are super important. Problems occur when we ingest more calories than we expend on a daily basis. A lot of people online will argue that a calorie isn't a calorie, which is a stupid argument. A calorie is a calorie. A calorie is just a unit of measurement. It's like saying a mile isn't a mile. A mile is always going to be a mile. The mile can be uphill; it can be downhill. Your experience of the mile changes drastically on whether or not it's uphill or downhill, or if it's raining out or if it's sunny out, but a mile is always going to be a mile.
- 2.9** Similarly, a calorie is a calorie. That doesn't mean that a calorie from French fries is the same thing, provides the same value, or it's going to have the same effect on your biology as a calorie of eggs. A hundred calories of French fries versus a hundred calories of eggs—very different. We just have to be really clear with that—that a calorie is always going to be a calorie, but where those calories come from matters.
- 2.10** Now, before we break down all the different ways in which those foods are different, here's a very useful way to think about your own daily calorie needs.
- 2.11** Any way that we're able to figure out today is going to be an estimate. Just an estimate. Calorie needs do vary widely from person to person. The only way to know with crystalline certainty what your daily calorie needs are would be through a process called indirect calorimetry, where you basically go into a lab, and they measure the amount of oxygen and heat that your body—the oxygenate that you're inhaling, the carbon dioxide that you're exhaling—and they figure out what your metabolic rate is.
- 2.12** Short of that, which is something that very few people get to do, we have different methods. I actually left a method out here that—I'll talk about that—is actually probably one of the best ways to estimate your calorie needs. But essentially, these are some of the ways that you can estimate your calorie needs on a daily basis.
- 2.13** There's a Mifflin-St Jeor method. It's like an equation that takes into account your weight, your height, your age, your gender to establish your basal metabolic rate. Your basal metabolic rate is basically what your body is utilizing at rest from a calorie standpoint. Your organs, as I mentioned, they're all utilizing calories. Calories are required to sustain life. So, your basal metabolic rate is the sum total of calories that your body is using completely at rest. If you were bedridden, that's the amount of calories that your body would require to sustain itself.
- 2.14** Now, your total daily energy expenditure—that's different. That is how many calories your body is using going about the day and doing all the things that you do on a day-to-day basis. That's referred to as your total daily energy expenditure or TDEE.

- 2.15** So, you could use this method, this calculation of figuring out your basal metabolic rate and then ultimately your total daily energy expenditure, depending on how active you are. There are calculators that you can use online to do that. I didn't provide the formula here, but you can easily search a calorie calculator online. Google has tons of them. They're all typically the same. They all use this method.
- 2.16** Another really easy way to go about this, and again, this is just an estimate, is to multiply your weight by 15. That gives you a general sense of what your daily calorie needs are going to be. That is, again, for energy balance. That 's a rough estimate of how many calories your body is burning over the course of the day. Concurrently, it's a general sense of how many calories you should be ingesting over the course of the day so that you maintain weight homeostasis, assuming you're not trying to gain or lose any weight.
- 2.17** Another way—probably the most accurate way and easiest, a way that checks all the boxes being easy and accurate—you could just take a food journal of what you eat over the course of the week, assuming that your weight is stable, and generally figure out how many calories you're eating over the course of the day. If your weight is stable, you average that out over the course of a week because your calorie intake is going to vary day to day. But if you're really diligent about it and you get an average, again, assuming your weight is stable, that's a very good sense of what your daily calorie needs are. Every day, you journal every single day what you're eating. You can use an app to do this. There are apps that can help you track calories, and that's going to be a general sense of how many calories that you're consuming over the course of the day.
- 2.18** The weight by 15 method works, again, as an estimate, but these are all just ways to get a general sense of energy balance. Aside from energy needs and macronutrients—we already touched on this;
- 2.19** we talked a little bit about protein quality. When it comes to measuring protein quality, how do we measure protein quality? I've already talked about one important way that we can assess protein quality: via its constituent amino acids because protein is amino acids. All proteins get broken down into their constituent amino acids. We see that from an amino acid standpoint, animal protein is the highest quality. That's not controversial. What is controversial is how much that matters in the context of a varied diet. The reality is it doesn 't matter so much in the context of a varied diet where you're consuming adequate protein, that amino acid issue.
- 2.20** The other aspect of protein quality that becomes important is just purely its digestibility, like how easily we are able to digest protein. There are two gold standard ways in which we measure protein digestibility.
- 2.21** The first one that was devised is actually the PDCAAS, which stands for Protein Digestibility Corrected Amino Acid Score. Basically, they use, I believe, pigs to see when they feed them a certain bolus of protein, how many of those amino acids are actually

digested, and how much are able to be recovered in the stool. We see that certain proteins are more well-digested than others. Animal protein—very well digested. In fact, if anybody has ever had in this room, we don't have to have a show of hands, but if anybody has ever had any bowel issue or any gastric issue, for a period, a physician might put you on what's called a low-residue diet.

- 2.22** Low-residue diets basically mean that of the foodstuffs one consumes, very little ultimately makes its way down, is able to make its way down to the large intestine. Animal proteins are very low residue, meaning we absorb them almost fully in the small intestine. So, the digestibility of animal protein is very, very high. The C in that acronym stands for corrected. So with that model, the maximum digestibility that a protein—the maximum score that a protein—could achieve is one. But in the newer model, the digestible indispensable amino acid score, we see that actually certain proteins are way higher than one and that animal protein, again, takes the crown for the most digestible proteins available.
- 2.23** Soy actually gets pretty high. So, soy protein like tofu, people on plant-based diets, tofu is something that becomes a staple in the plant-based diet. That's, I think, partly because of its amino acid profile, and it's also very digestible. But then you get proteins like peanuts, peanut protein, rice protein—not as digestible. The proportion of protein that you're actually yielding when you eat a bolus of peanut protein actually becomes less viable. Again, if you're eating a lot of protein, or I should say adequate protein, it doesn't seem to be a major issue. But again, for people on low-protein diets, this does become an issue.
- 2.24** There's also one cohort that I didn't mention, but another one of these groups of concern for whom protein quality and amino acid content really matters are people in low-income diets, people who live in food deserts, people in impoverished parts of the world. Protein quality becomes really, really important.
- 2.25** Carbohydrates—carbohydrate quality is very important. I mentioned earlier the simple carbs versus complex carbs. That designation doesn't really seem to play a massive role, but generally speaking, you want your carbs to come from whole foods. You want your carbs to come from plant material, whether it's fruits or vegetables. Generally, that's going to be the highest concentration of carbohydrate quality. The food matrix plays a role. The fibers and the water and everything that you find generally in a piece of whole fruit or a whole vegetable all play a role in how we digest carbohydrates and extract, in the safest way, the sugars and whatnot that come with those carbs. But carbohydrate quality is definitely important.
- 2.26** Conversely, we see that in a lot of these ultra-processed foods, the carbohydrate quality is very low. It's just very rapidly digested. No food matrix to help us better assimilate those carbohydrates. So, one way to increase diet quality is to increase carbohydrate quality.

- 2.27** Then, fats. In quotes, I put "healthy versus unhealthy." Generally, in whole foods and minimally processed foods, I'm not actually concerned about making a distinction between healthy versus unhealthy fats in the context of whole foods, particularly foods that are raised and grown responsibly. I think fat can be very satiating ; it's a part of a food matrix. It helps us to absorb fat-soluble nutrients, which is very important. It's another very important role that fats play in the diet. They help us to absorb fat-soluble and utilize fat-soluble nutrients. So, crucially important.
- 2.28** Then, we have our micronutrients. Of note, 90% of Americans are deficient in at least one essential micronutrient. So, we definitely need to be concerned about where we're getting our micronutrients. Our food supply has also become a bit impoverished. You can attribute it to food processing, the fact that our diets have become so dominated by ultra -processed foods, but also growing practices are now leading to a decline in concentration of certain micronutrients from our food, certain B vitamins, for example.
- 2.29** Of the micronutrients, we have vitamins, a ton of vitamins: B vitamins, C. We've got our fat-soluble vitamins: A, E, D, and K. And aside from the amount of micronutrients that we're getting, we also want to make sure that the micronutrients that we're getting are bioavailable. What use is it to have a food that's rich in vitamins if we can 't absorb them? We want to make sure that the micronutrients that we want are in their most bioavailable form. In general, micronutrients found in animal products are going to be the most bioavailable, and vitamins found in plant products are going to be less bioavailable.
- 2.30** For certain examples of this, we can look to vitamin A. In animal products, you get vitamin A in a form called retinol, which is the exact form of vitamin A that our bodies require to fulfill our needs for vitamin A. In plants, you get beta-carotene. Beta-carotene is actually a carotenoid, meaning it's a plant pigment that's referred to as pro-vitamin A. So, beta-carotene is not actually vitamin A ; it has to be converted in our bodies to retinol, and people vary widely in their ability to do this conversion. Genetics are involved ; it's hard to measure; there are certain genes, but generally, there is a variation in terms of person to person, in our ability to basically utilize vitamins when they're in their plant-based form, certain vitamins.
- 2.31** Vitamin C is most commonly found in plants. That 's pretty bioavailable. B vitamins are generally across the board. You'll find them mostly in animal products, although in plants, you find folate, which is super important. They're sort of like just another testament to the value in consuming a diverse diet. You 've got vitamins that are essential for good health in both plants and animal products.
- 2.32** And then, minerals tend to be more bioavailable in animal products, but that doesn't mean that you can't also get a significant amount in plants. Some minerals of note—magnesium is one of these very important ones that about 50% of the population doesn 't consume adequately.

- 2.33** Magnesium is actually a really important mineral to discuss. It's utilized as a co-factor in hundreds of processes in the body, about 300, if not more, processes in the body rely on magnesium. Again, half of adults don't consume it adequately. It actually ranges in its importance in the body from helping to synthesize ATP, which is energy, but also it's a crucial co-factor for DNA repair enzymes. So, DNA damage is at the root cause of cancer and possibly even aging itself.
- 2.34** There's a theory that has been posited by a noted aging researcher named Bruce Ames—he himself is in his 90s at this point—but he's called it the Triage Theory of Aging, which is really fascinating. You guys should check it out. The Triage Theory of Aging basically suggests that when micronutrients are in low abundance in the body, they'll essentially be triaged and basically shuttled to the processes that are most important for our immediate survival. That's really what evolution has programmed us to seek or to be most concerned with—our immediate survival. These long-term projects like DNA repair and making sure that we live a long time—not as essential as making sure that we can fight or flee from the predator that's chasing after us or reproduce so that we can perpetuate the species.
- 2.35** So, it's been thought that minerals, like magnesium, which is the perfect example of this because it's utilized in so many different processes that range in their importance in the body. Actually, by under-consuming it, it is actually handicapping our bodies in profound ways. So, generally, minerals are going to be most bioavailable in animal products. However, certain minerals are just not found in animal products.
- 2.36** Magnesium is a great example of this. It is found in bone, but unless you're consuming bone broth on a regular basis—actually, I'm not even sure if bone broth is a good source of magnesium. I know that bones are a good source of magnesium, but how many of us are eating bones regularly? It's generally going to be found most abundantly in plants. So, almonds are a great source of magnesium. In one handful of almonds, you get 20% of your daily magnesium needs.
- 2.37** Dark leafy greens are also a great source of magnesium. In fact, anything green in the supermarket is a good magnesium source. I should have had this on the slide—if you look at the chlorophyll molecule in Google Images, it looks a lot like the hemoglobin molecule, which is the protein in our blood that carries oxygen around. But at the center of hemoglobin, you see iron, and at the center of the chlorophyll molecule, you see magnesium. Anything green in the supermarket, for the most part, is going to be a good source of magnesium.
- 2.38** All that is to say is that micronutrients play a really important role in optimizing your health. This doesn't even include the non-essential nutrients like chemicals and plants that we see, which I'm going to talk about later, but various phytochemicals that we see are able to modulate our biology, whether they're plant pigments or plant defense compounds. These invisible compounds seem to be very beneficial from the standpoint of health, whether or not, even though they're not necessarily essential.

- 2.39** So, we've talked about macronutrients. We've talked about energy balance. We're going to go deeper into protein in future sections. But all that is to say protein, super important, provides the building blocks for your body. All the enzymes in your body, your musculature, and your neurotransmitters are all built on—they're all built from amino acids, which we ingest when we ingest protein.
- 2.40** Carbohydrates—carbohydrate quality is super important. This also includes fiber, which we know is beneficial, which I'm going to talk about, and fats. The concern about healthy versus unhealthy really only becomes relevant when we're talking about added fats. We're going to go deeper into that in the sections to come, but in the context of whole food, fat can be very beneficial. It's very satiating, and it helps us to absorb certain nutrients, like certain vitamins, for example, that are fat-soluble.
- 2.41** Just for the sake of clarity, the fat-soluble vitamins that you want to be concerned with ingesting regularly are A, E, D, and K. Those are the fat-soluble vitamins. You always want to be ingesting those with food or with fat, more specifically. But chances are, unless we're talking about supplements, you always are ingesting them with a fat source because nature has done that guesswork for you. So, fat-soluble vitamins tend to be packaged with fat.
- 2.42** For example, vitamin D is found in egg yolks. Egg yolks are a great source of fat. Vitamin A, retinol, found in animal products like grass-fed beef and beef liver, is one of the top sources of vitamin A. You also get fat. Vitamin K. Interesting about vitamin K—vitamin K is fairly hard to get unless you're consuming liver on a regular basis. It is in liver, but we actually produce vitamin K endogenously when we consume dietary fiber. Dietary fiber gets fermented to short-chain fatty acids. So, in a way, you are consuming vitamin K with fat when you consume a fiber source. Vitamin E is rich in nuts and seeds and things like that, so again, a good source of fat.
- 2.43** It's important to recognize that the daily needs that we have for micronutrients and macronutrients are not set for human optimization. This is crucial. They are set to avert population deficiency. As I mentioned, what makes a vitamin, or mineral, or macronutrient essential is that if you were to not consume them, you would, over time, develop a deficiency disease with vitamin D. These are, actually, how we've identified and recognized these vitamins.
- 2.44** Throughout history, people have developed these mysterious diseases like rickets, which we've had to figure out how to cure. With rickets, if you don't consume adequate vitamin D, then you develop that condition. Vitamin C, scurvy, is actually—I forget how it was initially identified, but these recommended daily intakes for these vitamins are set so as not to, with vitamin C, for example, so as not to encourage scurvy on a population scale, which is a pathologic inability to generate collagen because vitamin C, we know, is an important co-factor in the synthesis of collagen in our bodies. So, if you want to produce more collagen, consume more vitamin C.

- 2.45** But that's not to say that those levels are adequate for human flourishing, because vitamin C plays lots of roles in the body; vitamin D plays lots of roles in the body. Vitamin D, we see, is actually, we know, is a hormone that's involved in epigenetic signaling. We know that vitamin D modulates gene expression and regulates the expression of at least 5% of the human genome.
- 2.46** There's a lot that vitamin D does in the body; there's a lot that vitamin C does, but the daily intakes are not set to optimize all of those different processes. To be honest, the reason why they're not is because we don't know the appropriate doses to optimize for all of these different processes. And people vary widely in how much micronutrients they need. Person to person, again, there's no one-size-fits-all diet.
- 2.47** If you drink lots of coffee, there's evidence that you might need more magnesium, for example. If you're on a fat-loss diet, there's evidence suggesting that you might need more riboflavin. If you are a premenopausal woman, you're going to need more iron than a man, for example. So, there's a lot of considerations to make, but all that is to say, we should be thinking about how to optimize our own individual health, and again, that these guidelines are inadequate when it comes to optimization.
- 2.48** We've also seen that for protein, which we're going to talk about. But essentially, the daily recommended intake for protein is to prevent protein deficiency. There are very few people in the Western world walking around deficient in protein; you just don't see it. Does that mean that we're consuming adequate protein? No. You could argue that the obesity epidemic is a consequence of a food supply that's become essentially diluted of protein. We're going to talk about all these different things, but all that is to say, yes, again, no one-size-fits-all diet. Nutrition isn't a hat.
- 2.49** One of the most important features of a healthful diet is that it optimizes for satiety. Satiety is super important. There's actually, I think, a new way to think about optimizing diet and fighting obesity, which seems to be so common these days and is only increasing in its prevalence. One way to do that is to optimize for satiety per calorie. What that means is how can you essentially—another way to think about satiety is fullness.
- 2.50** When you're full, you're satiated. How can you get the most full with fewer calories at a general population level? At a population level that seems to be struggling with overweight, with obesity, how can we optimize for satiety with the least amount of calories? It's not to say that we want to underconsume calories necessarily, but people generally eat until they're full. In the free-living food environment, people will eat until they're full.
- 2.51** People currently overeat; we can see that manifest as the staggering obesity statistics that we see now at the population level. So, how do we optimize so that we are getting full from our food, so that we're not going hungry, which is not a good feeling, and

doing it with the most amount of nutrition, yielding the greatest degree of nutrition and a biologically appropriate amount of calories, right?

- 2.52** Well, we have to look to what I call the three satiety superstars. This is actually why just 'eat less' fails. This is why when we were talking about energy balance, it's not really helpful to think purely in terms of calories because people who are eating an obesogenic diet—and they just attempt to eat less of it—that fails because they were already eating that diet to satiety. So, now you're telling them to not eat to satiety. How do we replace the foods or replace that real estate that they have now freed up on their plate with foods that are more satiating, that are going to fill them up, but with fewer calories? We have to look to the satiety superstars to do that.
- 2.53** The three major satiety superstars of note: protein, fiber, and water. Protein is the most satiating macronutrient. Of protein, carbs, and fats, protein is the most satiating. Very few people overconsume chicken breast or lean fish. It just doesn't happen. It's very satiating. Has anybody here ever overconsumed turkey? Like lean turkey breast? I mean, it doesn't happen. What we do tend to overconsume is some combination of carbs and fat.
- 2.54** It's not that carbs are bad; it's not that fat is bad. I think we have this misconception that carbs are the devil. All of the most obesogenic fattening foods in the Standard American Diet are some combination of fat and carbs. People will cut out donuts, thinking that they've cut out carbs. No, donuts are fat and carbs. Doritos—fat and carbs. And that's because those foods are not satiating. Part of it has to do with the processing in general, but it's just that fat and carbohydrates, particularly when processed, tend to be the least satiating.
- 2.55** Protein is very satiating. In fact, there's this hypothesis—it's called the protein leverage hypothesis—that our hunger mechanisms are in many ways dictated by our necessity for protein. Protein is the most valuable macronutrient. We're made of protein, as I mentioned. It's the most important building block for the human body. It's the hardest to achieve in nature. You have to hunt for it, essentially, so it's not easy. It's also the most expensive to produce.
- 2.56** This is one of the reasons why our food supply has become essentially depleted of protein because carbs and fat are cheap. Carbs and fat—what are carbs and fat made of? Cheap oils, subsidy crops. Protein is the most expensive, and that's unfortunate because it is the most satiating. So, if you're hungry, reach for protein: grass-fed, or it doesn't have to be grass-fed, Greek yogurt, eggs, beef jerky, protein shakes, cottage cheese, make use of dairy—all really good at sniping your hunger like a boss.
- 2.57** Just, again, to reiterate, ultra-processed foods, which we tend to overconsume, which tend to be fattening because of their calorie density, but also because they tend to be not satiating, they all tend to be depleted of protein. There are very few ultra-processed high-protein foods. They're all carbs and fat, generally.

- 2.58** Another satiety superstar is fiber. Now, fiber is satiating for a few reasons. One, it slows down the digestion of food. Fiber slows the passage of stomach contents from our stomach to our small intestine. But also, fiber expands in our stomach. If anybody has ever taken a Metamucil shake, you can see it very clearly. Fiber absorbs water. It expands; it mechanically stretches out the stomach, which has the effect of modulating and turning off various hunger hormones, like ghrelin.
- 2.59** Ghrelin is a well-established hunger hormone. When it's secreted by the stomach, which happens when the stomach is empty, it hits the brain, and the brain goes, "Feed me!" When you stretch out the stomach, when you put food in it, ghrelin turns off. Fiber is great at turning off the release of ghrelin. It's another attribute of ultra-processed foods—they tend to be depleted of fiber. It's one of the reasons why our fiber consumption is so low. It's because we overconsume these processed foods low in fiber.
- 2.60** And then water. I learned this recently that just by drinking a glass of water before a meal, people tend to consume about 100 to 150 fewer calories during that meal. Water is hydrating. Sometimes, we can feel hungry when we're really just dehydrated. The reason for this is that for the vast majority of human history, we didn't have running water. We didn't have bottled water. We had to literally go and look for our hydration. Unless we were able to find a spring, where would the next best place be to hydrate ourselves? From food.
- 2.61** We all know that fruits and vegetables are rich in water. Even animal products—rich in water. We get hydrated when we eat it. I have a cat. Cats don't drink a lot of water. They get their hydration from food. We can see, again, we can look to ultra-processed foods. Ultra-processed foods tend to be dehydrated. Water moisture is one of the quickest ways to shorten a product's shelf life. These foods tend to be depleted of water. It's another contributing factor to their lack of satiety, to the fact that they are so not satiating.
- 2.62** All that is to say that we want to optimize for these three factors in our diets: protein, fiber, and water. Along with protein, you get great nutrition; you get animal-sourced micronutrients. Along with fiber, you tend to get great nutrition, vitamin C, magnesium, as I've mentioned. And then water, great to keep you hydrated for many reasons, to keep your metabolism running. So, these are the three factors that I think generally predominate most healthy dietary patterns. Protein, fiber, water.
- 2.63** Wrapping up this section, I do want to talk about the importance of fiber other than its satiety inducing effect, but we did touch on that. We've talked about the fact that it stretches at the stomach mechanically. It also helps to blunt the blood sugar impact of foodborne carbohydrates by slowing the transit of food and its constituent sugars to our small intestine, which is where we absorb our nutrients.

- 2.64** We see that dietary fiber consumption is associated with a reduction in inflammation, which shouldn't be misconstrued as me saying that suddenly eating a high-fiber diet and eating all the plants that you know that you're sensitive to is going to reduce inflammation. No, you have to assess for yourself what plants and what amount of fiber works for you. But you can generally increase fiber consumption if you do it slowly and cultivate the microflora, the bowel microflora, to be able to contend with that new dose of fiber.
- 2.65** Fiber recommendations are not one-size-fits all, but generally, we see that fiber consumption is associated with lower levels of inflammation. Part of it has to do with the fact that when we consume fiber, as I alluded to, we have this drug factory that churns out anti-inflammatory chemicals, one such being butyrate, as a response to fiber consumption. So, it seems to be beneficial; it seems to be associated with longer life and the like.
- 2.66** It's a great way to balance lipids and hormones. Fiber traps certain compounds in the body, like estrogen, like cholesterol, which can help our bodies balance those different compounds, which can be associated with disease, whether it's estrogen or other hormones with regard to certain hormone-sensitive cancers or cholesterol in the context of high cholesterol, which is something that we can talk about. There's a lot of misunderstandings there. But it 's super important.
- 2.67** You just want to make sure that you're increasing it slowly if you do decide to increase it because we 're not ruminant animals, so we don't absorb it. We literally rely on gut bacteria to break down fiber into these more bioavailable nutrients for us, whether they're postbiotics, like the short-chain fatty acids, like butyrate, which I've mentioned, or other such compounds. You want to increase your fiber consumption slowly.

Lecture 3: Nutrition Standards

At a glance — Nutrition standards and how dietary guidance is set.

- 3.1** What I've hoped to kind of illustrate, even thus far, is that nutrition really can be as simple or as complex as you want to make it. You know, you could distill nutrition recommendations to something as pithy as just 'eat whole foods.' But, again, everybody's different, and we come to this world of nutrition with different backgrounds, with different problems we'd like to solve. Ultimately, if nutrition were so intuitive, you wouldn't see the levels of chronic disease and sickness that we just see so much of today. So, hopefully, with this section, we can kind of dive in a little bit.
- 3.2** Also, to preface, I guess I should say that I'm not a medical doctor. Obviously, I didn't go to medical school. However, this has been my focus for the past ten years, and having books out that have reached the kind of audience that my books have been able to reach, has given me not a clinical lens, so to speak, but I've been able to see with, I think, fairly statistically significant sample size and effect size that a lot of kind of like what people are generally experiencing with all the varied diets that people tend to adopt these days, as well as the recommendations that I've made publicly, it's given me a pretty unique perspective. So, let's get into it.
- 3.3** So, I think first, it's important to contextualize the old and new models of disease and where nutrition relates to disease. I think for many years, certainly me growing up, I saw kind of the tail end of the demonization of fat, specifically saturated fat. I remember growing up in New York City; there was actually a supermarket close to my house called "Fat-Free Foods." It was like a supermarket; they only sold fat-free foods. It was kind of insane, and it didn't last very long, but it was definitely there, and I remember it.
- 3.4** And so, I think that still today, we see with various diet cults or tribes— however you want to refer to them online—we still see, again and again, the demonization of various nutrients. It tends to predominate saturated fat, but I think that it's not very helpful because, as I've mentioned, saturated fat— there's more nuance. There has to be more nuance brought to the conversation. Not all saturated fats are the same. There are saturated fats in coconut oil, there are saturated fats in dairy, and they all affect our bodies differently.
- 3.5** So now, with a more nuanced view, I think we need to transition from the old model, which is like 'single nutrients in isolation'—which few people consume in isolation; we eat foods; we've evolved with foods— to exactly that it's the 'foods.' It's the foods that we're eating that really matter. When it comes to the foods, most people are consuming the Standard American Diet, which, as I mentioned, predominates with these ultra-processed foods that are essentially made in a lab, made to drive—to

establish— loyalty and to yield repeat customers for these companies, and I think that it's a problem.

- 3.6** So, let's define the Standard American Diet. The Standard American Diet is typically high in added sugars, unhealthy fats, and refined grains. You could also say that it's very high in ultra-processed foods, but typically, when you break down what comprises these kinds of foods that we're talking about, it's an overabundance of added sugars, unhealthy fats, refined grains.
- 3.7** Standard American Diet, or the Standard Western Diet, is now linked to many chronic diseases, including obesity, type two diabetes, heart disease, and certain cancers, as well as dementia. And this is typically how we think about it—it's a mostly carbohydrate-based diet, about 45% to 65% carbs, 20% to 35% fat, and 10% to 20% protein. So, it's a high-carb, high-fat diet, generally. And though it contains adequate proteins so as not to drive population deficiency of protein, arguably, it's not sufficient in protein from the standpoint of body composition and metabolic health. We've already talked about why—the potential consequences of this.
- 3.8** If protein is the most satiating of the macronutrients, what would happen if the population, if we all ate a little bit more protein? The protein leverage hypothesis, which I alluded to, would stipulate that if we just ate, even incrementally, more protein, we'd be inclined to consume naturally because of just satiety, less carbs and fat, which clearly typify the Standard American Diet. I think what you can gain from this also is to see that it's not necessarily the carbs that are the problem. I think, arguably, our overconsumption of carbohydrates is a problem, but I don't think it's the problem, and I think it's highly context-dependent. We tend to overconsume fat, too.
- 3.9** It's the unhealthy added fats, it's the added sugars, it's the refined grains. So, I don't think that you can disentangle any one of these components from it. I think that is the milieu. That's the problematic milieu that we're contending with, and part of it has to do with the fact that carbs and fats are cheap. These oils that we make with soybeans, corn, and God knows what else—they cost pennies. Sugar has now gotten to a point where it used to be a luxury. It's now cheap enough where we can use it to sweeten up our foods and make those foods hyper-palatable so that people want to overconsume them.
- 3.10** And then it's these subsidy commodity crops, like the refined grains, corn, wheat, and rice. These are now 60% of the calories that your average person consumes. Sixty percent comes from, in fact, those three plants: wheat, corn, and rice, which, if you think about it, is really kind of funny because there are thousands of edible plants around the world, each with unique and distinct benefits. Most of our calories come from three nutrient-poor plants: wheat, corn, and rice.
- 3.11** It's not to say that ancient sprouted wheat that your grandma used to make sourdough bread is unhealthy. Or that corn on the cob comes summertime is unhealthy. I'm not

saying that, but the fact that our diets have become just so dominated by those three crops and the added sugar and the added—I mean, that literally is the problem right there. And they predominate in ultra-processed foods.

- 3.12** So, let's get a clear definition of ultra-processed foods because I've referenced them a bunch of times: foods that are industrial formulations that typically contain multiple ingredients, highly processed, often high in calories, added sugars, fats, and salt.
- 3.13** There's actually the definition of ultra-processed foods that came to us from a nutrient profiling system called NOVA. So, is anybody familiar with—recently, online, on social media, there was a viral thing, furor, surrounding a Tufts University nutrient profiling system that was devised called the "Food Compass." Some high-profile people on social media—okay, you guys are familiar with what I'm talking about—there was like a hierarchy of foods where this nutrient profiling system seemed to suggest that Lucky Charms, Frosted Mini-Wheats, and fake eggs fried in vegetable oil were somehow healthier than ground beef and poached eggs and things like that. It was like this crazy thing belched up from the halls at the Friedman School of Nutrition at Tufts University, and it just went viral.
- 3.14** And anyway, that was one example of a nutrient profiling system where there's great interest among regulators and the food industry to be able to promote food sales, but also to do it in a way where consumers get a better sense of what's better for them, what's in their best interests to consume more of, what's in their best interests to consume less of. So, this is like the latest offering from a highly respected university called, well, the university, Tufts University. And it was just like the latest example of a nutrient profiling system.
- 3.15** But all that is to say, the reason why I bring that up is because there have been other nutrient profiling systems, and I think the one that I've seen, that I'm most familiar with, that I think is the best of all of them, is called the NOVA. You can Google it. It was devised in Latin America, and basically, the way that this nutrient profiling system—this food profiling system—really worked was it just had four groups, and it categorized foods in accordance with the degree of processing that they underwent. So, it was either group zero or group one you had minimally processed foods. So, you could eat the foods in this category unlimited, unrestricted, and it was foods like meat, fish, eggs, fruits, vegetables, nuts, seeds, things like that. And then, it was level three or four, you would see the ultra-processed foods. So, the NOVA food scoring system actually coined the term ultra-processed foods.
- 3.16** So, these are the foods, essentially, that I think I have a definition. Yes, I do have a definition later, or an informal definition later on in my deck. But essentially, these are foods that you couldn't create in your own kitchen if you tried. You couldn't create these foods in your own kitchen if you tried. And so that's kind of like a good heuristic to use to help you better identify these ultra-processed foods.

- 3.17** But these really are—and again, these are foods that you can't necessarily say that they're all carbohydrate-based foods or they're all fat-based foods. They tend to be a combination of carbs and fat. They tend to be low-protein. The high-protein, ultra-processed foods tend to be the ones that, actually, I'm okay with, like whey protein. You could argue that Greek yogurt, or fat-free Greek yogurt, or some of these dairy products are ultra-processed—I have less of a problem with those. But in general, the ultra-processed foods that are the most harmful tend to be some combination of carbs, fat, usually with added salt.
- 3.18** Salt's not bad. Salt provides sodium, which is an essential nutrient, but we tend to overconsume sodium today because of our overconsumption of ultra-processed foods. So again, underscoring the fact that it's not the nutrients that are the problem. I think it's not helpful to fearmonger these isolated nutrients like carbs, fats, sodium. It's how we're getting them. It's the context that matters. Healthy—well, this should actually say 'Health Effects.' So, 'Health Effects of Ultra-Processed Foods.'
- 3.19** As I've mentioned, they do not promote satiety; they're not very effective at promoting satiety. They tend to increase mortality risk, they tend to increase dementia risk, they tend to increase cardiovascular disease risk, and they tend to increase cancer risk. That doesn't sound good, right?
- 3.20** There was a study that actually very clearly illustrated their lack of satiety-inducing effect. You guys can Google this. I think the year was 2018. The lead researcher was Kevin Hall, who's a well-known researcher in the obesity space. What they did was they basically took patients or subjects, they weren't patients, but they put them in an inpatient setting, in a metabolic ward, which is a very tightly controlled environment. It's very hard to do these studies, very expensive, very few people volunteer to do them, but they're very important because in nutrition, as I mentioned earlier, in the first, in the first hour, it's very hard to do controlled studies in the field of nutrition.
- 3.21** All we have, the vast majority of studies that we have are observational in nature, and that's because it's hard to motivate people. It's hard to get people to move into a hospital setting to do these kinds of studies, and there are just so many variables. It's just so hard to control them all. But this Kevin Hall study—it was a seminal study—got people to go into this inpatient setting, and they controlled everything that these subjects were basically allowed to eat. Everything was controlled for fat, calories, sodium, and they gave patients, or subjects, rather, access to a diet that was minimally processed. So, the kinds of foods that you could recognize, identify, probably cook for yourself in your own kitchen, prepare for yourself in your own home. When eating to satiety—the researchers, they call this ad libitum feeding. So, ad libitum basically means you could eat whatever you want.
- 3.22** And what they found in this study was that when you gave these subjects all-you-can-eat access to a minimally processed food diet for, I believe it was like a week, it was like seven straight days, if not more, could have been two weeks— and told to eat to a

point of fullness, like, "Don't restrict yourself; just eat until you feel like, I'm good, I'm going to leave the table now," they ended up eating effortlessly to a calorie deficit of about 300 calories. So, meaning, they were eating fewer calories than they were coming under their energy balance. So, they were eating at a calorie deficit, which is, all things considered, at least for a short period of time, maybe they weren't these subjects' favorite foods per se. It was being prepared for them. They ended up eating at a calorie deficit, which goes back to the satiety per calorie. So, they were eating clearly foods that had a very high satiety per calorie quotient, if you will.

- 3.23** And then—what they did in this trial, so it was like a crossover trial—then, they switched them to ultra-processed foods. So again, foods that you couldn't create in your own kitchen. Things like hot dogs with a bun, refined wheat bun with an ultra-processed meat product—I'm trying to think what else they gave them—basically, things like that, things where you couldn't possibly identify all of the different ingredients utilized. And they told them just to basically eat again to satiety. Like, "Eat to the point where you feel just as you did in the first arm of the study, to the point where you're just going to eat naturally and push yourself away from the table." They ended up eating at a 500-calorie surplus in this study, in that arm of the trial.
- 3.24** So, what that showed was that that was an 800-calorie swing, essentially: 300-calorie deficit in the first arm, 500-calorie surplus in the second arm. That was an 800-calorie swing determined purely by the quality of the foods that these subjects were eating. So, all that is to say, when we were talking about energy balance earlier, a lot of people will focus primarily on calories, which, as I've said, and as I will repeatedly say, calories matter, but they're not all that matter. And if that is where you focus your efforts, especially if you're on the Standard American (ultra-processed-food-dominant) Diet, you're putting the cart before the horse because the quality of the food in many ways dictates the quantity of the food that you subsequently consume.
- 3.25** And so that's why these foods are so dangerous and why they're so foundational to the obesity epidemic. And again, that's not to say that one here or there, or occasionally, or if you're a dessert person—having dessert once or twice a week—in the form of these ultra-processed foods is going to be a problem, but it's when we overconsume them; it's when they form the bulk of our diet.
- 3.26** We see that all of these problems, and granted, many of the problems associated with them, could potentially be tied to the fact that they are obesogenic, and weight—carrying an excess of body weight—really is problematic from the standpoint of chronic disease. But that's where we see that it's just again and again seemingly associated with all of these poor health outcomes. So, it's not one thing; it's many things.
- 3.27** Some of the characteristics of ultra-processed foods that make them harmful is the fact that they contain ingredients that you typically wouldn't use in your own kitchen. They contain industrially refined seed oils, which, just to define them, there are many different types of seed oils in the supermarket. We have sesame seed oil, which has

been used in Japanese cuisine, among other types of Asian cuisine, for millennia; that's not what I'm talking about when I refer to seed oils.

- 3.28** The kinds of seed oils that are typically found in ultra-processed foods are actually ultra-processed themselves. You couldn't make them in your own kitchen because they're not produced via pressing seeds. They go through a number of industrial processing steps; they're actually referred to as RBD oils: refined, bleached, and deodorized oils. Some of them include soybean oil, canola oil, corn oil. Again, you couldn't make them in your own kitchen.
- 3.29** And they do get a pass. These are very controversial these days. The medical establishment, the nutritional orthodoxy, they still consider these oils as being healthful because relative to saturated fats, in general, generally speaking, they will lead to a reduction in LDL cholesterol, but that doesn't abdicate them from all of the other problems that they potentially can cause in the body, which, I think, it's also important to recognize that we don't have long-term data on their utilization, but they are very oxidation prone because they're predominantly comprised of polyunsaturated fats, which I talked about earlier as being the most prone to oxidation.
- 3.30** And yes, they provide empty calories, they provide no nutritional value, and we don't actually have long-term data on what these oils do to our health, on other indicators of human health outside of the world of blood lipids, which seems to be the only thing that they—that that seems to be the one thing that the medical establishment seems to be so myopically focused on, what these fats do relative to other fats, more natural fats, to our blood lipids—that that's why they seem to get a hall pass.
- 3.31** But of course, there's a whole continuum of human health independent of blood lipids and the like. So, I think we need to really reassess how we think about these oils, and I'm going to talk more about them in a few, but ultra-processed foods also tend to contain emulsifiers, some of which have been indicted in promoting gut inflammation in animal models, animal models primarily, but we're now seeing the role that gut health plays in systemic health. And so, certain synthetic emulsifiers, I think, are, you know, we're probably better off avoiding.
- 3.32** Artificial sweeteners—this is sort of an area of debate. I'm not convinced that artificial sweeteners are toxic or the poison that many people make them out to be. But, I think, at the very least, all of these ingredients are essentially proxies for ultra-processed foods. And so, it's just 'the dose makes the poison' with all of them. And it might not be necessarily one aspect that is driving disease, but it's the sum total aggregate exposure that could potentially be harmful, and we just don't have long-term data. So again, we see that these kinds of foods are related to disease, and it could be potentially any one or any number of these aspects that is contributing to those observations.
- 3.33** Added sugar is one I think that is particularly problematic. The most problematic aspect of added sugar, I think today, in the context of the Standard American Diet—one is the

insidious nature of it. So, what 'insidious' means is that it just kind of creeps in, and in aggregate, it adds up to a lot, even though it doesn't seem like a lot as we're ingesting it. A little bit here and there over time, over the course of the day, over the course of the week, over the course of the month, adds up to a lot of sugar. Today, your average adult consumes about 77 grams of added sugar every single day, which, just to visualize, that's about 20 teaspoons of added sugar. This is not sugar from fruit. This is not sugar from fresh produce. This is the sugar that food manufacturers will add to their products, usually with the explicit intent of driving overconsumption.

- 3.34** We see that there are a number of problems associated with added sugar. The biggest problem is that it's empty calories. That, again, we have no biological necessity for it. We live in, contextually, we live in a world where many people are struggling with issues related to their weight. It's empty calories; it contributes to the hyper-palatable nature of food; it displaces the calories that we ingest in the form of added sugar, displaces more nutritious calories—added sugar provides no nutritional value, so it displaces more nutritious foods—and there's some evidence that the higher glycemic, meaning, the more sugary our diets are, the more hunger we tend to feel.
- 3.35** So, people might be familiar with blood sugar. When you're on a blood sugar roller coaster throughout the day, it tends to perpetuate feelings of hunger. Being able to identify added sugar in your diet and minimize your consumption accordingly, I think, probably very important in terms of helping to regulate your hunger levels, which is super important.
- 3.36** There have been a number of studies that have shown us that even single high-sugar boluses can have a health impact. There was a study that found that a single high-sugar bolus led to an elevation of systolic blood pressure in a way that sustained for two hours post-ingestion. We also saw in another study, I believe in men, that a single high-sugar bolus led to a 25% drop in testosterone, and that persisted for two hours.
- 3.37** Now, to be clear, these studies tend to use pure glucose and a pretty high dose, about 75 grams of pure glucose, reason being that they will usually do this as part of what's called an oral glucose tolerance test. In those tests, they use 75 grams of glucose, typically, and that's not typically how people ingest sugar. We're very seldom ingesting pure glucose—even table sugar is about 50% glucose, 50% fructose—but that is roughly the amount of sugar that we each consume over the course of the day. As I mentioned, 77 grams of added sugar. So, it's likely not doing our health any favors.
- 3.38** But in general, I would say the biggest problems with added sugar, to be clear, it's not inherently fattening, it's not inherently toxic, the dose makes the poison, but it's the insidious nature, it contributes to the hyper-palatable nature of foods, and it's empty calories, and the fact that it can perpetuate hunger. Those are, I would say, the biggest issues with regard to added sugar.

- 3.39** So, we talked a little bit about seed oils. Again, these are the refined, bleached, and deodorized fats. They're very easily oxidized, which is a problem, and I'll talk about some of the conditions that can catalyze that process. Generally, they're a good proxy for ultra-processed foods. Do you guys know what that means? When I say it's a proxy, it's basically like a surrogate marker. So, generally, the more seed oils in a person's diet, the more ultra-processed foods they tend to be consuming. So, it's a good indicator of how processed one's diet is, generally speaking. So, it's not a foolproof indicator, but probably at the population level, can be used to predict just the degree of processing in one's diet because ultra-processed food manufacturers tend to use seed oils because they're cheap.
- 3.40** So, earlier, I talked about trans fats and how the most nefarious source of them were these partially hydrogenated vegetable fats, which have been outlawed for the past ten or so years. But trans fats—man-made, artificial trans fats—still exist in the human food supply, and that is in the form of these refined, bleached, and deodorized seed oils. The production process actually generates a small but significant amount of trans fats when these oils are exposed to high heat during the production process. And a little bit here and there, I mean, we know that there's no safe level of trans fat consumption, but ultimately, the dose makes the poison, which is something I've said a number of times, but it tends to be true.
- 3.41** That being said, with your average person now overconsuming these fats, I mean, our consumption of them has increased for some of them a thousandfold compared to 100 years ago. We generally are ingesting still a significant proportion of these fats, which we know are very damaging to our cardiovascular systems and also associated with worse brain health and brain function, even among the young and healthy.
- 3.42** The setting in which we really want to be diligent in reducing our consumption of them, I think it's worthwhile to not bring them into the home, but they're particularly harmful—we have good evidence on this, or at least better evidence on this—when they're used in the restaurant setting and particularly in fryers. So, nobody out there—medical orthodoxy, nutritional establishment—nobody is out there saying that fried foods are good for you; fried foods we know are not good for you. The reason for that is that these easily oxidized oils are kept at a temperature in the fryer setting for, at the very least, hours. What this does is, it not only oxidizes the oils, but it sets the stage for various oxidative byproducts to be created. Compounds like aldehydes, known carcinogens, are formed in the fryer setting of these oils.
- 3.43** Now, I did an experiment recently where I was at a restaurant, and I saw the fryer. I saw a deep fryer setup, and I posted it on Instagram, and I asked my followers how frequently they thought this oil gets changed. I'm very grateful to have a lot of people follow me at this point, and many people who follow me—who'd have thought—work in the restaurant setting, and so what a lot of people told me is that these oils, even though they might have guidelines for them to be changed at a certain interval in the

real world setting, these oils sometimes don't get changed for weeks. They get strained, but they don't get changed sometimes for weeks. These oils are toxic in that setting. So, you really have to be careful with fried foods. I think that probably, if you can just do your best to minimize your consumption of fried foods, if not altogether cut them out, which is, you know, that's a heavy lift today. I get it. French fries are delicious. I recognize that. Falafel—great.

- 3.44** But what they've shown is that when taking these oils out of that fryer setting, it wouldn't even be ethical to do this study in people. They've done it in mice, where they will give mice these oils, and they see that it upregulates markers of inflammation. In mice that are genetically prone to developing tumors, they see an acceleration of tumorigenesis. They see an increase in leaky gut in these mice compared to other oils, I believe, olive oil they've used. They used olive oil as a control. So, incredibly unhealthy, particularly in that setting.
- 3.45** Now, are these oils in the form of salad dressing—if some of them make their way into your house—are they dangerous? Again, I think the dose makes the poison, but you're probably better off avoiding them because you don't know how they're stored, how they're shipped, what degree of heat or other chemical exposures they may have had over the course of the production process. So, generally, you want to do your best to minimize these oils.
- 3.46** Also, I'll add, we don't know the long-term impact of these fats on our brains, which is something that nobody has stopped to question along the way—that our brains are made of fat. Our brains are made of polyunsaturated fats, for the most part, these very vulnerable, damage-prone fats, and not every nutrient that we ingest has access to the brain. The brain is protected by what's called the blood-brain barrier. But polyunsaturated fats, of which these oils are largely comprised of, have easy entry into the brain. So, if we're consuming these damaged fats, it stands to reason that they could be, in an insidious way, damaging our brains, and we just don't have any long-term data to say otherwise. So, I think skepticism and caution is very important.
- 3.47** Just to give you a sense—this graph, I hope you can see it. I know it's a little bit small, but this is how these refined, bleached, and deodorized oils are made. So, this gives you a sense of the processing that they all undergo, which isn't to say, again, what I don't want to imply is that all processing is bad processing. Just the fact that something is processed doesn't necessarily make it healthy. But if you take something like olive oil, which you can see on your right, we've been making olive oil for thousands of years because the way to make olive oil is you crush olives. So, you crush olives, you package it, and ship it. Canola, corn oil, soybean oil, grapeseed oil, sunflower oil, safflower oil—it goes through all of these myriad industrial processes, and these oils are made of these damage-prone fats. So, we've got the refining, pre-bleach, some of these oils are destined for hydrogenation.

- 3.48** You can still find fully hydrogenated oils on the market, but people often have this misconception—it's actually the partially hydrogenated fats that are the most damaging. The fully hydrogenated fats, to my knowledge, are not as damaging; they behave more like saturated fats. They actually aren't trans fats. But still, so, many are still fully hydrogenated. And then they continue to undergo this processing, de-waxing, and the n, down here, you see deodorization. Deodorization they all undergo, and that's sort of like the food industry's equivalent of the Witness Protection Program.
- 3.49** So, what deodorization does is it takes these oils, no matter where they come from—corn, soybeans, sunflower seeds, and the deodorization step basically makes them bland, removes any aroma, gets rid of any bitter flavors, basically makes these oils so that food manufacturers can use them to roast nuts in, to use in salad dressings, to use in spreads, to fry foods in, and this exposes them to very high heat. And it's that step in the production process, the deodorization step, that generates trans fats. So, they all undergo this step, generally. Whether the oil is expeller, cold-pressed, organic—they all undergo that deodorization step.
- 3.50** And this is another reason when you look at this hierarchy of processes, why these oils are essentially ultra-processed because you could not make them in your own kitchen. You could hypothetically squeeze soybeans to get the oil, but you'd have to take pounds and pounds and pounds of soybeans, and you would not end up getting the soybean oil that I'm talking about when I say soybean oil. So, generally, you just want to be very, very careful. If I recall correctly, peanut oil smells like peanuts. So, I would assume that it probably doesn't undergo the deodorization step. It's, in general, the ultra- processed foods.
- 3.51** So, I want to be clear in that I think that seed oils are a problem, but I don't think that they are the smoking gun, the root cause source of all of our problems. I think, in general, it's the foods. So, I don't want to be guilty of making the same mistake that our regulatory agencies have made over the past few decades of isolating out single nutrients. I think it's the entire milieu. So, as best as you can avoid these ultra-processed foods, the better.
- 3.52** One of the other reasons that I mentioned that they're so harmful is that they basically subjugate the food matrix and they extract all these nutrients, and we lose a lot when we disregard the food matrix. We miss out on the fiber, which is satiating; we miss out on the hydration, which is satiating; we miss out on protein; we miss out on various compounds in foods that amplify the effects of other compounds that they're co-packaged with. We call this the entourage effect, actually.
- 3.53** So, the entourage effect suggests that what nutrition science has done really well is identify many of the essential nutrients that are found in food, but something that has paradoxically arisen in the literature is that when you give the constituent nutrients that we think are the more beneficial aspects of these foods to people in supplement form, we don't always see the benefits that we would anticipate. I think that has to do

with the fact that it's the food matrix that's really responsible for a lot of these benefits that we see when people regularly eat healthy foods. So, you can't necessarily distill what we're seeing, the health benefits of regularly eating blueberries to a supplement. It's that there are other compounds in blue. There are myriad compounds in blueberries that have an entourage effect, a synergistic effect, that ultimately are what seem to be responsible for the health benefits of foods like blueberries.

3.54 So, I had Marissa Pier, who's a therapist, on my podcast, and I really loved the way, this heuristic, that she came up with for how to recognize real food. So, this is a credit to her, but I really love this. And so, these are the five R's of real food. How do you recognize real food? So, the five R's:

3.55 Does it Roam or Ripen? Is it an animal or has it been grown?

3.56 Does it Rot? Most ultra-processed foods that you see on the food shelves—they don't rot; they have infinite shelf lives. So, that's a good way to recognize food that you probably should be minimizing. You want foods; you want perishable, fresh foods, the foods that hang out around the perimeters of the supermarket.

3.57 Can I Recognize it? If you were to look at the food, could you recognize all of its various ingredients? One thing that I like to say is real foods don't have extensive ingredient lists; they are the ingredients. Can you recognize all of those different ingredients? If I go eat at a restaurant and I enjoy a nice Mediterranean meal, I can visually identify all of the components of that meal: I can see the olive oil on the plate; I can see the olives; I could see the piece of salmon; I could see the beans that they use; I could see the tomato sauce—I can literally, with my own two eyes, identify all of the ingredients that meal.

3.58 Where things get murky when you go to restaurants, and you see these mysterious sauces that cover our food, and you can't, you don't know what's in it. Is there soy sauce in it? Is there, is it gluten? Is it cream-based? Once you start to lose track of that connection with what it is that you're eating, I think that problems arise. In the ultra-processed food setting, very obvious—you couldn't possibly recognize, just looking at an ultra-processed food, all of the ingredients that it took to create.

3.59 Could you eat it Raw? Now, this one actually confused a lot of people because they were like, well, you can't eat meat raw, but, actually, if you go to parts of the world, like the Mediterranean region, if you go to Japan, where people live a long time, they're eating all kinds of things raw. They're eating beef raw. I've had a delicious Italian dish made of raw beef called Carne Cruda. Obviously, Carpaccio is made using raw meat. Raw fish is commonly consumed in sushi. In Japan, they even have a form of chicken sashimi. They eat raw chicken. You can't, I mean, I'm not telling anybody here to eat raw chicken.

3.60 No, raw chicken, you can, actually, there is nothing—I learned this from somebody, a farmer, that I'm friends with. The reason why, and again, do not go and eat raw

industrial chicken, but the reason why chicken is so dangerous to eat raw—it's because the chickens have tiny bodies, and when they're mechanically—t his sounds horrible—but when they're mechanically eviscerated, there's just cross contamination of fecal matter everywhere, just in the industrial production chain, but if you were actually a farmer producing your own chickens, there's nothing implicitly dangerous about eating raw chicken. Is eating raw industrial chicken dangerous? Absolutely. I wouldn't do it. But, all that is to say, you can essentially eat natural foods raw, in their raw state. So, that's another, I think, important aspect of this.

3.61 And then, could you Recreate it? I mean, generally, if you could recreate something in your own kitchen, that's a pretty good indicator that it's minimally processed food and thereby being implicitly better for you than one of these ultra-processed foods. I love that. Five R's of real food. I wish I came up with that.

3.62 So, the big wins. We generally want to achieve a healthy body composition. That's super important. We want to eat foods that are satiating and that provide fewer calories— more satiety per calorie, generally. Those kinds of foods are going to be the minimally processed foods. We want to make sure that our foods are nutrient-dense so that we can attain nutrient repletion with an abundance of highly bioavailable micronutrients. We want to ingest macronutrients that support our body composition and ultimately support our musculature because what we're starting to see is that muscle is one of the often overlooked vital signs of vigor and of robustness. And so, in the sections to come, we're going to talk about how to best support your musculature, which is crucially important. One of the primary macronutrients to do that is protein. And so, we're going to get into the nitty gritty there.

Lecture 4: Nutrient Timing

At a glance — Nutrient timing — when to eat for best effect.

- 4.1 Part four: "Nutrient Timing". Okay, so this is kind of like where we diverge a little bit from nutrition and nutrients, and we start to talk about the more extrinsic variables surrounding eating, which I think is super exciting. So, we will talk a little bit about circadian biology—is anybody here unfamiliar with the term 'circadian biology' or 'circadian rhythms?' Okay, cool. So, that basically, what that implies is that our biology, in many ways, is influenced by a clock that functions on a 24-hour timer. And so, we're going to talk a little bit about that.
- 4.2 Time-restricted eating, intermittent fasting—I'm sure you guys have plenty of questions about that, but we will unpack each. The form of time-restricted eating that I think there's a lot of, there's a growing body of evidence supporting, has been referred to as Early TRE—TRE stands for time-restricted eating.
- 4.3 We want to talk about peri-workout nutrition. So 'peri' means 'around the workout.' We've got pre-workout, post-workout, peri-workout. Nutrient timing and sleep, which we already kind of alluded to in an earlier Q&A, and then fasting as a hormetic stressor. So, these are all worth touching on.
- 4.4 So, circadian biology—basically what this implies in this field of research that is really exciting—we are programmed to respond to cues in the environment that alert our genome as to what time of day it is. This affects everything from when we feel at our most alert, when we feel at our most coordinated, when we feel sleepy, when we metabolize our food most effectively, when we feel most energized, when our bodies heal most efficiently. I mean, every, nearly every aspect, every organ system is responsive in one way or another to time of day.
- 4.5 What's so exciting about this is that, well, there's a few things. For one, how do our bodies know what time it is? I mean, time is actually a human artifact, but nonetheless, our bodies need to have a window into the time of day, and the primary time setter that our brains and bodies use is light.
- 4.6 So, when we perceive a certain degree of light intensity, it triggers a little part of our brain called the suprachiasmatic nucleus, abbreviated the SCN, and that sets off this 24-hour cycle, which, again, influences when we feel our most alert, digestion—our digestive system is primed to receive food earlier in the day, and it's less efficient at assimilating and metabolizing nutrients later in the day—nearly every aspect of how we feel is influenced by light.
- 4.7 And so, upon waking, getting in that bright light through our eyes—crucially important. A lot of people are talking about this today, but our circadian rhythms are as old as time. So, it's wonderful that science is now catching up to illustrate all the many

different ways that we have a synergistic symbiosis with the sun, which is really what it comes down to.

- 4.8** Even though light is the primary time setter, it's not the only time setter. So, we also have a periphery to our central clock. We have what are called peripheral clocks, and the most important peripheral clock is essentially our digestive system. So, when we eat. We're starting to see that food eaten at certain points in the day can affect everything from glucose homeostasis—meaning how we respond to sugar— but then also our metabolic rate, our subjective feelings of hunger. So, what if you could eat at a certain time of day that makes you less inclined to feel hungry after that meal? So, this is where nutrient timing comes in and plays a role.
- 4.9** Now, just to be super clear, what you eat still matters more than when you eat. So, when you eat is important, but what you eat is more important. So, I think that in the hierarchy of things that matter, it's sort of like Pareto's principle, meaning 80% of what you should be concerned about should be the foods that you're eating, but then if you've got that 20% at your disposal to play with, then when you eat does matter. I don't think anybody would argue that. We just want to be clear in terms of where we should most spend our efforts. After this lecture, you guys are going to have a good sense of what to eat. And so when you eat, it can play a role in terms of affecting your health, subjectively, how you feel, and the like.
- 4.10** So, among the different paradigms that we have at our disposal for meal timing, a few have really become, I guess, popular these days, and not just popular on social media and in books, but in terms of what scientists are now studying. So, among the different paradigms that we have, there's intermittent fasting and time-restricted eating.
- 4.11** Now, intermittent fasting is basically the umbrella term under which we can put time-restricted eating—time-restricted eating is a form of intermittent fasting. Intermittent fasting is really very broad. So, intermittent fasting can be anything from going on a 24-hour fast once a week to fasting on a 24-hour period. Some have referred to it as 16:8, where you fast for 16 hours every day, and then you ingest your calories, and you get your nutrition in a span of eight hours. So, intermittent fasting is essentially very broad.
- 4.12** But time-restricted eating, what that refers to, is restricting your food consumption to a certain number of hours each day. Some will do eight hours, as I mentioned; some will do 10 hours; some will do 12 hours. Now, to give you context, your average person today is consuming food 16 hours a day. So, from the moment they wake up to just before they go to sleep, they are digesting and metabolizing food. And you're actually, you're metabolizing food—if you're eating food for 16 hours a day, you are digesting food for even longer than that because you digest food for an hour or two after you eat it. So, if you're eating something up until the point at which you go to bed, you're still digesting that food hours into your sleep, which is probably a time in the day when you should allow your body's resources to be utilized by other functions.

- 4.13 So, there are many different ways to practice time-restricted eating, and I think earlier, when this started to be a focus for research, it wasn't clear the best or most biologically appropriate or relevant way to practice intermittent— time-restricted eating. So, they've done studies where they've looked at people who delay their first meal until 2:00 p.m., and they don't eat their last meal until about 10:00 p.m. And so, they've done a number of studies where they've compared these different paradigms, and what they've seen, or what we're starting to see, is that early time-restricted eating seems to yield the most benefits. Now, why would that be? And first of all, what is time-restricted eating, early time-restricted eating? So, early time-restricted eating is essentially eating,
- 4.14 beginning your calorie consumption for the day, your food consumption for the day, about an hour after you wake up. So, we'll just say you wake up at 7:00 a.m., eating breakfast at around 8:00 a.m., and then, typically, within a general time-restricted eating window, you might eat from 8:00 a.m. until 8:00 p.m. So, that's a twelve-hour feeding window. So, already, you're cutting down from the average, your time-restricted eating compared to your average person. But, it seems to be the case that when people eat earlier dinners—so, from 8:00 a.m. to, for example, 6:00 p.m.—there seems to be an improvement of various indices of metabolic health, various metabolic health biomarkers, whether it's blood sugar or blood pressure—the people who eat earlier dinners, even when all calories are controlled.
- 4.15 You take the same amount of food that you would eat otherwise over the course of 16 hours, and you concentrate it to that eight- hour window skewed earlier in the day, that there seems to be better control over glucose levels and that it seems to coincide with better blood pressure as well, systolic blood pressure, which I think is really exciting because, at the very least, not everybody has access to super healthful food. As I mentioned earlier, we have people that still live in food deserts. And so, something as simple as just eating an earlier dinner might be a really potentially helpful intervention for people that don't necessarily have access to healthier foods or for people—a lot of this nutrition information that we put out is put out in English, for example, what about the massive global population of non-English speakers? You hear something as easy as 'eat an early dinner.' That's a really easy modification that doesn't require a whole ton of nuance and could be potentially very beneficial for some people. So, there's a lot of exciting, I think, potential here with this line of research. And so, it seems to be beneficial from the standpoint of these indices.
- 4.16 Now, how does it affect weight and body composition? It doesn't seem superior to just regular old calorie control. So, it's not that intermittent fasting is like this magical means of controlling calories or for losing fat or anything like that. Unfortunately, I think, at times, it can get marketed as such. The research doesn't support that. But from a body composition standpoint, it does provide an effective way of providing parameters to just control calories in a fairly effortless way.

- 4.17 Also, I'll add that many people feel that they're able to not eat, but then, when they have that first bite of food, that's when their hunger gets kicked off for the day. There's a reason why in French cuisine, you have a *muse-bouche*, like an appetizer that's meant to literally stimulate hunger. It's why they bring the bread basket at restaurants. Many people feel like once that they're able to have a more binary approach to food, that it's helpful to them. Not everybody feels that way, but some people do.
- 4.18 The other thing that makes time-restricted eating very effective from a calorie control standpoint, and just from a body composition standpoint and health standpoint, is that after dinner, what are the kinds of foods that most people end up consuming after dinner, particularly when sitting around the TV? Sweets. So, nothing good happens after dinner from a nutrition standpoint, typically. So, if you're able to just integrate a parameter or a guardrail and use that as a tool, then usually skimming off calories at the latter end of the day that tend not to be the most nutritious calories to begin with—very effective. So, as a tool, I think, probably worth experimenting with.
- 4.19 There have also been studies that have shown that late-night eating increases subsequent day hunger and, to a small but significant degree, decreases metabolic rate the following day. So, when we talk about calories in, calories out, these are variables that we both have agency with regard to controlling, and the calories outside of the equation might actually be affected by late-night eating in the sense that we might actually expend fewer calories by eating late at night. There was a study that came out last year suggesting this.
- 4.20 So, as I mentioned, calorie control, circadian alignment, may have the effect of aiding with digestion, potentially increasing metabolic rate, helping to better regulate hunger hormones, hormones involved in hunger. But there are some potential downsides.
- 4.21 So, one downside, an obvious downside, is that intermittent fasting, time-restricted eating can be socially difficult, whatever your fasting paradigm may be. We can't overlook the crucially important role that social connection plays in good health. We see that social isolation is a risk factor for depression, a risk factor for dementia, for all-cause mortality, and eating an earlier dinner—dinner tends to be our most social meal. It tends to be a lot easier to—we tend to eat breakfast in isolation, but dinner is a social meal. So, these are considerations that need to be made.
- 4.22 The other thing that's really interesting that we're actually starting to see in the literature is that intermittent fasting might not be ideal from a muscle composition, body composition standpoint. It can help us lose fat, certainly, but the research on the maintenance of lean body mass and muscle accrual—you generally want to have your protein, it's ideal to have your protein intake spread out over the course of the day, an evenly distributed protein intake across the span of the day. So, if you're curtailing your consumption of food to 6:00 p.m., then that's a window of opportunity, potentially, that you're missing out on to ingest protein later in the day. Now, as I mentioned, most people are not consuming high-protein foods at the end of the day; they tend to be

consuming mostly ultra-processed foods comprised of carbs and fat, but we do tend to see that when we spread our protein intake out evenly over the course of the day, that tends to be the most optimal from the standpoint of lean body mass accrual and maintenance. Does that all make sense? Cool.

- 4.23 There was another study that found, I believe it was in men, that intermittent fasting was associated with a reduction in testosterone. So, let me just recap and make sure I didn't miss anything. So, I think the way that I like to think about time-restricted feeding—I don't eat for an hour after I wake up, generally, and I don't eat for two to three hours, generally, before I go to sleep. I think the body likes consistency. So, whatever it is that you decide to do, I would pick that.
- 4.24 And then, the whole protein issue—protein timing, definitely, it doesn't not matter, but it is secondary to making sure that you're consuming adequate protein over the course of the day. So, the number one most important thing is making sure that you're consuming adequate protein, and I'm going to talk about how much protein you should be getting over the course of the day.
- 4.25 So, protein timing does matter, but it depends on your goals. So, if you are a bodybuilder, if you are an athlete, then it might behoove you not to necessarily practice early time-restricted eating and to make sure that your protein intake is consistently spread over the course of the day. I'm personally not an athlete, although body composition and fitness does matter to me, but I feel content that by consuming adequate protein over the course of the day, I'm basically, again, Pareto principle, getting myself 80%, if not more, of the way there.
- 4.26 And also, digestion occurs better earlier in the day as well, and so sometimes people find that eating late at night—your peristalsis is not what it is late at night, it's just not as effective—and so people can have digestive symptoms, side effects from eating late at night. So that's just not something that doesn't sound good to me. A lot of bodybuilders and stuff like that—they contend with digestive issues and things like that. So, there is pluses and minuses to each approach, but generally, it's important to note that we're diurnal creatures; we're meant to eat largely during the day, not at night, and then to not eat too late at night.
- 4.27 The other potential benefit of fasting is that it can be utilized as a form of hormetic stress. Has anybody not heard of the term hormesis? Okay, so just to give a 30,000-foot overview: a hormetic stressor is basically a type of stress or exposure to something that, at a high dose or at a high level, might be potentially deadly, but at a low dose, actually provides a strengthening effect to the system, and we see this in many different areas.
- 4.28 We see this with exercise—exercise is the perfect illustration of a hormetic stressor. Exercise is good for us in part because it's bad for us. If you were to over-exercise,

you'd probably compromise your immune system; you'd experience excessive wear and tear of your muscles and joints.

4.29 We see this with the data on sauna use. It applies a form of heat stress to the body that could potentially kill you, but because your body knows that this is something that it's going to need to adapt to, it actually elicits a cascade of really beneficial adaptations to the body that make you more resilient.

4.30 There are foodborne compounds that actually serve as hormetic stressors. A lot of the so-called plant defense compounds that are found in fresh produce actually have a beneficial effect, in part because they're toxic. Our bodies sense those toxic compounds and put up essentially a scaffolding and institute an effort to defend itself against these compounds or these activities or these exposures. That actually leaves us better off. It's literally the biological equivalent of 'What doesn't kill you makes you stronger.'

4.31 And so fasting—fasting is a form of hormetic stressor. We're depriving our bodies of food during a certain window, and this has the ability, the effect of actually activating certain genes, certain pathways that are clearly involved in this strengthening process.

4.32 There's a researcher at the NIH named Mark Mattson, who's done a lot of this early seminal work showing in animals that fasting increases compounds in the brain, proteins in the brain that promote neurogenesis, that promote neuroplasticity, one of them being BDNF or brain-derived neurotrophic factor. We see that fasting activates gene pathways that are involved in calorie restriction, which makes sense. I mean, we're not ingesting calories, but some of these pathways are involved in longevity. So, there's been a lot made of fasting as a longevity and calorie restriction, as a longevity-promoting intervention. A lot of this has been done in animals.

4.33 Also, we have to take note of the overall degree of stress in our bodies and in our lives before just adding more stress to the system. A lot of people that have chronic stress, drink a lot of coffee, perhaps are underslept, are exercising regularly—we have a finite capacity for dealing with all of these different stressors, and so we just have to be mindful of this. But, yes, fasting is another form of hormetic stress.

4.34 Now, again, it comes back to individual goals. For somebody who is looking to reduce weight or build muscle—is hormetic stress something that you need to be concerned about if you're exercising regularly? We have to prioritize and triage our approach, I think, but it is worth talking about this research because it's pretty compelling. The brain health benefits, I think, are pretty interesting.

4.35 Time-restricted eating seems to help us get a better handle on hunger. So, if this is a tool that works for you, I think it's something that you should experiment with. At the end of the day, we want the ultimate outcome to be better body composition. We want to be able to better have a handle on our body composition, our waistlines—crucially, crucially important.

- 4.36 So, one of the most effective tools in our fight for better body composition is protein. Now, there used to be this concept called the anabolic window, which is worth talking about because we still see some people promoting this concept online. This is a concept that has been largely debunked with some stipulations, with some important nuances.
- 4.37 So, the idea of the anabolic window used to stipulate that in the post-workout setting, you needed to run to get protein, run to ingest protein as soon as your workout ended, or you may as well have not worked out. Gains lost. That was sort of the dogma in the fitness space for a long time, but I think, in part, that was driven by the producers of protein powders because if they can tie their consumption of a product to a particular ritual, in this case being the post-workout setting, then it basically inspires greater adherence to the product. But we now know that the anabolic window is a lot wider than was previously thought, and generally, if you're consuming enough protein over the course of the day, those are the big wins. That's the big win. It's sort of like if you were going to fill up a jar with rocks. You put the big rocks in first and then worry about all the little gaps where you could fill it with sand and whatever, but it's the big rocks that we're going after.
- 4.38 This is relevant to early time-restricted feeding and intermittent fasting because, again, spreading your protein intake out throughout the day is definitely going to be optimal. But for the vast majority of people who are not athletes or bodybuilders, the number one variable to get a handle on is your total protein intake for the day. Now, when is there potentially a better time to consume protein compared to others? The one time when you would want to consume protein very soon after your workout would be if you are working out fasted.
- 4.39 So, generally, if you consume breakfast or if you've got food in your stomach—protein, to be specific, circulating amino acids—in the post-workout setting, there's no need to run to go and consume protein post-workout. But if you work out fasted first thing in the morning, like many do, like I used to, that's when post-workout you want to consume protein right away, and the reason for that is that when you exercise, you stimulate, well, you're stimulating muscle protein synthesis, but you're also stimulating muscle protein breakdown. And when you consume protein in the post-workout setting, you're basically pumping the brakes on muscle protein breakdown, and you're providing your body with the nutritional stimulus and the building blocks to support muscle protein synthesis, which you want. Your body's muscle, the ability of your body to accrue muscle, depends on this balance between muscle protein synthesis and muscle protein breakdown. So, you want muscle protein synthesis to be high and you want muscle protein breakdown to be low, and consuming adequate protein is the best way to do that.
- 4.40 When it comes to the amount of protein that you should be consuming every day, well, the RDA is 0.8 grams per day kilogram. Okay. To find out your weight in kilograms, I

think you just multiply by 0.45—I could be wrong, but I think that's what it is—and that's your weight in kilograms.

- 4.41** As I mentioned, the RDA is not calibrated for optimal health. So, what the data suggests is that from a body composition standpoint, and to support the effort that you're putting into your fitness, which you should be putting in effort into your fitness, you actually want at least double that. If you're not practicing a fitness regimen for any reason, 1.2 grams of protein per kilogram seems to be effective. But anybody that I'm speaking to that I'm advocating to should be resistance training, and so this is what I try to get every single day: at least 1.6 grams times my body weight in kilograms for optimal body composition. You'll see bodybuilders go even higher than that, up to like 2.2. So, that's double the RDA. In pounds, just to do a rough and dirty calculation, you can do 0.7 times your body weight in pounds to get a rough estimate.
- 4.42** So, protein is really important for providing the raw materials to create new muscle and to sustain the muscle that you have. The most important amino acid, the necessary but not sufficient amino acid in protein, in whole proteins, is leucine. And so you get the highest proportion of leucine in animal proteins compared to plant proteins, and that's one of the reasons why animal protein is so beneficial. But you'll see people that are dieting down to fitness, like competition level, body fat levels, that actually increase their protein because it helps you to sustain your lean mass.
- 4.43** For people who are on weight loss diets—if you're eating more calories, like more carbohydrates, carbohydrates are protein-sparing. So, the more calories you're consuming, the less protein you can get away with. But for people that are dieting down to fitness levels, definitely try to consume more protein as opposed to less. So, that's generally what you want to ingest on a daily basis. Again, ideally, spread evenly throughout the day, but it doesn't matter if you want to break it down into three meals or even two meals.
- 4.44** There was also a myth, it was a myth for a while, that we only utilize 30 grams of protein per meal. Not true. Your body will utilize any amount of protein that you give it. And so, crucially important—protein—should be like, you'll hear a lot in the fitness world to prioritize protein, and this is why. It can be difficult to get because it's so satiating, as I've mentioned. So, this is where protein shakes and things like that come into play and can be very, very useful.
- 4.45** Nutrient timing and sleep— all I'll say about this: you don't want to consume food too close to bedtime, generally speaking, because it can potentially interfere with sleep. We know that anti-inflammatory diets, diets that are not super-processed, seem to be associated with better, higher-quality sleep.
- 4.46** Now, if you really are trying to maximize your muscle accrual, protein before bed, there is data that suggests that protein right before bed seems to enhance recovery compared to no protein before bed. But, again, I think that's splitting hairs for not just

your average person but your average person who's interested in fitness and wants to optimize body composition. I don't think that you need to consume protein right before bed to achieve a fantastic body composition, but if you really were splitting hairs and wanted to and were a fitness competitor, then eating protein, like a protein shake, before bed seems to facilitate muscle recovery.

- 4.47 The one issue with protein before bed that I can see, and I had a conversation recently about this with Satchin Panda, who's one of the world's leading circadian biologists, is that protein has a thermogenic effect. So, we actually burn calories when digesting protein. All foods have a thermic effect, and protein has the highest. So, about 30% of the calories that we ingest via protein are actually burned off via the digestive process, and so that generates heat.
- 4.48 Now, when we want to go to sleep, our bodies actually want to cool down. So, this can actually be potentially disruptive to sleep, it's thought, at least Satchin thought so. I haven't seen any evidence on this, but it makes sense to me that if you are generating a lot of heat, given this digestive process, and you're consuming a lot of protein before bed, it's a problem. And then also, as we mentioned, consuming food late at night seems to be able to impair next-day hunger and hormones that regulate metabolism.
- 4.49 So, again, these are more granular details, but something that I always say is, "You don't want to major in the minors." At the end of the day, I think approaching quality is super important. The overall diet quality is super important. Fasting—if this is a tool for you, if it's a tool that works for you, then, by all means, use it. It's not necessarily superior to other forms of calorie control, but it exists. So, make use of it if you so choose, and I think that's pretty much it for this portion of the seminar.

Lecture 5: Dietary Tailoring

At a glance — Dietary tailoring — adapting nutrition to the individual.

- 5.1** Part five. First of all, good morning. How's it going, everybody? Okay, in this section of the seminar, we're going to talk about how to tailor the recommendations that I've made thus far to your own unique individual goals. We're going to talk about the different dietary patterns that are out there, the most popular ones. We're going to talk about the levels of evidence supporting them. We're also going to talk about some of the dietary patterns that I see as being potentially helpful in certain contexts, that are more popular in the zeitgeist, some dietary patterns that you guys are likely familiar with. We are going to talk about why 98% of diets fail. We already touched on that in a Q&A yesterday, but we'll do a bit of a recap and expansion of those ideas. And then, as I alluded to, we're going to talk about tailoring your diet.
- 5.2** As always, keep in mind any questions that you might have, and feel free to stop me if you want something clarified or if you want me to repeat something or what have you.
- 5.3** Just an overview before we get into it. Again, as I mentioned yesterday, no single food has any—first off, food doesn't have moral value. You're not good or bad; you shouldn't feel any guilt about eating foods outside of a healthy dietary pattern. No single food is capable of swaying your health in the direction of health or disease. It really, ultimately, comes down to the dietary pattern as a whole, the kinds of foods that you eat day in and day out for the bulk of your life. That's really, from a dietary standpoint— that's where the power is in terms of your dietary pattern.
- 5.4** Just to recap, the Standard American Diet is high in added sugars, unhealthy fats, refined grains, ultra-processed foods, which we've talked about quite a bit, linked to many chronic diseases: obesity, type 2 diabetes, heart disease, and certain cancers, dementia, of course, and it tends to be a diet that is high in carbs, high in fat, low in protein, but not to the point of being protein deficient, but it's generally a high-carb, high-fat diet.
- 5.5** When it comes to dietary patterns that are generally considered to be healthy, well, there's a continuum here because, obviously, I listed out the carnivore diet. I think it's particularly relevant to— anybody who might gravitate to this course is probably going to be familiar with the carnivore diet. I'm not anti the carnivore diet, although it's important to point out that there's very little long-term evidence on its impact on human health. Certainly, we see a lot of anecdotal evidence suggesting that the carnivore diet can be helpful as a therapeutic diet; it's a very strict elimination diet of sorts. You're cutting out quite a bit of foods on a carnivore diet, but people seem to see a reprieve from symptoms with regard to various types of autoimmune conditions, allergies, and the like.

- 5.6** There's a paleo diet. The paleo diet was first popularized by Loren Cordain. Lots of reviews and writing in the medical literature discussing paleo diet principles. In general, I've always said that if somebody were to ask me in an elevator and I had 10 seconds to give my view on the healthiest popular diet out there, I would generally say that it was a modified paleo diet. I think the paleo diet, one thing that I really appreciate about it, is that it tends to focus on food quality. Now, this is obviously changing. As it's become more popular, the market has bent to, cater to paleo enthusiasts, and so there's now a lot of ultra-processed paleo-approved foods on the market. But in general, it's a diet that has long been hailed for its appreciation for food quality, which I'm a big fan of, personally.
- 5.7** We've got vegan and vegetarian diets, the opposite end of the spectrum where you have carnivore on one end. Vegan and vegetarian diets, obviously, cut out all animal products. When discussing vegan diets, specifically, vegetarian diets typically will integrate eggs and dairy and things like that.
- 5.8** From an evidence standpoint, the Mediterranean diet, since about the 1950s, has been lauded in the medical literature. Ancel Keys was one of the initial scientists to go over to the Mediterranean region of the world and write about the dietary pattern that he observed there. He stationed himself on the island of Crete. As lore has it, it was during a particularly sparse time, and so there wasn't a high degree of animal product consumption.
- 5.9** So, for some reason, the Mediterranean diet is often described as being a low-meat diet, but actually, when you go to various parts of the Mediterranean and the neighboring countries, you see that the Mediterranean diet is not a low-meat diet. They eat lots of lamb; they love even cured meats like soppressata and prosciutto and things like that. People that adhere more closely to Mediterranean dietary patterns tend to have reduced risks for various chronic diseases: cardiovascular disease, neurological disease, certainly obesity, type 2 diabetes, and the like.
- 5.10** Then we have a keto-style diet—'keto' is short for ketogenic. I've been interested in the ketogenic diet well since I was about 16, I was familiar with the ketogenic diet. The ketogenic diet was initially designed as a therapeutic diet, and it's been used for over a century at this point to treat certain types of epilepsy. It's a diet that completely restricts carbohydrates, dietary carbohydrates, particularly glucose-yielding carbohydrates. So, we talked a little bit about carbs. Fiber is a type of carbohydrate. Fiber is not a glucose-yielding carbohydrate. It's inaccessible to us. So, it typically will restrict sugars, starches, and things like that.
- 5.11** There are various ways to iterate on a ketogenic diet. If any of you guys have questions about keto, feel free to ask. There's what's sometimes referred to as 'dirty keto,' where people will just calibrate their diets to fit their macronutrient requirements. But then, now I think as information about the microbiome and the importance of dietary fiber comes into the fold, we're seeing that now you can create a new version of the

ketogenic diet, which integrates a lot of these learnings about the value of plants. So, fibrous, low-starch vegetables have been utilized in more modern iterations of the keto diet.

- 5.12** But what's interesting about the ketogenic diet, again, it's a therapeutic diet, so I often talk about it. I'm not on a ketogenic diet, and I don't think you need to be on a ketogenic diet for good health, and certainly not for weight loss, but it is the only dietary pattern that significantly changes or modulates the neurochemistry of the brain. That is because it switches the brain to be able to utilize an alternate fuel source, which are ketone bodies. So, for people that have neurological problems, traumatic brain injury, and various neurocognitive conditions like Alzheimer's disease, it's a diet that is now the focus of lots of research to see if it can essentially help those conditions the same way that it helps patients with epilepsy. When I think about it, I think of it more as a therapeutic diet and a diet that is specifically relevant to brain function, brain health.
- 5.13** Then, we have the MIND diet and the DASH diet. The MIND diet is a hybrid diet between the Mediterranean diet and the DASH diet. The DASH diet is a dietary approach—DASH stands for "Dietary Approach to Stop Hypertension." So, the DASH diet is a diet specifically designed to reduce hypertension, which is high blood pressure. We'll talk a little bit about what the DASH diet entails.
- 5.14** The MIND diet is a hybrid diet between the Mediterranean dietary pattern and the DASH dietary pattern, specifically catered to help the brain. So, it's not a ketogenic diet, but it has been shown to quite dramatically reduce risk for the development of dementia for people who adhere to it. For even mild adherence to the MIND diet, we've seen something like a 35% risk reduction for the development of dementia, and moderate adherence to it coincides with about a 50% risk reduction. So, the MIND diet, basically, we'll get into it, but it's more brain-centric. The DASH diet is more about hypertension, which is a major problem these days; it puts you at risk for stroke, for dementia, and others. Any diet that's going to improve blood pressure is definitely a good thing. Here, we just have
- 5.15** a bit of an overview. The carnivore diet allows meat and fish, animal products, excludes plant-based foods. I added the Lion diet. The Lion diet was actually coined by Mikhaila Peterson, and it is her take on the carnivore diet, which excludes chicken and fish, so poultry and fish, and it is a red meat-only diet, so beef, lamb, bison, salt, and water. Very next to no research on it, but lots of anecdotes and anecdotes are, I think, important. We need to be able to contextualize where anecdotes sit within the evidence hierarchy. But nonetheless, for people with significant health issues, always worth a shot, I think.
- 5.16** Paleo, as I mentioned, generally, you avoid grains, processed foods. The diet, for whatever reason, doesn't endorse dairy products. I think dairy is great. I present these

diets so that you can be able to get a sense of them, but then, once you know the rules, that's when you're, I think, best able to break them and see what works for you.

- 5.17** Vegan diets, I think, are pretty suboptimal. Vegetarian diets—significant risks associated with those diets, which we'll get into. Mediterranean diet emphasizes whole foods, vegetables. Olive oil is the primary oil used. The keto diet, as I mentioned, tends to be high in fat, moderate protein, very low in carbohydrates. Then, we have the DASH diet and the MIND diet. What's interesting about the MIND diet, as I mentioned, it's
- 5.18** a hybrid between DASH and Mediterranean. A researcher named Martha Clare Morris at Rush University— I've met her a number of times. She actually passed away a couple of years ago, sadly, but she's done a lot of really great research on dietary patterns as they pertain to brain health. The MIND diet was her brainchild, no pun intended. What's interesting about the MIND diet—it only advocates for the consumption of extra virgin olive oil. So, for whatever reason, while the medical and nutritional orthodoxy still promotes these highly refined grain and seed oils, in the Mediterranean diet, olive oil is the only oil used. Similarly for the MIND diet—it only makes the recommendation for extra virgin olive oil, essentially, because the evidence for extra virgin olive oil is just so strong relative to the other types of oils that are out there.
- 5.19** It also, interestingly, doesn't make a general recommendation for fruit. The only fruit that it recommends are berries, which is, again, very interesting, but I think it's important to point out that just because we have lots of evidence or moderate evidence on the impact that berries have on the brain, that doesn't mean, in any way, that other fruits aren't beneficial to the brain. There's this saying, "Absence of evidence is not evidence of absence," meaning that just because we have an absence of evidence for a brain benefit from other fruits doesn't mean that those fruits aren't beneficial to the brain. We just don't have evidence of it. So, what that's saying to me really suggests is that we need to have an open mind about these things and not take any single dietary pattern as dogma.
- 5.20** So, ultimately, why do 99% of diets fail? It's because people don't adhere to them; they adopt very black-or-white perspectives about these diets. If they can't adhere to the keto diet, if they can't adhere to the Lion diet or the MIND diet, then they've failed. The reality is diets do work. You just have to adhere to them. Ultimately, the best diet for you is going to be the one that you can adhere to; that's going to help you achieve the goals that we discussed at the beginning of the seminar. The goals should be the purpose of good nutrition: to nourish your body, to provide you with proper energy balance, and the like. That's why, ultimately, you have to just figure out what works best for you.
- 5.21** When it comes to tailoring your diet, there are certain salient goals, as I mentioned. You want to make sure that you're optimizing for protein, energy balance, nutrient density, and the like.

- 5.22** Now, we can talk about dieting for weight gain, but generally, fat loss is something that, I think, tends to be more relevant for people. We live in a time where one in two people are almost obese, unfortunately. About 66% of the US population is either overweight or obese. By the year 2030, one in two people are going to be obese, not just overweight, but obese. Generally, we have an overweight problem in this country and increasingly around the world. In fact, this is the first time in human history that there are more overweight people walking the Earth than underweight. So, this is a big problem, and that's why this becomes super relevant from a public health standpoint. But certainly, everybody knows people—it's hard to maintain a healthy weight in the modern food environment due to the nature of modern ultra-processed hyper-palatable foods.
- 5.23** Some general points about dieting for fat loss, and I make the distinction between fat loss and weight loss, very important, which I will unpack in a bit. But generally, slower is better. So, slower weight loss is better. A lot of people want to lose weight as fast as they can. They're preparing for a trip or a wedding, and so they want to lose weight fast. But when it comes to losing fat, slower is better. Rapid weight loss sometimes can coincide with the loss of lean mass, lean tissue; specifically, I'm talking about muscle, but slower is better, and I will explain why that is in a few.
- 5.24** As I mentioned, we want to be able to disentangle weight loss from fat loss. You can go on a ketogenic diet starting today and see a pretty dramatic weight loss by the end of the week, but most of that is going to be water. That's not the weight loss that we aspire to, I don't think. We can modulate our water fairly easily with our carbohydrate intake. Carbs cause you to store water in your muscles. It's not a puffy look; it actually helps you perform better. We want to be able to know, we want to be able to cultivate the skills of losing fat—very valuable.
- 5.25** This is something that has become actually relevant for me lately or as of the past couple of months, just as an experiment, so that I can speak to this topic in a more credible way, I actually, over the past couple of months, modulated my own body fat by utilizing some of these principles, which I can unpack. I didn't start out as overweight, but I've, in the process, learned quite a bit about this, which I'm excited to share with you guys.
- 5.26** Of course, strategies to protect lean mass are super important. Muscle tissue is crucially important to mobility, to injury prevention, to metabolic health, to hormone health. Muscle really is crucial. When it comes to losing fat and preserving muscle, we have to take stock and do a bit of an inventory and assessment and see where we're coming from. So, generally speaking, as I said, slower is better when it comes to fat loss, but it ultimately depends on where you're starting from. So, people who are very overweight, so people who have more weight to lose, can actually lose fat more rapidly than people who are starting out lean. The leaner you get, the slower you want your

weight loss to be. The more rapid your weight loss when lean, the more you risk losing muscle.

- 5.27** Now, the reason for this is that there's a maximum fat transfer, essentially. A pound of fat mass can only release about 32 to 34 calories per day (per 24-hour period). So, the more fat you have, obviously, the more calories you have available to you when in a calorie deficit. If you have a lot of fat, if you're carrying many, many pounds of fat mass, then you can hypothetically release a lot more calories from that fat mass over a 24-hour period. If you don't have enough calories to supplement your calorie deficit, your body is going to start looking elsewhere for that energy to make up that energy.
- 5.28** So, the more obese you are, the more rapid your fat loss can be, meaning if you choose to enter a calorie deficit, the wider your calorie deficit can be before you risk losing muscle mass. People who are very lean should aim to lose fat much slower. This is why you'll see bodybuilders when training, especially in the latter end of their cutting phases, be very slow with regard to a very narrow calorie deficit. That's where actually tracking becomes essential. The leaner you get, the more useful, it's not necessary, but it becomes increasingly useful, the leaner you get, to actually track your calories so that you can ensure that you're creating as narrow of a calorie deficit as possible.
- 5.29** Also, the leaner you are, the more protein you require to sustain lean mass. So, bodybuilders who are getting extremely lean, up their protein. We discussed protein, generally, you want it to be around 1.6 grams per kilogram of lean mass, but the leaner you get, the more—you'll see bodybuilders go even higher than that, up to double that. Now, for most people who are not bodybuilders, you just generally want to make sure that your protein is at the level that I discussed. So, 1.6 grams per kilogram or about 0.7 to 0.8 grams per pound of body mass.
- 5.30** General fat loss—a more obese person can aim to lose a pound, a pound and a half, two pounds a week. The leaner you get, the less you should aspire to lose on a weekly basis. I'm talking about pure fat, so I'm not talking about water weight or anything like that. But generally, somebody who's not overweight, trying to get really shredded, we're talking about half a pound a week, which doesn't seem like a lot, but you really want to protect that lean mass. So, slower is always going to be better.
- 5.31** It doesn't matter the ratio of carbohydrates to fat. That's what we're starting to see in meta-analyses on low-fat diets versus low-carb diets. There doesn't seem to be much of a benefit. In the short term, low-carb diets do tend to lead to more weight loss, but that's primarily because people tend to lose water weight when initiating low-carb diets because, as I mentioned, carbohydrates cause your muscles to store water. So, low-carb diets, ketogenic diets, in the short term, have shown greater degrees of weight loss. In the long term, the fat loss is equivalent, essentially.
- 5.32** There are certain benefits to either, which might influence which dietary pattern you choose. So, for example, with higher carbohydrate diets when dieting—this is actually

something that I learned to be true just in my own process, but the evidence backs this up, and you can see it anecdotally within the fitness community. It's actually interesting — the fitness community, there's a lot of what's called 'bro science.' Has anybody heard of that term? There's a lot of bro science in the fitness community, but the fitness community, in many ways, has pushed this research forward. The same way that the porn industry, which sounds really weird to say, has actually pushed forward the technology with regard to streaming, and they were one of the first industries to embrace DVD technology. It's weird that one of these fringe industries can ultimately influence the world at large. But fitness, in many ways, bro science, does stimulate research so that scientists can validate some of these ideas.

5.33 The thing about carbohydrates is that when you're dieting, resistance training becomes crucial to send the signal to your body to hold on to that muscle. If you want to lose primarily fat, you have to continue resistance training. Don't change your resistance training. You want to continue training as intensely as you would when trying to gain muscle, to preserve that muscle. The same resistance training paradigm that you would utilize to maximize muscle growth is also going to be the best for muscle retention. The beautiful thing about carbohydrates in this context is that carbohydrates are what your muscles use as fuel during intense resistance training. So, it actually benefits you not to cut carbohydrates completely when on a fat-loss diet.

5.34 If you look at any bodybuilder that's getting, and this is just a microcosm, I'm not saying that the average person needs to eat or even think about food like a bodybuilder, but they tend to cut fat. Fat tends to not be as satiating as carbohydrate-containing whole foods because carbohydrate-containing whole foods tend to be produce, which contain fiber and water. So, actually, there's been a lot of work done by evidence-based practitioners to undo the stigma that carbohydrates have as being inherently fattening. They actually support your workouts, and your workouts become crucially important to be able to sustain energy in the gym when dieting so that you could preserve your muscle.

5.35 Resistance training, as opposed to cardio and all the things that the general population thinks you need to do to lose weight, it's resistance training that sends the signal to your body that you need to hold on to the muscle and cut, essentially, the fat. Resistance training can come in many forms. It can be pushups—there's all different ways to do it. But generally, I enjoy having a gym membership. Resistance training and weight training are not one and the same. You could resist against gravity, and that's a form of resistance training. Isometric training, doing pushups and holding yourself in a certain position, pullups—all great forms of resistance training.

5.36 A lot of people will actually change their resistance training because they believe that dieting is going to make them weaker, but it doesn't necessarily. It might not necessarily make you weaker, provided you're keeping your carbohydrate intake up. So, train as if you were not dieting, essentially, is the way to approach it.

- 5.37** How do you know how many carbohydrates you need? Well, so yesterday, I shared how to get a rough estimate of your maintenance calorie needs. A very rough way to gauge how many calories you might need over the course of the day, you can multiply your weight by 15. There are more accurate ways of doing this. Indirect calorimetry, I mentioned, but that's going to be inaccessible to most people.
- 5.38** So, your weight times 15 is a good way to get a sense of your maintenance calories. To lose about a pound a week, you need to be in a 500-calorie deficit. Okay, 500 calories. So, if you multiply your weight by 15, you subtract 500 calories from that. That gives you a general starting place to figure out what it would take, what calorie level you need to be at, to lose about a pound a week.
- 5.39** Now, let's just say that equates out to 2,000 calories. I don't know what it is for you individually. I don't have a calculator on me, but let's just say that a 2,000-calorie, 24 hours for you, 2,000 calories would be your 500-calorie deficit. You want to aim for, as I mentioned, at least 1.6 grams per kilogram of body weight or 0.7 to 0.8 grams per pound of body weight. So, you would try to figure out how many grams of protein that you need. Protein has four calories per gram. So, then you multiply your protein needs by four. We can pull up a calculator and actually try to figure this out, just to make it easy because I'm not that great at math on the fly. So, we can use myself.
- 5.40** I'm about 185 pounds, so I would multiply that by 15 to get a general sense. Again, this is an estimate. The best way to figure out your maintenance calorie needs, generally, as I mentioned, would be to do a seven-day food journal, assuming your weight's stable, and then to get an average of the amount of calories that you're consuming on an everyday basis. But this is just a rough and dirty guide. So, I multiplied my weight of 185 by 15, and I arrived at 2,775 calories. So, I would want to subtract 500. Now, the reason why this would yield a pound of fat loss per week is that a pound of fat contains about 3,500 calories. So, one pound of stored fat yields about 3,500 calories. If you divide 3,500 by seven, seven days in a week, to lose one pound of fat, you would need a calorie deficit of about 500 calories a day. It's like transcribing what I say for some reason. Does that make sense? Okay.
- 5.41** For me, a calorie deficit at 185 pounds—again, this is all based on that estimation, so this could be off by 200 calories in either direction. We don't know for sure. Our metabolisms are influenced by many things, hormones. But generally, this is thought to be a good starting place. So, that would be a calorie deficit starting place for me. So, I would, basically, begin aiming for this as a target to lose a pound of fat a week. So, let's keep that in mind.
- 5.42** Now, from a protein standpoint, I'm 185 pounds, and I would want to get about point—it's a little bit higher than 0.7, so let's just say 0.8 for now—I should be aiming for roughly 150 grams of protein a day. Now, the reality is you can go higher than that, and the leaner you are, it probably behooves you to go even higher than that. Some bodybuilders will go up to a gram of protein per pound of body weight. So,

hypothetically, I could even go to 185 grams, but this is just the minimum. So, about 148 grams of protein. I would want to multiply that by four because there are four calories in a gram of protein. So, we got 592. Let's round up to 600. So, 2,275, which is my daily calorie intake on a weight loss diet, minus 600. That gives me 1,675 calories to play with. Those calories are obviously going to come from carbs and fat because we've already accounted for our protein.

- 5.43** Generally, you don't want to be on a low-fat diet because that can affect your hormones negatively, and it can also impact your ability to extract the full nutrition from your food. We've talked about fat-soluble vitamins like A, E, D, and K, which are very important. We've talked about various, well, I guess we haven't covered them yet, but there are various fat-soluble plant compounds like phytochemicals, polyphenols, carotenoids, like pigments that are super important to brain health, to eye health. You need fat to absorb these compounds. You don't want to go on a very low-fat diet. But, essentially, you've got almost 1,700 calories to play with. So, as long as you're not going very low-fat, it really is up to you in terms of how you want to modulate this diet or modulate these calories and make use of them.
- 5.44** As I mentioned, what's the number one variable for diet success? Adherence. So, with this number, provided you're keeping your food quality high because health matters, it really is up to you to integrate the kinds of foods that you enjoy. That's what it all comes down to. You want to make sure that you're enjoying the process. At the same time, choose foods that are high-satiety because you don't want to be hungry on a diet. That spells doom for any diet, like hunger. Maybe you'll be able to contend with hunger for a day, maybe you'll be able to contend with hunger for a few days, but ultimately, it's going to catch up to you, and that's where bingeing happens. So, you want to optimize for satiety. Food quality plays a huge role in that. Actually, another way to increase satiety is just to increase protein because we know that protein is the most satiating.
- 5.45** So, these are all the variables that you can play with, and as I mentioned, there's no one-size-fits-all diet. It really is up to you to figure out how to best fill this calorie budget with your needs. If you find yourself lacking energy in the gym, maybe you need more carbohydrates. If you find yourself, maybe your skin is starting to feel dry, that can maybe be a sign of not enough fat consumption. So, you've got to figure it out for yourself. This is not meant to advocate for a low-fat diet, but generally, fat being the most calorie-dense, especially oils and added fats, those are usually the first to go.
- 5.46** That was a fun little experiment. That's, essentially, what I did. I started doing that about two months ago, and in the process, I lost about 8 to 10 pounds of pure muscle. I haven't lost any strength. Of pure fat, rather. I haven't lost any strength in the gym. I dropped my body fat percentage. This wasn't a health experiment. This was mostly just an experiment to see what it would take to drop a few body fat percentage points. I went down—I started at about 14 to 15% body fat, and I landed at around 11.5% body

fat, which was pretty easy. I'm not going to lie. It wasn't really that hard when you take these principles and you cater them accordingly.

- 5.47** A lot of people make the mistake of exercising to burn calories. Generally, that 500-calorie deficit, you can create by eating 500 calories less food, or you can split the difference so that you can have a little bit more food. You can burn 250 calories in the gym and consume 250 fewer calories. Any way you slice it, you just want to make sure that you're in that deficit, and increasing your activity is a great way to do that.
- 5.48** Now, as I mentioned, you want to continue resistance exercise training. So, I'm actually not a fan of cardio. I'm a fan of cardio for health, but when it comes to burning calories, a lot of people will get on the treadmill and try to run. You're not getting a lot of value from that, from a calorie standpoint. The greatest value that you get from a calorie burn energy expenditure standpoint is from something called NEAT, which stands for non-exercise activity thermogenesis. Collectively, over the course of the day, these non-exercise activities burn a hell of a lot more calories than the exhausting effort that you might put towards running on a treadmill. Also, running on a treadmill, particularly when dieting, is going to take energy away from you that would be better put towards resistance training. So, again, for the sake of dieting and burning calories, I am not a fan of doing cardio.
- 5.49** It also isn't great for preserving your muscle in the context of dieting. You want to resistance-train, and you want to increase your NEAT. So, NEAT is just walking. Just walking on a treadmill is incredible. You get a lot more bang for your buck walking than you do running. But it also includes things like dancing or playing with your cat or chores around the house. As I mentioned, low-intensity walks. Even fidgeting is a form of non-exercise activity thermogenesis. Collectively, you burn way more calories with NEAT than you do with running on the treadmill, and just the amount of effort that it takes—it's just not worth it. You'd be much better off putting that energy towards resistance training.
- 5.50** When it comes to consistently being on a weight loss diet, a few things happen, but they're really not anything to be concerned with for most people. There is a degree with extended dieting of metabolic adaptation, meaning your body senses the calorie deficit, and obviously, your body not wanting to starve to death might titrate down your metabolic rate to some degree, but the risk of that is higher with very lean people and more extensive calorie deficits. Again, you can, for the most part, prevent this by losing weight more slowly.
- 5.51** There's also what's called adaptive thermogenesis, which happens. Basically, you'll notice that on a calorie deficit, especially on a wider calorie deficit, you're just going to have less energy, which is your body's way of conserving energy. You can avoid that by just keeping your non-exercise activity up, keeping your NEAT up.

- 5.52** Also, there is what is called, particularly the leaner you get, there's this concept of refeeds. The benefits of refeeds are primarily psychological. So, taking a break from your diet. There's absolutely no harm in if you've been on a diet for a couple of months for beginning to take a day off or two days off a week or even taking a week off. The benefits that you get to your psyche— those are the primary benefits of refeeds.
- 5.53** There's some thinking and maybe a little bit of evidence that taking a day away from your calorie deficit and bringing your calories up, back towards your maintenance levels, can upregulate some of the hormones involved in your metabolic rate, that it can help boost your metabolism a little bit. Probably beneficial to have a higher carb day. Also beneficial from the standpoint, again, of refilling your muscles with glycogen, which is that energy that your muscles need when exercising vigorously. So, anything you can do to keep your exercise up in the gym so that you can continue to lift heavy is going to be great for your diet in terms of preserving your muscle mass. Make sense? Cool.
- 5.54** There are lots of myths that circulate on the internet. I'm here to bust those myths. So, this concept of metabolic damage for people that have dieted extensively, that they've damaged their metabolisms. It's BS. You can't damage your metabolism. It just doesn't happen. You can always come back, essentially. Your body does adapt, but it's not a permanent thing. So, we can't think about this damage as being permanent. We do adapt to extensive dieting, but as I've mentioned, there are ways that you can counteract those adaptations with a smaller calorie deficit, increasing protein, and continuing to resistance- train. You never damage your metabolism, so don't worry about that.
- 5.55** There's also very little evidence that metabolism slows with age. A lot of people expect that as they get older, their metabolism slows down. It's generally unfounded. We just tend to be less active as we get older, so we burn fewer calories. It's less neat. We're more active the younger we get. Maybe it's due to injury, maybe it's due to life obligations. But, generally, there's very little evidence that metabolism slows significantly enough to account for the weight loss many people pin to their advancing age. It might slow somewhat, but generally, not enough to be used as an excuse, generally speaking.
- 5.56** I already touched on exercise to burn calories. Exercising to burn calories— you're not getting a lot of bang for your buck unless you consider walking on a treadmill, which I love to do. That's a great way to burn calories. It's easy. You can listen to podcasts. You're not compromising the energy that you have to give towards your resistance training. You burn a lot more calories over the course of the day with NEAT.
- 5.57** Fat-burning foods—no such thing. There are foods that support fat loss, meaning foods that are more satiating, foods that are higher in protein, foods that give you the energy to work out in the gym. Those are the foods that I would consider to be most supportive of losing fat, but there's no such thing as a fat-burning food.

- 5.58** And then, finally, that whatever diet is best. That's a myth. The diet that's going to be best is the diet that you can most closely adhere to, that nourishes your body, that keeps you full, doesn't keep you hungry, and, importantly, that doesn't compromise your relationship with food. We don't want to do any crazy fad diets, binge diets, or anything like this. You can achieve all of your wildest dietary body composition goals with myriad permutations of dietary patterns. So, it ultimately is about finding what works best for you and thinking both in terms of the short term as well as the long term.
- 5.59** Before we wrap up this portion, it's also important to mention that fat loss isn't everything. You want to maintain a healthy degree of muscle in your body. About 20 to 33% of the population, depending on what studies you look at, are normal weight but yet metabolically obese. So, under the hood, have signs that are associated with metabolic syndrome. So, there is a problem with being undermuscle in this country. So, I don't want you to be thinking purely in terms of fat. We want to make sure that we take what my friend, Dr. Gabrielle Lyon, calls a muscle-centric approach. You want to be doing what you can to optimize your body composition, and building muscle to the best of your ability plays a huge role in that as well. Normal weight, metabolically obese, is a real thing. A lot of people refer to it as 'skinny fat,' the phenomenon of being skinny fat. You don't want to be undermuscle.
- 5.60** I guess one more myth to add on to the top fat loss myths, but it's more relevant to muscle growth—a lot of women are scared of getting too bulky. It's not going to happen. I've been lifting weights for 20 years, and it's just hard to put on muscle. It's very hard to put on muscle for men and for women. So, you should be doing what you can to reach the peak of your genetic potential. Male, female, no matter what age. Crucially, crucially important.

Lecture 6: Nutrition Controversies

At a glance — Nutrition controversies — separating fact from fiction.

- 6.1 "Separating Fact From Fiction: How to Think About Nutrition" Over the next hour, we're going to talk about controversies in nutrition. We're going to talk about meat. We're going to talk about veganism. We're going to talk about the danger of nutritionism. We're going to talk about the benefits of animal-source foods, plant-source foods. I believe, in this section, we're going to talk about hormesis. And then, we're going to talk about whether or not obesity is genetic. I love controversy.
- 6.2 So, meat equals bad. This is something that we've heard for decades. It's important to understand that much of the advice that gets filtered down to the general population comes from a form of evidence called epidemiology.
- 6.3 Now, to understand why epidemiology is not sufficient to inform dietary decisions, we have to understand how nutrition science is performed. Without getting too into the weeds, what's always important to keep in mind is that there's a difference between correlation and causation.
- 6.4 So, correlation—you could argue that with the proliferation of smartphones, there's been a rise in obesity. Is it because smartphones are causing obesity? Is obesity causing smartphones? Or is there some unknown variable that's influencing both? So, that's where correlation comes in. Correlation means that there's an association, there's a relationship, there's a risk associated. A lot of the time, you hear the word 'risk.' The word 'risk,' whenever you hear the word risk, that implicitly implies correlation.
- 6.5 Causation is when we know that something causes another thing. So, I press the down button on this little device, it causally advances to the next slide. That's a direct causal relationship.
- 6.6 Now, you can do observational studies and see very strong correlative links that will strongly suggest causation, but there's still just that: correlation. Now, as that relationship grows stronger, we can say with greater certainty that there's a causative link there. As in the case with cigarettes—people who smoke cigarettes have a 2,000% increased risk of developing lung cancer. Still, many people smoke cigarettes that don't develop lung cancer. So, it dramatically increases your risk and is causal in those instances. But we can compare that to a lot of the weaker links that we see in the field of nutrition— red meat consumption increases risk of X cancer by 30% or something like that. Comparing 30% to 2,000%, that's a lot different. That gives us a lot less certainty.
- 6.7 The goal of science is to reduce uncertainty; that's what it's all about. There are very few knowns and unknowns. There's just degrees of uncertainty throughout science. When it comes to epidemiology, what epidemiology is, it's looking at populations, and

it's seeing how they live, how they eat, and then it's looking at their health, and it's trying to make inferences about their activities, their behaviors, their habits, and their health outcomes.

- 6.8** There's also the issue of—I didn't actually write this in the slide, but this is also pretty important—the difference when looking at these links between absolute risk and relative risk. So, let me try to come up with an example on the fly. You might see something that a study that finds that egg consumption increases risk of type 2 diabetes by 25%. The issue is that that is generally an increase in relative risk. And so, if your lifetime risk of incurring type 2 diabetes was— your absolute risk was—5% or 4%, rather. A 25% increase in relative risk takes your absolute risk from 4% to 5%. That's not a very—we can't say with great confidence that eggs are causally having an impact there. So, that's always the difference between relative risk and absolute risk.
- 6.9** Generally, whenever you see these kinds of studies, they're talking in terms of relative risk. You can go to the research papers, and you'll usually see relative risk, abbreviated RR. There are different ways to indicate risk. There's hazard ratios, there are odds ratios. Generally, they all say the same thing.
- 6.10** When it comes to epidemiology, there are what are called confounding variables. So, when looking at people who consume meat at the population level, because meat is something that we've been told for so many decades is unhealthy, it takes a certain person to—or meat consumption generally coincides at the population level with other lifestyle habits, and because the more health-conscious eaters for the past few decades were told to avoid meat, generally, we see something emerge again and again in the nutritional epidemiology known as healthy user bias, which I also should have added to the slide because healthy user bias is quite important as well.
- 6.11** But generally, we see that people who eat more meat tend to also be more sedentary; they tend to be more likely to smoke. So again, it's going back to the smartphone and obesity example: is it the meat that is responsible for the poor health outcomes that we see in some of these people when they're looking at the population s of meat consumers, or is it the other unhealthy lifestyle and dietary habits that they've adopted that are responsible for those poor health outcomes?
- 6.12** We can see this with plant consumption. Everybody knows that fruits and vegetables are healthy. Also, fruits and vegetables—eating fresh fruits and vegetables tends to be a privilege. There's a socioeconomic variable here that is a potential confounder. People who have access and regularly eat fruits and vegetables, they might be more educated; they might have access to farmer's markets.
- 6.13** So, we see at the population level that people who eat more fruits and vegetables tend to have better health. But again, is it because of the fruits and vegetables? It may be. It very well may be, but it can also be—that observation might be significantly contributed to by, again, healthy user bias, that people who are eating more fruits and

vegetables are higher on the socioeconomic spectrum. They might have other healthy habits. They might be more likely, for example, to have a gym membership, to take a multivitamin, and have a higher degree of education. So, healthy user bias is always one of these confounding aspects.

- 6.14** We tend to see that also with saturated fat, saturated fat intake. Saturated fat, we've been told for decades, is not good for us. So, people that are generally more saturated fat-conscious might actually have better health. But is it because of the saturated fat consumption or the low levels of saturated fat consumption, or is it in spite of that? As we talked about yesterday, there are many different types of saturated fat. Saturated fat is found in— any natural, healthy fat-containing food is going to contain some proportion of saturated fat. So, always crucial to keep in mind.
- 6.15** The vast majority of nutrition research that we use to base our decisions on and that generates clicks on social media comes from this type of research, whether it's epidemiology or other forms of observational research, which cannot, by its nature, establish causation. There's only one type of trial that can establish firmly causation. That is the randomized control trial, where we control for all of these different variables. We change one variable, we compare that to a placebo, and we see whether or not a variable has a causal connection with the outcome that we observe. So, randomized control trials, those are the only kinds of studies that can show cause and effect. Super, super, super important.
- 6.16** Then, dietary cholesterol. I feel like this is starting—we're becoming a bit wise to this fact, but dietary cholesterol, for most people, very little impact on blood cholesterol. Dietary cholesterol actually isn't very bioavailable. When we—cholesterol in food is essentially in a form of chemical packaging that makes it not very bioavailable to us. We don't absorb very much of it. There are always going to be exceptions to these general rules. There are hyper absorbers, there are hyper responders, but they tend to be the minority. For the most part, dietary cholesterol is not something that you need to be concerned with.
- 6.17** Just to illustrate for you guys, this is from the USDA. So, these are the top sources of saturated fat: again, the type of fat that we've been told for decades is no bueno for us. Most Americans are eating quite a bit of saturated fat, more than what the guidelines suggest, but it's very important to realize that only 3% of the saturated fat that your average American ingests comes from meats. This is excluding mixed dishes, and deli sandwiches, and the like. So, when it comes to foods like steaks, ground beef, and the like, only 3% of the saturated fat in your average American diet is contributed to by those completely natural, healthful foods.
- 6.18** A significant proportion does come from dairy fat, but dairy fat, as I mentioned yesterday, it's wrapped in this bubble, milk fat globule membrane, with the exception of butter, that, for the most part, is beneficial from a cardiovascular standpoint. So, we

have to stop fearing saturated fat. This just goes back to the point that I tried to make yesterday, that fearmongering certain nutrients makes very little sense.

- 6.19** On the other hand, foods like pizzas, mixed dishes, like fast-food burgers, lasagnas, and things like that—all very tasty, granted. But the vast majority of unhealthy fats that we get tend to come from these mixed dishes, not steaks, not ground beef, not whole food sources of myriad bioavailable micronutrients, protein. Eleven percent comes from—we demonize meat, but what about desserts and sweet snacks? Almost four times the amount of saturated fat. This comes from the USDA. So, we just have to be very mindful when we see people fearmongering certain nutrients. Does that make sense? Did I make the case? All right. I rest my case.
- 6.20** Veganism. So, a big push towards plant-based diets. I am not against people adopting vegan diets for whatever reason they choose, but my whole thing is to help people make these decisions with full, informed consent. There's little to no evidence that the vegan diet is a superior diet in any way. It's not. In fact, the advocates for the vegan diet will use this cobbled approach of very weak evidence, very low risk ratios, and things like that. Mechanistic word salads to suggest that veganism is ideal, but it's not.
- 6.21** We know that it's a highly restrictive diet. It's nutrient-poor. People tend to be deficient in a slew of really important nutrients when they adopt these diets. It's associated with nutrient deficiencies, lower protein intake, lower muscle mass, and, of course, as I mentioned yesterday in one of the Q&A sections, a higher intake of ultra-processed foods. So, I know, to some degree, I'm preaching to the choir in this room, but I don't think it's an ideal dietary pattern. And yet, nonetheless, it seems to get a pass from the nutritional establishment. It's very weird.
- 6.22** Okay, so I did get ahead of myself. I did touch a little bit already on how to interpret research, but just to recap a little bit: there are different types of studies. There's observational research, which establishes correlation. We look at populations, we don't tweak anything, we see their health, and we generate hypotheses based on what we observe. We can determine correlations from that type of research. There are also some types of observational research that are better than others, like cross-sectional evidence. For example, we look at a population, and we see how much cancer, and we see what they're eating. That's not a very strong form of observational evidence.
- 6.23** On the other hand, there are what are called cohort studies, where you take a large group of people and follow them over time. You're not intervening in any way. You're just following them over time to see how certain preset variables influence their health over the long term. Those are somewhat better. One example of that would be the Framingham Heart Study, which is an ongoing study in the northeastern US, where it's a multi-generational study. It's called a cohort study. They're not tweaking any variables. They're just observing, but they're observing over the long term. A prospective cohort study is what that's called. But again, these are only observational.

They cannot determine causation. So, they're very good for generating hypotheses that can then be tested experimentally.

- 6.24** But with regard to nutrition, there are very few actual experimental long-term, randomized large population trials. It's just there's not a lot of money to do these kinds of research. So, that's why we tend to lean a little bit too much on this observational level data. Unfortunately, that's why you tend to be able to prove whatever it is that you want with the current state of nutrition science. You can go and find studies that prove that meat is bad. You can go and find studies that prove that or suggest, rather, that meat is good. That's just the nature of the beast.
- 6.25** But then, occasionally, you got studies like that Kevin Hall study, which I mentioned yesterday, that obesity study, and that was an example of an interventional experimental study, and that's why those are so great.
- 6.26** Even among experimental studies, we've got human and animal studies. Animal studies don't always translate to humans, but it is a lot easier to control variables in an animal setting. It's also a lot easier to see how certain variables impact, from a lifespan perspective, an animal because mice live for two years. You can learn a lot about how to modulate the lifespan of a mouse, but it's a lot more difficult, in fact, impossible, to do that with humans. So, it is a very difficult science; as I mentioned, it's much less well-funded than drug research. And so, that's where I think the limitations of nutrition science really become apparent.
- 6.27** We also have meta-analyses. Meta-analyses are considered to be at the top of the evidence hierarchy because they're not just one study; they are studies of studies. So, we take all of the different observational studies that have been done on meat, and we try to ascertain the collective conclusion of all of them. We take all of the randomized control trials that we have on berry consumption, for example, all of the experimental interventional trials that we have on berries, and we try to see whether or not the evidence is strong enough to make recommendations for berry consumption. So, meta-analyses are thought to be the top of the evidence hierarchy, and that's why we rely on them. It's a great place to start, actually, when doing your own research, so to speak.
- 6.28** You always want to look at the year of the study. You want to look at the methodology of the study. Population size is always important. Whether or not the findings are statistically significant is super relevant. But generally, I think that this is something that anybody can really grasp. Also, whether or not you choose to read what's called the primary literature, you can use this when reading articles about research. When words will start to stand out, like 'risk' or 'correlation.' It'll give you a better sense of whether or not any one study should cause you to change your diet in any significant way. Generally, the answer to that is going to be 'no.' No single study. I don't see studies, and then, suddenly, from one day to the next, change my diet.

- 6.29** We saw this recently with the erythritol study. Did you guys see that on social media? So, erythritol is a sugar alcohol that a lot of keto products will use. I actually really enjoy erythritol, so I was really bummed when I saw those headlines. But when you actually looked into the methodology, it was purely correlational and just a perfect example of why we can't let headlines get the better of us.
- 6.30** One theme that I've gone back to again and again, is that we can't be so focused on these isolated nutrients. When you see people focusing on these singular nutrients, it should be a big red flag. Usually, it's a diet tribe, primarily vegan diets. They love to demonize saturated fat, dietary cholesterol, and things like that. There's a danger in reducing food to its constituent nutrients, and this has been referred to as 'nutritionism.' We see there's a product on the market called "Soylent," which basically is the brainchild of a bunch of Silicon Valley guys who were like, "Let's take all of the vitamins and minerals and macronutrients that we know that humans need to live, and let's just put them all in an ultra-processed, ready-to-drink meal." That's what this product is, and it's essentially like a canola oil-sugar bomb with a bunch of micronutrients in it. I know that I've singled out Soylent, but this happens all the time in the world of nutrition.
- 6.31** We say, well, I can go on a vegan diet as long as I'm supplementing with vitamin B12, which is the only nutrient that we know currently is essential that we get from animal products. But we've co-evolved with our food. There's a whole slew of nutrients that—foods have innumerable nutrients, nutrients that we have yet to name, that we've co-evolved with, and trying to reduce food down to its constituent nutrients, I think, is a big and underappreciated mistake. It's a huge mistake. We're learning all the time about nutrients that might not be essential in all kinds of food: animal-source foods, plant-source foods that benefit human health, that augment aspects of our physiology. And so, again, to reduce food down to vitamins, to minerals, it's a huge mistake. But, of course, a lot of people get paid a lot of money to do that. That is like the foundation of nutrition science, to break everything down.
- 6.32** And it worked. At the dawn of nutrition science, it worked. We figured out that rickets could be cured by giving a patient vitamin D, and that was an amazing finding. We figured out that scurvy could be cured by giving somebody vitamin C. So, at the dawn of nutrition science, I mean, nutritionism really was this incredible thing, but I think we've taken it now too far. We've forgotten, again, that we've co-evolved with foods, with whole food.
- 6.33** To recap a little bit, we know that there are significant benefits associated with animal-source foods. There was a paper published in, I believe, 2020 by a researcher named Ty Beal, who looked at the most—who basically organized the foods with the highest nutrient density, looking at certain nutrients, particularly nutrients of concern, nutrients that people tend to be insufficient in. Animal products took all of the top spots, with the exception of dark leafy greens, which were found to be very nutrient-

dense. Animal-source foods provide our most nutrient-dense foods. The nutrients that they can

- 6.34** contain tend to be highly bioavailable. There are certain bioavailable nutrients in animal-source foods as well, to be clear, but there are certain nutrients that are—in, I mean, in plant-based foods—but there are certain nutrients in plant-based foods that are not as bioavailable, like minerals. Generally, the micronutrients that we see in animal-source foods are highly bioavailable. Animal-source foods don't tend to contain anti-nutrients, chelating agents. What's in animal foods, it's like what you see is what you get, plug and play, pristine quality, and that is not to be underappreciated.
- 6.35** Animal-source foods also, as I mentioned—very high-quality protein. We talked a little bit about how protein quality is graded. We talked about protein digestibility. Animal-source foods provide highly digestible protein and a wonderful amino acid profile. I talked a little bit about leucine. Leucine is the rate-limiting amino acid involved in muscle protein synthesis. Whenever you're eating the skeletal muscle of animals, you're getting an appropriate and optimal amino acid profile to support your own skeletal muscle. So, it just makes perfect sense that that would be a benefit, particularly in today's undermuscle world. Very satiating, owed in part to the protein that animal-source foods contain.
- 6.36** And then, this is also something that tends to be underappreciated as well. There are various phytochemicals in plants that are beneficial to us. Many of them provide a hormetic effect. But we get a lot of phytochemicals—and 'phyto' means plants. So, you have to ask yourself: what are phytochemicals doing in animal-source foods? Well, animals, herbivores, that eat plants, store phytochemicals in their fat. So, in the fat of grass-fed beef, for example, you have a significant beta-carotene content, for example, which is pro-vitamin A. You get carotenoids. You get catechins, which are compounds otherwise found in green tea.
- 6.37** There's a slew of phytochemicals found in animal-source foods, some even in higher levels than the plants themselves. Generally, plants are a better choice for just the sheer quantity that you're getting of phytochemicals. I wouldn't prioritize animal-source foods for their phytochemical content but note that you are getting significant phytochemicals, so plant-source compounds, in animal-source foods. It's pretty incredible. Then, we've got our plant-source
- 6.38** foods. So, we've talked a bit about fiber. The best available evidence suggests that fiber is beneficial. Now, there are going to be people with various GI disorders that don't, at a given point, tolerate fiber. Everybody's fiber tolerance is different. There's no one-size-fits-all recommendation to make. I'm going to talk about the different types of fiber in a later presentation. But in general, know that fiber is beneficial, in part, because it helps not just its satiety-inducing effect but it supports the gut microbiome, which we know is like a drug factory in our large intestines that generally yields very positive postbiotic fermentation products.

- 6.39** Many of the hormetic phytochemicals in plants are not actually all that bioavailable. Fiber is not a nutrient that we derive energy from. So, you might think, "Oh, well, it's a waste of my—why should I fill up any real estate on my plate with something that I'm not yielding any benefit from?" Well, we do actually extract energy from fiber. This is one of the reasons why fiber on nutrition facts labels is counted towards a food's calorie contribution.
- 6.40** Have you guys ever wondered why? I've wondered, but I'm a nerd. Fiber still has a calorie contribution, so you have to ask yourself why. If it's a carbohydrate that's inaccessible to us, well, it's because our gut microbiota ferment the fiber that we consume, and they output short-chain fatty acids and other fuel substrates that we actually do derive energy from. The consensus is a bit divided in terms of how we should actually count this as an energy contribution. But it is interesting that you might eat a high-fiber food and actually extract the calories from that food as fat because some of the postbiotic compounds that we derive from fiber consumption are actually short-chain fatty acids like butyrate. So, very interesting stuff, and I'm going to talk more about fiber.
- 6.41** Highly bioavailable micronutrients—not all micronutrients are bioavailable. It depends on the nutrient, but, for example, folate is one very important B vitamin. It's actually named after foliage, and the best place to find folate is in dark leafy greens. You can also find it in liver, but dark leafy greens are a great source of folate. Vitamin C—you don't find much vitamin C in animal-source foods. You get it in fresh fruits, vegetables, things like that.
- 6.42** We talked a little bit about hormesis, beneficial stress. It's true; there are defense compounds in plants. Plants are not able to fight off predators with their teeth or lack thereof. They're not able to get out and run away from predators. So, what they do is they become brilliant chemists, and they create compounds that literally serve as antifeedants to ward off smaller critters, animals, prey, fungi, insects, and the like, bitter-tasting compounds. That's why dark leafy greens sometimes taste bitter. Herbs and spices, which are actually really concentrated source of these plant defense chemicals, are very bitter. They're known to be bitter. Tea, the tannins in tea, coffee—rich sources of compounds that have a hormetic impact on us, which essentially means that it's a low dose of stress.
- 6.43** It's a low dose of toxic stress that in us isn't toxic at all; it actually provides a stimulus to adapt and to grow stronger. We call this 'symbiotic phyto-adaptation.' This is a term coined by physician Mark Hyman, who runs the Functional Medicine Center at the Cleveland Clinic. I love this term because it gives you a sense of our relationship with these plants. It's a symbiotic relationship. We adapt when we ingest these plant compounds that are not essential but seem to benefit human health.
- 6.44** Another benefit of plant-source foods—they contain pigments and other unique bioactive compounds. I'm a huge fan of carotenoids like lutein and zeaxanthin, very

abundant in kale and Swiss chard. We know that these two compounds—if you look at any eye health supplement, you'll see lutein and zeaxanthin. We've known for decades that lutein and zeaxanthin can help protect our eyes from aging; they help prevent age-related macular degeneration.

- 6.45** But we now also know that our eyes are an extension of our brains. Our eyes are neural tissue. We're starting to see that these same compounds help to support brain health. They help to optimize brain function and accumulate in the brain, where they help reduce oxidative stress, help prevent things like dementia, and the like. These are found primarily in plant foods. They're also fat-soluble. As I mentioned, this is another benefit of not going too low fat. You want to optimize for these compounds.
- 6.46** Anthocyanins are just another example of plant pigments. They're found in blueberries, red onions, purple potatoes, all the same compound. Wherever you see reds and purples in the produce section, it's attributed to anthocyanins, which we know are very beneficial to brain health. They accumulate in the hippocampus, actually, which is the memory center of the brain.
- 6.47** And then, we have compounds called glucosinolates. You might be familiar with a compound called sulforaphane, which has been talked about a bit in books and on social media. But sulforaphane is a compound generated when we chew cruciferous vegetables. They help to stimulate because, again, they're these hormetic compounds; they have a hormetic effect; they stimulate our body's own defense pathways. And so, glucosinolates, in particular, help us to detox. They've done studies utilizing sulforaphane in China, where they see that people who are given sulforaphane supplements more effectively excrete certain toxins, carcinogens, like benzene and acrolein. Those are found primarily in cruciferous vegetables: broccoli, broccoli sprouts, Brussels sprouts, cauliflower, kale is a cruciferous vegetable, I believe arugula is a cruciferous vegetable.
- 6.48** A lot of people are concerned about these defense compounds with the popularity of the carnivore diet. Not every compound is going to sit well with every person, but in general, they seem to be very beneficial.
- 6.49** In fact, one study that I cited recently found that flavanols, which are a type of plant defense compound, when subjects were given high-flavanol foods compared to low-flavanol foods, the foods that were high in these flavanols actually improved cognitive function and led to an increase in BDNF, which is a brain-derived neurotrophic factor. It's thought to promote neuroplasticity, be very beneficial to brain-aging. That was an RCT. High-flavanol foods were compared to low-flavanol foods. We didn't see an effect with low-flavanol foods. It was very likely the plant defense compounds that was responsible for that benefit. There are, obviously, lots of compounds in food, so it's not a perfect study, but it does show us that if flavanols and these compounds were generally so bad for us, why was it that the high-flavanol-containing foods, we saw

actually a very desirable benefit? In general, these are thought to be very good for you, and I agree.

6.50 Finally, is obesity genetic? I think I put this slide here because people tend to be talking about this a lot lately. Genes are not the primary cause of obesity, although genes do play a role. But as with every chronic disease, genes play a role, but it's not ultimately up to genes. It's about the interaction that our genes have with our environment, and obesity is no different than cardiovascular disease or any of these other conditions.

6.51 However, it is highly responsive. It is a highly responsive condition to our choices. We can't say the same thing for cardiovascular disease. Once these conditions have already set in, it's a lot more difficult to undo them, but obesity is one of these things that is very responsive to our diets and our lifestyles. The genes of obesity— it's complicated, and it's not fully borne out the full impact that genes play, but genes can influence things like satiety, reward-seeking behaviors, things like that. But in general, even somebody with genetic obesity, when adopting the principles that I've discussed, can shed the weight. There's lots of evidence of that. You just have to find a dietary plan that works the best for you.

Lecture 7: Genius Foods

At a glance — 'Genius foods' and achieving antifragility.

- 7.1 "Achieving Antifragility" Let's get into it. So, we're going to talk about establishing metabolic health, which is super important. We're going to talk about the genius foods, and we're going to talk about the benefits of exercise.
- 7.2 The genius food section is—I've adapted a talk that I often give. Part one, we're going to go over metabolic health. We're going to define metabolic health. We're going to talk about the seeming lack of metabolic health amongst the US population and how to better procure metabolic health.
- 7.3 Today, unfortunately, 93% of adults have some form of metabolic anomaly. Now, these five items are components of what is sometimes referred to as metabolic syndrome, which puts you at increased risk for type 2 diabetes, cardiovascular disease, dementia, early mortality, pretty much everything bad that you can think of. To make a diagnosis for metabolic syndrome, you have to have three of these, but 93% of adult Americans have some form of metabolic anomaly.
- 7.4 This is really a constellation. A lot of these features tend to ride together. You might have heard this statistic referenced over the past couple of years, particularly with reference to COVID-19, because it seemed that people with poor metabolic health fared less well, had poorer outcomes with that condition. So, metabolic health, really, should also suggest to you, when you hear it, resilience, strength, resilience, robustness. And yet, unfortunately, most of us are sick.
- 7.5 Now, you should also keep in mind that, as I mentioned, about two-thirds of adults are either overweight or obese. So, there's obviously discordance between the number of Americans who have some type of metabolic degree of unwellness and those overweight statistics. So, clearly, people who are not necessarily overweight are still suffering from some component of metabolic illness. So, we can talk a little bit about what these are.
- 7.6 High triglycerides—essentially, fat in the blood. Fasting triglycerides— typically, you get when you go to get a physical at the doctor's office; they're looking to make sure that your triglycerides are not low. You, generally, will see a spike in triglycerides after you eat anything fatty. Actually, it's interesting. After eating something fatty, very fatty, your blood takes on the appearance of pink cream. That's just to show you how much fat is present in the blood after a high-fat meal, but then that fat typically subsides, it goes back down. Similarly, when you consume something with sugar in it, you have a physiologic normal blood sugar spike, and then it goes down. Same with anything fatty. What doctors are really looking for are your fasting triglycerides. How much fat is chilling in your blood when you're fasting? Not as a result of some postprandial post-meal blood fat spike.

- 7.7** HDL—sometimes erroneously referred to as good cholesterol. HDL is a form of lipoprotein that is produced by your liver, circulates in your blood. Generally, it's thought that high HDL is better. We used to think that raising HDL deliberately improved health, but there have been a number of drugs that have been designed that have succeeded at raising HDL, and they don't seem to improve cardiovascular outcomes.
- 7.8** So, the new way of thinking about HDL is more that it is reflective of good health. So, you, generally, want to do things that lead to high HDL, not because it's going to prevent cardiovascular events or mortality or anything like that, but generally, things that raise HDL are thought to be good for us. Interestingly, saturated fat raises HDL. It also raises LDL, but it raises HDL. Exercise raises HDL. Excessive alcohol consumption lowers HDL. Generally, we know that high HDL is a good thing. It's reflective of good health.
- 7.9** Abdominal obesity. There are two primary types of fat that we carry. We carry fat subcutaneously, so fat underneath our skin. We have fat arms, fat legs. That's subcutaneous fat. That's fat; that's stored energy, essentially. But then, visceral adiposity is fat that we store in our midsection. Some degree of fat storage in our midsection is important. It's important for women. It protects the reproductive organs. It protects our vital organs. So, again, being super lean shouldn't be a health goal for anybody. Some degree of fat cushioning our internal organs is important, but too much, and that's when it becomes a problem.
- 7.10** We know that visceral adiposity is not just an inert storage site. It secretes compounds that are pro-inflammatory. We know that visceral adipocytes, so fat cells, are very sensitive to the hormone cortisol. Sometimes, when you see people who are chronically stressed out, they have an apple-shaped body—that apple-shaped body of skinny arms and skinny legs, but lots of abdominal obesity? It's because the cells are very sensitive to the hormone cortisol, which is chronically elevated in the chronically stressed person, and so they tend to accumulate more fat there.
- 7.11** Of course, fat storage is ultimately dictated by energy balance, as we mentioned, but hormones, like cortisol, can influence where we store that fat. So, a gain, in the chronically stressed out person, people with cortisol dysregulation, there are a number of pathologic conditions characterized by cortisol dysregulation; you see excessive adiposity of the midsection. So, that is a sign of metabolic illness. Many people carry excessive visceral fat.
- 7.12** Elevated blood pressure—this could be due to chronic stress, could be due to excessive sugar consumption. In a small percentage of people, I believe it's about 15% of people are sodium-sensitive. You can actually do a very simple test to find out whether or not you're sodium-sensitive. You just eat a lot of sodium and see if it raises your systolic blood pressure by five points or more. It's usually an indicator, but minerals other than sodium, like potassium, are also important with regard to regulating blood pressure.

But, in general, blood pressure is very important for keeping your brain healthy, preventing stroke. You want to make sure that you have a nice healthy blood pressure.

- 7.13** High fasting blood sugar. So, a lot of people these days are using CGM monitors. I think the jury is still out ultimately on their utility for somebody who is normal glycemic. So, for somebody who has normal metabolic health. After you eat something with carbs, you should expect a blood sugar spike, and then it comes back down.
- 7.14** Also, sometimes blood sugar spikes are helpful, like pre-workout or even post-workout sometimes. I think, again, the jury is still out on whether or not we should be pathologizing physiologic blood sugar spikes, but generally, high fasting blood sugar is no bueno. So, you want to make sure, generally, that your fasting blood sugar is below 99 milligrams per deciliter. Ideally, it should be even lower than that. Above that range, you get to prediabetes and then, ultimately, type 2 diabetes.
- 7.15** So, all of these components—what's so interesting about all these components is that muscle, exercise, really solves for all of them. We see that when we exercise at a low intensity, it reduces triglycerides in our blood. Low-intensity exercise preferentially uses fat as a fuel source, and, as I mentioned, triglycerides are fat in the blood. Physical exercise—crucially, crucially important for reducing triglycerides.
- 7.16** We see that exercise, I think primarily resistance training, if I'm not mistaken, raises HDL. It can help dissipate stress, which we know contributes to abdominal obesity. Meta-analyses show that exercise is as effective as medications for helping to reduce hypertension, to help normalize blood pressure, which is so empowering to know that exercise can do that.
- 7.17** Then, of course, high blood sugar. Resistance training— you get the benefit of something called insulin-independent glucose uptake, which is a fancy medical way of saying that after a workout session, your muscles are literally like a sponge for sugar. They soak up sugar from your blood. So, physical exercise is the way to correct all of these features, particularly resistance training. So, it is crucial. If there was a drug on the market that could do what exercise would do, it'd be worth billions and billions of dollars. But you just got to do it. You got to do the work.
- 7.18** We see that exercise boosts mental health. Mental health is distinct from brain health, so mental health, depression, anxiety. There's lots of good data on this. It helps to boost serotonin, feel-good chemicals. It also helps to strengthen brain cell networks. I already talked a little bit about BDNF, brain-derived neurotrophic factor.
- 7.19** It helps to normalize blood pressure. High blood pressure is a major risk factor for cognitive decline, dementia. The second most common form of dementia in the United States is vascular dementia, which is basically created by elevated blood pressure. Not necessarily to the point where you might have a stroke, but it's dysfunction of the little blood vessels that supply fuel, antioxidants to the brain. So, having healthy blood pressure is crucial.

- 7.20** I actually think that people should have—they have these really easy-to-find cuffs that you can just buy and keep in your house to measure your blood pressure when you're home chilling, which is the best way to really measure it. A lot of people have experienced white coat syndrome, where they see an increase in their blood pressure when they go visit a doctor's office; they're stressed out. Getting a sense of your blood pressure at home is actually a more accurate reflection of your fasting blood pressure. So, those are super easy to find.
- 7.21** Exercise improves endothelial function. Your endothelium is a one-cell-thick membrane that basically protects—it's the sort of inner lining of your blood vessels, is the way to think about it. It's one cell layer thick. Your blood vessels need to expand and contract in accordance with environmental demand, meaning if you want to go to the gym, if you want to have a workout, if you need to run from a predator, if you're standing in the street and you see a car coming, you've got to jump out of it, your vasculature needs to cater on an instantaneous basis to environmental needs.
- 7.22** So, having your endothelium function properly, being able to expand and contract, to vasodilate and constrict, super important. Exercise helps to work out all those little micro-muscles that basically facilitate that process. When we look at long-lived populations, cardiorespiratory fitness—the ability of your circulatory system, your respiratory system to be robust and resilient— that's where cardiorespiratory fitness comes in.
- 7.23** Hormone health. I talked a little bit about insulin, cortisol, and testosterone. Physical exercise helps to boost hormone health in a profound way. It's one of the best ways, actually, to regain insulin sensitivity, which is the inverse of type 2 diabetes. It helps to sensitize your cells to the hormone insulin.
- 7.24** And then, of course, optimizing for muscle hypertrophy or muscle growth and strength, which is really important. There's some crazy statistic, I forget what it is, but it's very stark. Somebody who's in advanced age who falls and breaks a hip, I think 50% of people don't make it past a year beyond when that occurs. So, making sure that you're able to—It gets harder and harder to maintain muscle as you get older. This is due to a shift in the hormonal milieu, many things, but making sure that you're optimizing for strength, for hypertrophy.
- 7.25** Ultimately, whatever form of exercise you enjoy, it's good to sustain that, but generally, people who prefer cardio ought to do more resistance training. People who only do resistance training could benefit from a little more cardio. But I would say as a whole, from what I've seen and the literature and just engaging what appears, if you had to pick one, I think resistance training would be the one to pick. And then, getting your cardio-respiratory fitness from walking and NEAT. Resistance training to me, really, it's like the Holy Grail, really, of whole body vitality.

- 7.26** Now we're going to talk about the genius foods, which are, again—so, 'genius foods' is not a scientific term; it's like superfoods. Superfoods has no scientific meaning, but it's a term that I use to describe foods that give you a general sense of how to eat rather than overcomplicate things, although you now, obviously, have the framework to understand why I'm making these recommendations. I generally like to talk about genius foods so that next time you're pushing your cart through the supermarket, you know basically what to reach for and why, which is the most important thing.
- 7.27** So, these are all the foods that stood out to me when I started doing my research. And again and again, I would just see them come up in the medical literature, whether it was in the MIND diet, which I've already described to you, or the DASH diet. Part populations in the world that seem to live a long time tend to integrate these kinds of foods, which really, ultimately, should represent categories.
- 7.28** Berries, obviously, a category. I think in my first book, "Genius Foods," I highlighted specifically almonds, and I could talk about those for nuts, for example. But, ultimately, what I meant to really describe, and I did in the text, was to describe categories. So, if you don't like blueberries, strawberries are great. If you don't like almonds, macadamia nuts are great.
- 7.29** But, essentially, we'll start with berries. Berries, whichever type you like, whether they are blueberries, strawberries, or goji berries. Berries tend to all be low in sugar and calories. They contain myriad hormetic plant compounds. We do have evidence that blueberries and strawberries are associated with reduced brain aging by up to two and a half years. I believe it was the Nurses' Health Study where we saw that. The dose is only about a serving or two a week of blueberries and strawberries.
- 7.30** Of course, that's a correlational finding, just to be clear. So, people that eat fresh berries on a weekly basis probably are doing other things. They're probably health conscious. They probably are well off to some degree. Berries are a fairly expensive fruit relative to others. But that's where other types of research, like mechanistic studies, come into play, and we see, actually, that compounds in berries have a longevity-inducing effect in smaller organisms. They're great, associated with reduced risk of various chronic diseases. One to two servings a week is generally the recommendation that I make. Also, as I mentioned earlier, berries are the only fruit that has a recommendation in the MIND diet. This is a diet that is, obviously, well known for the research that's been done looking at how it relates to cognitive health. So, berries are really wonderful.
- 7.31** Nuts. Nuts of all kind are great. All kinds. They're high in healthy fats. They also, because they're so high in healthy fats, they contain fat-soluble antioxidants. You won't find very many water-soluble antioxidants in nuts, which makes sense because nature tends to package fatty foods with fat-protecting antioxidants— antioxidants to protect those fats. Why is this relevant to us? Because the brain is made of fat. So, the brain is

made of fat. Fat-soluble antioxidants are particularly important when it comes to preserving brain health, and vitamin E is one of the most important of them.

7.32 Now, vitamin E is an interesting vitamin. You might be inclined to think that it's a singular compound, but actually, vitamin E represents about eight different forms of vitamin E. It's like a troupe of superheroes that are found in nature wherever you find polyunsaturated fats, which, if you recall from my first lecture, from the first part of the seminar, is the most vulnerable to oxidation. In nature, wherever you find more polyunsaturated fats, you also tend to find more vitamin E. Concurrently, the brain is primarily composed of polyunsaturated fats, and vitamin E seems to be highly protective with regard to brain health. I wouldn't supplement with it because supplements tend to only have one form of vitamin E, whereas whole foods tend to have all the different forms, which is why I'm a huge fan of nuts like almonds, which are a rich source of vitamin E.

7.33 Different nuts have different benefits. Almonds are a great source of magnesium. In one handful of almonds, you get about 20% of your daily magnesium needs, which is great. But then, you can look to pistachio nuts, which pistachio nuts—their unique color is attributed to the presence of carotenoids, lutein and zeaxanthin, which are those two compounds that I talked about earlier today as having a protective effect on eye health and brain health. So, pistachios are the only nut that contain appreciable amounts of lutein and zeaxanthin. So, no matter what nut you choose, they're all beneficial.

7.34 The one caveat that I would add with regard to nuts is that among whole foods, they are highly calorie-dense. So, unless you're shelling them yourself, I tend to use them more for recipes where I can portion-control them, because I've seen, at least with me, it's very easy to consume 1,000 calories of nuts in one sitting. It's just very easy when you buy them shelled and then throw salt into the mix. So, nuts are one of these foods where there's a little asterisk. You just want to be careful because they are very calorie-dense.

7.35 That being said, interestingly, a USDA study found that when you chew whole nuts, about a third of the calories that you ingest from nuts are actually excreted in your poop, which goes to show you the value of the food matrix. That's not the case when you consume nut butter or when you buy ultra-processed foods made with nuts. But when you chew foods, when you chew whole foods, there's a portion of calories that are simply—you're not even ingesting because of the food matrix and how difficult it is for your body to actually access those calories when you're swallowing whole particles of foods like nuts, for example. So, whole nuts are great. Portion them out. Use them in recipes. Super, super, super healthy. Trying to think if there's anything else—also a good source of prebiotic fiber, which I'm going to talk about later.

7.36 Cruciferous vegetables. So, I've already talked about some of these hormetic compounds. Cruciferous vegetables are rich in them. They're also high in fiber. Vitamins, like vitamin C—broccoli contains a ton of vitamin C. Minerals—there is some

concern over the bioavailability of minerals when present in plants, but in the context of an omnivorous diet, a diverse diet, not a concern. They still contribute to nutrient repletion. We see that they're good for heart health. They are associated with lower risks of certain types of cancers, I think partially because of their ability to help us detox but then also because they trap certain hormones in the gut. That might be relevant to certain hormone-sensitive cancers.

- 7.37** As I mentioned, they help stimulate body detox processes, and they provide compounds that our bodies require to create some of these detox compounds, like sulfur. You might have heard of glutathione, which is the body's mother of all antioxidants, it's been referred to as. We require sulfur. I, occasionally, will take a glutathione supplement that is a liposomal glutathione supplement that I sometimes will take, mostly for mental insurance. I don't have any evidence that it—I just take it sometimes if I've spent a lot of time in airplanes, on the tarmac inhaling jet fumes, I just feel like it's a little bit of added insurance.
- 7.38** The interesting thing about it is that it smells like rotten eggs, and that's attributed to the sulfur that it contains. You just get a sense of how important sulfur, that rotten egg compound, is in the production of these very important antioxidants that our own bodies produce. Broccoli, cruciferous vegetables, if anybody has ever worked in an office where somebody has microwaved broccoli, you can smell the sulfur, and as unpleasant as that is, it's quite important from the standpoint of whole body vitality because it provides this very important compound for detoxifying compounds, like glutathione. So, sulfur-containing foods. You also get sulfur in eggs and beef and things like that. Whey protein is another great source. These are all compounds that really help with our own detoxification.
- 7.39** Dark leafy greens. As I mentioned, dark leafy greens, in general, which comprise kale, spinach, arugula, Swiss chard, romaine lettuce. They're all great, whichever ones you like. I'm not a stickler for any one in particular. I occasionally enjoy kale. If you are at risk for kidney stones, you're going to want to watch your spinach consumption. Spinach is a very rich source of oxalates, calcium oxalate. But, in general, kale is totally fine to consume moderately. I love arugula. Arugula is actually the top source of compounds called nitrates.
- 7.40** All dark leafy greens are rich in nitrate compounds, with arugula being the top, which are very important for helping to maintain healthy blood pressure. So, I've talked about blood pressure, how important that is, and how we want to avoid hypertension. Dark leafy greens—a wonderful source of compounds called nitrates that boost nitric oxide in the body. So, nitric oxide is this gas that we produce in our blood vessels that literally works to vasodilate. There are pharmaceuticals on the market, like Viagra, which work by modulating this nitric oxide pathway to increase blood flow in certain parts of the body. Well, dark leafy greens function similarly in that they provide nitrates which then get reduced by oral bacteria to nitrite, which then enters this nitric oxide pathway. One

single high-nitrate meal, one single salad, actually, can benefit cardiovascular health with a few certain important stipulations that I'm going to get to later, but just know that dark leafy greens are great from a cardiovascular health and brain health standpoint, ultimately.

- 7.41** They're also a top source of pigments, like lutein and zeaxanthin, which I've already talked about. Kale is the top source. In one cup of cooked kale, you get about double the dose required on a daily basis to help prevent age-related macular degeneration, I believe. This is one of the reasons, potentially, why Martha Clare Morris at Rush University, who devised the MIND diet, if you recall, published a study finding that people who eat a bowl of dark leafy greens every day have brains that perform up to 11 years younger. Eleven years. So, potentially, more than a decade of age reversal from the standpoint of the brain with regular dark leafy green consumption. And this isn't necessarily a ton. This is like a cup and a third, if I recall correctly, from that research, was the serving, the dose of dark leafy greens on a daily basis. I'm a massive fan of dark leafy greens.
- 7.42** Some caveats. We're going to talk about the nitrates a little bit more, but a lot of these compounds tend to be fat-soluble, so you definitely want to integrate a form of fat. You can use extra virgin olive oil, which is my go-to. I love extra virgin olive oil, but if you were sauteing your dark leafy greens, whatever oil you use is fine. If you are eating a fat source concurrent with the salad, or if you had some fat an hour or two prior, you don't necessarily need to worry about it as long as you have fat in your digestive system at the time of consumption.
- 7.43** Ruminant meat. Grass-fed, grass-finished beef is my go-to. Obviously, high in protein, bioavailable nutrients, we've talked about that, and also conditionally essential nutrients. So, what 'conditionally essential' means, we've talked about essential nutrients, nutrients that you might develop a deficiency disease or become deficient in, but there are a whole host of nutrients that are conditionally essential, meaning our needs for them increase, or that we may produce some degree of these nutrients endogenously in our own bodies, but perhaps not necessarily enough for good health. Good examples of this are creatine and taurine.
- 7.44** Red meat is very high in a compound called creatine, which weightlifters have been familiar with for years. There's a ton of evidence on creatine, both from a brain health standpoint as well as from a physical performance standpoint. Creatine helps to boost physical performance. It's one of the most well-studied sports supplements on the market and one of the very few with a solid evidence base regarding efficacy and safety. And red meat is a natural source. In fact, when they give vegans and vegetarians supplemental creatine, they see a boost in cognitive function. So, it seems like the creatine that we get from meat actually helps to boost the way that our brains work, which suggests that it's a conditionally essential nutrient and that we should be ingesting it from our food.

- 7.45 Then, taurine is an interesting amino acid. It's not essential unless you're a cat. It's essential for cats and, I believe, dogs. But taurine—there's a lot of good randomized control trial data to suggest that it helps to lower blood pressure and have all these other interesting, beneficial effects.
- 7.46 The other thing I'll add that is a non-trivial data point is that ruminant meat— there's low toxicity to it, and what I mean by that is that I'm a massive fan of fatty fish, but depending on where the fish that you consume is raised, it can harbor environmental pollutants. A lot of fish these days have microplastics. There are various environmental pollutants and carcinogens, like PCBs, that have been identified in farmed salmon, for example. I still think that the benefits, to be clear, of eating farmed fish outweigh the downsides, but with ruminant meat, very low risk of any form of toxicity. To my knowledge, there's been very little—you don't get that environmental exposure, exposure to pollutants in the fat or in the flesh of ruminant meat. So, it seems to be extremely healthy from that standpoint.
- 7.47 Eggs. We've already talked a little bit about the fears around dietary cholesterol and how they are unfounded. For the most part, dietary cholesterol does not affect blood cholesterol. Eggs are a wonderful source of protein and highly bioavailable essential nutrients. Choline is an example of a conditionally essential nutrient found in eggs, which we know directly supports brain health. And also, there are compounds that support heart and eye health. When an embryo is developing, the first structure to assemble is the nervous system, which includes the brain. An egg yolk literally contains everything that nature has deemed important to grow a brain. A chicken brain, but a brain nonetheless. So, it is one of nature's multivitamins.
- 7.48 Eggs are on the list of foods that tend to induce reactions in people that are allergic to eggs. But it is worth noting that people who are sensitive to eggs tend to be sensitive to egg whites, not the yolk. Everybody's different, but generally, the sensitivities are to egg whites. You want to make sure that your egg whites are fully cooked through. You don't want to consume raw egg whites in any circumstance, generally, especially not chronically. Egg yolks are a different story. You can consume raw egg yolks; it's where all the nutrition is.
- 7.49 Dark chocolate. I love dark chocolate. Loaded with antioxidants, minerals, like magnesium, it's a wonderful source of magnesium. Contains many beneficial phytochemicals. I talked a little bit in the last session about a study, an RCT, on high-flavanol foods. Dark chocolate is rich in flavanol compounds, and we see that people who habitually consume dark chocolate, or chocolate in general, actually, tend to have better memory function. Lots of great compounds, lots of minerals that we don't get in high amounts in other foods, like copper, can be found in dark chocolate. It's also a rich source of nitrates, which means that it can help support cardiovascular health, lower blood pressure, and improve mood.

- 7.50** Olive oil. I've talked about olive oil. It's made simply by pressing olives. It is rich in heart-healthy monounsaturated fats. It's chemically stable. You can cook with olive oil. It's a huge misconception that you can't cook with it. They cook with it in the Mediterranean region of the world. It has a flavor profile, so if your goal is to use a more neutral oil, then I would reach for avocado oil or macadamia nut oil. There's a number of really great safe oils on the market to use for high-heat cooking.
- 7.51** Meta-analyses show that olive oil does have an anti-inflammatory effect. That's very important. Olive oil has an anti-inflammatory effect. Many of the kinds of chronic diseases that we're now seeing overtake society are characterized by rampant inflammation. Olive oil, actually, might share a mechanism with non-steroidal anti-inflammatory drugs like Advil. It's that powerful of an anti-inflammatory. You're not getting the same effect, the same degree of an effect, with olive oil, but we do see that it has anti-inflammatory properties due partly to the presence of these phytochemicals, these compounds in extra virgin olive oil.
- 7.52** Its consumption is associated with reduced risk for chronic disease. It helps us to absorb fat-soluble nutrients from greens, and there's lots of evidence. So, you can look up and down the evidence hierarchy, and we see meta-analyses that support it. We see, observationally, people who use more extra virgin olive oil tend to be protected to some degree from chronic disease. There are animal studies that point to a beneficial effect. There are randomized control trials that point to a beneficial effect. So, it is really great. In Mediterranean kitchens, you're not going to find canola oil, you're not going to find soybean oil, it's always olive oil.
- 7.53** Wild fish. Again, aquaculture has had to scale up a lot over the past few decades, and mistakes have been made, as it has with nearly every aspect of the food industrial complex. But to date, it's still well-accepted that the benefits of consuming fish outweigh any risks. We see this with regard to pregnant women and their offspring, that women who consumed more fish seemingly have smarter kids. We see that fish consumption, one to two servings per week, even for people who are genetically predisposed to Alzheimer's disease, have a reduced risk for developing dementia owed, primarily, it's thought, to the omega-3 fatty acids that fish contains, which are one of the most important structural building blocks of the brain.
- 7.54** So, omega-3 fatty acids are highly abundant in fish and are most abundant in fatty fish: salmon, sardines, mackerel, herring. Those are the four more widely available fatty fish: salmon, sardines, herring, mackerel. Sardines and salmon are my personal favorites. Great source of omega-3 fatty acids and a whole host of other important nutrients, like selenium, and various unique antioxidants, like astaxanthin. So, wild fatty fish is great. We see again and again, that fish consumption is associated with reduced risk for chronic disease. It could be healthy user bias. People who consume fish tend to be more health conscious. You always have to remember that, but it seems to be a very helpful food.

- 7.55** Then, we have avocados. Avocados are really a wonderful food. We talked a little bit about fat-soluble antioxidants and how important they are for the brain. The reason why I've singled out avocados is that they have the highest proportion of fat-soluble antioxidants of any fruit or vegetable. You get a ton of lutein, you get a ton of zeaxanthin, you get a ton of vitamin E, all the different forms of vitamin E in avocados. They also are very high in heart-healthy, monounsaturated fats, a great source of fiber, and a medium-sized avocado has twice the potassium of a banana. Bananas tend to get all the love for the potassium that they contain, but avocados have double that amount.
- 7.56** Potassium is really important. We're starting to see that potassium and sodium have inverse effects. They work in tandem with one another on blood pressure, but, generally, it's not necessarily—the idea that sodium is the bad guy with regard to our blood pressure is not the whole picture. Part of the problem has to do with the fact that people underconsume potassium today. Where is potassium typically found? Well, actually, wild salmon is a great source of potassium. You also get potassium in red meat. So, it's found pretty abundantly in whole foods. Fruits and vegetables are the most well-accepted and most well-defined source of potassium, but avocados are a great source. Focusing on getting more potassium in your diet seems to be quite beneficial from the standpoint of blood pressure.
- 7.57** We know that avocados are very satiating. They're mineral-rich. The fats are satiating. It's a really great food. I feel like it's like dropping a bomb. I think about them as dropping a bomb on inflammation and oxidative stress. But yes, a wonderful food.
- 7.58** Those are the genius foods, and, essentially, part of the reason why I decided to, basically, identify all of these foods and then list them out like this is I came across a study from Tufts University that found that people who adhere to the often-repeated advice to 'just eat all things in moderation' actually used that advice as an excuse to consume more unhealthy foods. People who had greater degrees of dietary diversity, in the general sense—so, you have to remember that the population that they were studying was just the Standard American Diet; this wasn't people who were eating a diverse array—weren't prioritizing a diverse array of plants necessarily. They were just eating more of everything. What they found was that more dietary diversity equated to the greater consumption of confectionary products, sugar-sweetened beverages.
- 7.59** Conversely, what this Tufts study found was that the healthiest people, actually, just bought a narrower range of foods and just bought those foods on loop. So, it was my hope that by coining this list, people would just see it and buy these foods on loop. If I could just buy these foods on loop and then keep some nice seasoning spices, salt, pepper, garlic in my kitchen, this would basically be all you need. Then, you can garnish around it with whatever other foods you choose to integrate. But these foods can get you a very long way. They're pretty accessible. They can be generally low cost. Now, you can find them at Costco, Walmart, and other national retailers.

Lecture 8: Microbiome Mastery

At a glance — Microbiome mastery, with a recap of the course.

- 8.1** I want to do a bit of a recap but also talk about some things that we haven't yet gotten to.
- 8.2** So, I've definitely referenced a few times the microbiome in reference to the gut, but it's important to recognize that we have a microbiome all over us. The microbiome is a term that we give to the universe of microorganisms: bacteria, viruses, fungi. Interestingly, the collection of fungi that we coexist with has been referred to as the mycobiome. But we have evolved with this invisible universe of microorganisms. We have bacteria in places of us that were once thought sterile. Mammary tissue harbors bacteria. We have bacteria in our pancreas. It's like we coexist with these microorganisms, and so we ought to consider them when making dietary and lifestyle choices.
- 8.3** The three most important microbiomes that I'm going to talk about, well, I'm going to focus on two of them. But just in general, it's important to note that the microbiomes that you have the ability to, in a sense, manipulate, you have your oral microbiome, your gut microbiome, and your skin microbiome.
- 8.4** The oral microbiome, as you probably can infer, is the collection of microorganisms that live in your mouth. Now, we've been told—I remember from my first dentist visit thinking that bacteria in the mouth are not good. Ever since the germ theory of disease presented itself in the field of medicine, we've always thought that bacteria are bad up until a few decades ago when it started to dawn on us that these bacteria are playing a symbiotic role in our health, a synergistic symbiotic role. So, we know that our oral microbiome is crucially important. And so, we're going to talk about that.
- 8.5** Gut microbiome. I'm referring to the microorganisms that live primarily in our large intestine. The alimentary canal comprises the mouth, and it goes all the way to the other end: you've got your stomach, your small intestine, and your large intestine.
- 8.6** Your stomach is where it's a very volatile place. After you eat, your stomach fills itself with hydrochloric acid. It's a very unfriendly place; it doesn't harbor very many bacteria. The small intestine is also quite volatile. The small intestine is the first part of the GI tract after the stomach, and it is an active site of nutrient absorption. So, in the small intestine, that's where all the proteins, all the sugars, all the fats, they're all getting absorbed. That's where the host, you, essentially get fed. It's where the rubber meets the road in terms of nutrition, digestion. It's the small intestine.
- 8.7** Then, you go further south, and you hit the large intestine, also known as the colon, and things get a little more chill. It's like the Miami of the GI tract, actually. The

climate's really nice. Bacteria love to set up in the large intestine. That's where the bulk of the gut microbiome is focused.

- 8.8** Also, interestingly, that's where the majority of our body's immune system is really poised to focus on. If you think about it—a lot of people, when asked where they think our largest interface with the environment is, most people would probably answer our skin because our skin is on the outside. But actually, what's in your digestive tract is also outside of your body. We haven't yet absorbed the contents of what's in our food, and you can swallow ultimately anything. It's still outside of your body until it gets into circulation.
- 8.9** Our gut, actually, if you were to take out your digestive tract and sprawl it out, it would probably take up the square footage of this floor that I'm standing on right now. It's much larger than your skin, if you would do a comparable thing to your skin, as much as that sounds like it would be out of a horror movie, a bit morose.
- 8.10** But all that is to say, our gut is our largest interface with our environment. And so your immune system is really poised to focus primarily on what's happening there. It's got sensors. It's constantly surveying and surveilling the environment. Also, because you've got all of these foreigners stationed in your large intestine, in a sense, your immune system is learning about the outside world through its interaction with these gut microbes. That's why I've often described the gut microbiome as being a training camp, like a boot camp for your body's immune system.
- 8.11** Lo and behold, we see a lot of people that have immune dysfunction, whether it's autoimmunity or allergies. The root cause is thought to be gut dysbiosis, a problem with the gut microbiome, as well as, of course, perhaps a genetic predisposition, but ultimately, some impaired relationship between the gut microbiome and the host immune system. So, the gut microbiome is super, super important.
- 8.12** I've also, aside from the immune function, referenced the fact that we extract very important chemicals from the gut microbiome, whether it's vitamins or short-chain fatty acids like butyrate, which we know have a powerful anti-inflammatory effect. There's a lot of potential with regard to the gut microbiome. Even after everything that I've just said, we are at the very tip of the iceberg in terms of understanding how to manipulate it, whether or not we should be attempting to manipulate it. It really is like we're still in the relatively dark ages, but what we do know is that it's important and that the best we can honor it with whole foods and essentially allow it to do its thing, the better off we will be.
- 8.13** And then, the skin microbiome is also very important. I think a general rule of thumb is to, generally, not put too many things on your skin that you wouldn't consider ingesting. We know that the scents that our bodies give off are in many ways dictated or at least influenced by the bacteria that live on our skin. And so, that is massive. But,

in terms of influencing our systemic health, we really want to focus on the oral microbiome and the gut microbiome.

- 8.14** So, one of the most important aspects of the oral microbiome is that when I mentioned that dark leafy greens contain compounds called nitrates, I wasn't kidding, but also that humans lack the enzymes required to convert nitrate to nitrite, which nitrite is what ultimately enters that nitric oxide pathway. Without oral bacteria, we would not be able to make use of that nitrate in dark leafy greens like arugula, kale, also beets. So, we rely on oral bacteria for that conversion.
- 8.15** People who frequently use antiseptic mouthwash seem to have an increased risk for the development of hypertension, which is high blood pressure. By frequently nuking the bacteria that live in your mouth, you are possibly not extracting the complete picture of cardiovascular benefits from an otherwise healthy diet. Meaning, if you're regularly eating a salad in hopes of helping to reduce your blood pressure, but you're nuking your oral bacteria on a regular basis, you're not yielding the full benefit from that salad. You're still getting some benefit; you're getting those carotenoids; you're getting the dietary fiber. But we, again, rely on oral bacteria to convert some of these compounds. Again, it just shows you that synergistic effect. We need them there.
- 8.16** So, frequent antiseptic mouthwash users see an increased risk of hypertension. One study saw a doubling of risk. Another study saw that frequent antiseptic mouthwash use was associated with a 50% increased risk of the development of type 2 diabetes because nitric oxide is not just involved in blood pressure homeostasis or balance. It's also involved in insulin signaling, so making sure that we're staying insulin-sensitive is really important for glucose utilization, fuel partitioning, and the like. More research needs to be done in this area. There haven't been a ton of studies, but we do see that the oral microbiome does play an important role.
- 8.17** And how could it not? Again, it's been millennia that we've had this cohabitation with these microbes. In a way—and we've seen this with the gut microbiome as well—the human genome has actually become more efficient as a result of our, as a species, being able to offload some of these responsibilities to our gut microbiome.
- 8.18** There was this revelation back in the early 2000s at the completion of the Human Genome Project, when scientists realized that the human genome is actually very rudimentary. We have extremely rudimentary genomes. Scientists wondered, well, then, how do we explain all of the capability and all of the diversity that we see among the human population? Well, it's because we, essentially, carry genetic material in the form of these microbes. The amount of genetic material that we carry in the gut microbiome and in, ultimately, the oral microbiome outnumbers the number of our own genes, something like 10 to 100 to one, orders of magnitude.
- 8.19** And so, we've become actually more efficient as a species by having this auxiliary set of genes that we can then use to convert nitrate to nitrate and to harness the power of

dietary fiber and the like. I mean, it's a beautiful illustration of the symbiosis of all things. That's why I think integrating fresh fruits and vegetables and fibers, which I'm going to talk about, is so important because we're basically paying homage to them, and we're helping to support them with those choices.

- 8.20** So, essentially, anything that you eat or consume changes your microbiome. We don't have the sophistication yet to know whether or not these changes are always good or always bad. But all that is to say that you want to be really cautious with synthetic compounds, compounds that are not a part of the food matrix and yet have been shown to alter the oral microbiome or the gut microbiome.
- 8.21** Obviously, we know that full-spectrum antibiotics, unless medically necessary, are bad for the microbiome. Antiseptic mouthwash—you're essentially nuking a bacteria-rich cavity in your mouth, which doesn't make any sense to do frequently. Certainly, if medically necessary. With regard to artificial sweeteners, there is some evidence that they can alter the gut microbiome. Again, we don't know whether or not these alterations are positive or negative, but a lot of gums, for example, are made with artificial sweeteners. All of the evidence has really asked the question of whether or not these sweetening compounds might affect the gut microbiome. In the microcosm of the mouth, I think it stands to reason that artificial sweeteners might not be the best thing for the oral microbiome, but I think the jury is still out on that.
- 8.22** In general, how I approach this—I generally don't use antiseptic mouthwash unless I'm at risk of getting infected. Even at the dentist's office, when she offers me Listerine after a cleaning, I don't use it. I pass. I don't use fluoride in my toothpaste. The reason for that is that I spoke to a nitric oxide expert, and fluoride affects the oral microbiome, obviously. That's part of what we hope: fluoride helps us to reduce cavities. I think that the way to really look after oral health is to not eat foods that are karyogenic, not eat foods that get stuck in the gum line and are easily fermentable by oral bacteria, to floss regularly, and then to brush. But what toothpaste—toothpaste is like a manmade thing. I think toothpaste is great. It freshens up your mouth, but do you need to constantly be using fluoride, which can disrupt oral bacteria? I don't think so. You just want to be mindful of the fact that you have an oral microbiome. In the same way that you wouldn't every week take full-spectrum antibiotics, you just want to be cautious about how you treat your mouth as well.
- 8.23** So, again, just to reiterate, when it comes to the gut microbiome, there's a lot of unanswered questions, but one of the things that we do believe to be true at this point, generally, is that consuming a diverse array of plant fibers seems to be beneficial. All plants generally contain some degree of fiber, and there are many different types of fibers. As I mentioned in the first session, carbohydrate isn't carbohydrate. Protein isn't necessarily protein. Fiber isn't necessarily fiber. There are different kinds of fibers, which I'm going to talk about.

- 8.24** In general, the more diverse the array, the more bacterial diversity you are cultivating because what you feed, you breed. So, you want to make sure that you are just consuming a diverse array. If all you're doing every day, as we talked about, is eating a big kale salad, well, then you're selecting for bacteria specifically that enjoy kale. But like with a city, our gut microbiome seems to benefit from a plurality of voices, and so that's where diversity really is key. You want to think of it almost like a city, like a city that you are architecting.
- 8.25** Consuming fermented foods seems to be extremely beneficial. This ranges from fermented dairy to raw sauerkraut, pickles, kimchi. There's a Japanese fermented soybean called natto. Dark chocolate, actually, is a fermented food. Fermented foods, generally, seem to be beneficial from the standpoint of the microbiome.
- 8.26** We want to avoid full-spectrum antibiotic drugs unless medically necessary. I forget the exact statistics, but oftentimes a doctor will prescribe an antibiotic drug because they think that patients want to be prescribed drugs, even when those drugs aren't medically indicated. So, we just want to be careful about the overuse of antibiotics.
- 8.27** Go organic when possible. I'm not a big, just to be clear, 'organic-is-the-best-and-only-way' guy. I think that depending on access, conventional, can be just as good. It really depends on the person. I choose to eat organic when I'm eating the rind or the peel of the produce. So, if it's an apple or a dark leafy green or a berry, I am generally buying organic. If I'm out at a restaurant and they serve berries that are not organic, I'll still eat those. But in general, when you go organic, you're reducing your exposure to synthetic herbicides and pesticides, which by their very nature are antibacterial. They're antimicrobial. We also see that herbicides like glyphosate can affect certain pathways for gut bacteria. We just don't have a long-term sense of what health effect that has, and I would rather play it safe than sorry.
- 8.28** You want to avoid ultra-processed food additives. This, generally, ranges from the ultra-processed seed oils, which I've talked about, to emulsifiers. There are various synthetic emulsifiers that are used in processed foods like almond milks, nut milks in general, ice creams that help to lend pleasing mouthfeels to these processed foods. Preliminary research, generally in animals, suggests that they have an inflammatory effect on the bowel. You, generally, want to avoid synthetic emulsifiers, in particular. Two that I would point out that tend to be most common in these foods and where there is some research is polysorbate 80 and carboxymethylcellulose. I talk about this a bit in "Genius Foods," my first book. The dose makes the poison, of course, but ultra-processed foods—the gut microbiome does not seem to like ultra-processed foods in general.
- 8.29** Then, finally, when increasing dietary fiber consumption, you really want to go slow. A lot of people think that fiber is the enemy when they, overnight, start eating a lot more of it, and they get really bloated and gassy; that's not meant to happen. It means you haven't cultivated the gut bacteria to contend with that sudden onslaught of fiber. If you go slow and attempt to increase your fiber consumption to a point of comfort, you

should always be comfortable; you can cultivate the bacteria to extract the full nutrition from those greens. That is super, super important.

- 8.30** Also, fermented foods can help to encourage bacterial diversity. So, it can help you further increase your tolerance for these fibers, which, the research seems to suggest, are beneficial to health: anti-inflammatory, associated with better cardiovascular health, and I'll explain why, and so much more.
- 8.31** As I alluded to, fibers and fiber. There are different kinds of fibers. There is soluble fiber, and there is insoluble fiber. Soluble fiber is, generally, the type of fiber that gut bacteria ferment. So, your gut bacteria ferment soluble fiber, meaning they eat soluble fiber, and they churn out all of those beneficial compounds, whether we're talking about vitamins or butyrate, powerful postbiotic compounds—there's a compound now being studied called urolithin A, which is produced by some species of gut bacteria when they ferment pomegranate, polyphenols, compounds in pomegranates. So, there are all these wonderful chemicals that we get, and, generally, soluble fiber is the type of fiber that yields those compounds.
- 8.32** Insoluble fiber is another type of fiber. Insoluble fiber: I want you to think more like roughage. It helps to move things along and add bulk to stool. It can help exfoliate, slough off dead epithelial cells throughout the digestive tract. Generally thought to be very healthy as well. However, it can be harder to digest. Brown rice is a good example, and the fiber in grains. I don't actually advocate for grain consumption, but I listed it here just so you guys can get a sense of where you can find these different types of fiber, but all of these foods are generally thought to be very beneficial.
- 8.33** Oats are a good source of soluble fiber. Psyllium husk—I'll talk a little bit about that in the forthcoming slide. Chia seeds, legumes. You can just get a sense. Chia seeds—you get that gel forming; that's the soluble fiber. Insoluble fiber, again, roughage, doesn't break down, doesn't break down, doesn't get fermented, keeps things moving. So does soluble fiber. There's overlap. And generally, wherever you see soluble fiber, you're also getting insoluble fiber. So, it's a little bit moot to try to categorize foods as being a good source of either or. They all tend to contain a little bit of each in varying proportions, but I just wanted to provide this definition for you guys.
- 8.34** Another type of fiber is known as resistant starch. It's not technically listed as a fiber on nutrition facts labels, but green unripe bananas are a good source. Well, before we get to that, so resistant starch is basically a form of starch. Starch is otherwise a chain of glucose molecules. White rice is a starch. But resistant starch is starch that we can't easily break down in the small intestine; instead, it makes its way down to the large intestine where it gets fermented.
- 8.35** Gut bacteria love resistant starch. They love it. In fact, they have a feeding frenzy when you eat a lot of it. Eating too much resistant starch right away can make you very bloated and very gassy, so be careful. You don't want to, from one day to the next,

rapidly and dramatically increase your consumption of resistant starch, but a little bit here and there is very beneficial. So, unripe bananas are a great source, green bananas. Also, unripe mango—I happen to love unripe mango. I get into fights with my friends all the time about this. They argue that the riper, the better. I think it's way tastier when it's unripe, but I'm a weirdo.

- 8.36** You can cook and cool tubers, like a potato. A raw potato is a rich source of resistant starch. You don't see many people sitting around eating raw potatoes. When you cook a potato, it basically unlocks all of that starch so that we can digest it. So, a cooked potato is a great source of starch. So, you go from resistant starch to just starch. If you cool off a potato to room temperature, or if you put it in the fridge, as is the case with the leftovers, you're creating what's called retrograde resistant starch. The potato starch, in a way, goes back; it becomes retrograded to its original form, which is resistant starch, but it's still edible. So, it's actually a very interesting hack. If you cook and cool a potato, you can dramatically reduce its glycemic impact, and you convert some of those otherwise highly bioavailable sugars found in potatoes to a form that is resistant to digestion and in fact, very beneficial to gut bacteria. You can do that with a white potato.
- 8.37** It happens to a smaller degree with sweet potatoes. White potatoes really are the best source of resistant starch. And then, interestingly, you can also do the same thing with other starches. You can do the same thing with rice. It's one of the reasons why I have no concern over sushi rice, really, because sushi rice, which has been cooked and cooled, is a good source of resistant starch. I'm not a big pasta eater, except for occasionally these legume-based pastas, but if you were to cook and cool pasta, the same thing happens: you create resistant starch, whether it's legume-based, whether it's wheat-based. So, that pasta salad or whatever Americans love to consume come summertime—I was never a big fan; that's why I said that—you're actually getting a good source of resistant starch, which is very beneficial to gut bacteria.
- 8.38** Now, we come to viscous fibers. So, some soluble fibers are categorized as viscous. This is one way of categorizing fiber: soluble fiber and insoluble fiber. But you can also categorize fibers in different ways; you can categorize them as viscous and nonviscous fibers. Viscous just means gel forming, essentially, and psyllium husk and oats are two great sources of viscous fibers. The reason why I bring them up is because there's a lot of evidence on the ability of viscous fibers to improve blood sugar and also to help reduce lipids like LDL cholesterol. I'm not saying that LDL cholesterol is bad or that everybody needs to reduce it to the floor, nothing like that, but it is interesting how it works.
- 8.39** So, psyllium husk, when we ingest it, or oats—when we ingest fats in general, at the top of our small intestine, we release bile acids to help us break down and emulsify these fats so that we can ingest them. Bile acids are made in the liver. We store them in our gall bladder, and they're squirted out at the top of the small intestine. These

acids are made of cholesterol. We use them, again, to assimilate fats, and then, those bile acids at the latter end of our small intestine get reabsorbed, those bile acids.

8.40 That whole cycle is super interesting. What happens when we consume psyllium husk is that it traps those bile acids. It traps the bile acids. When we're not able to reabsorb them at the latter end of the small intestine, our liver actually pulls cholesterol from circulation to create new bile acids. So, that 's one of the mechanisms by which psyllium husk and oats have these well-documented cholesterol-lowering effects. You end up pooping out cholesterol, and so your liver will suck more from circulation to create more of these of these bile acids. You can see up to a 20% reduction in total cholesterol from regular use of psyllium husk at a dose of about 10 grams a day, which is, that's not a small dose. It's a lot to consume every single day, but it's super effective.

8.41 Oats are the same. Oats contain compounds called beta-glucan, which not only have an immune modulating effect, but they can also—that's why you look, on a package of oats, you see these cholesterol claims. That's actually legit. So, fiber seems to be beneficial for a number of different reasons. I happen to be a big fan of psyllium husk.

8.42 Got to talk a little bit about digestion. When do you guys think digestion begins? When do you think digestion starts? Chewing? Well, actually, it's incorrect. Close, but incorrect. Digestion begins when you sit down. We have what's called the cephalic phase of digestion. 'Cephalic' means your head. So, the mere perception of the food that you're about to eat stimulates the flow of digestive juices. Twenty percent of total acid secretion occurs before food enters the stomach—the smelling of the food, the anticipation of the food. Chewing was a really good guess, though, because digestion does occur when we chew, and that is precisely why you want to chew your food slowly. That is super, super important. I'm super happy that you guys said chewing. It was a trick question, but you really want to make sure that when you're sitting down to eat, you are in a relaxed state, that you are appreciating your food, that you're smelling your food, that you're looking at your food.

8.43 Eating on the go is responsible for so many digestive issues. It's a huge problem. A lot of people eating on the go—you experience digestive problems. People get bloated, experience diarrhea. You're not extracting the full breadth of benefits that you could otherwise be getting from your food. And so, that's why when you first sit down to eat, you want to behold your food. You want to take a few deep breaths and really make sure that you are activating your parasympathetic nervous response, which is that rest-and-digest mode of your nervous system that's going to set the stage for optimal nutrient absorption and assimilation. It's a beautiful thing.

8.44 If you're spending all this money on healthy food and really taking the time to dial in your nutrition, what a shame it would be to not be extracting the full breadth of nutrition. What a shame to be eating that salad every single day, as I tend to recommend if you're rinsing with mouthwash frequently every morning and not extracting the full cardiovascular benefit from those nitrates. What about if you're not

fully relaxed when you sit down at the meal table to extract the full benefit, and you end up causing digestive upset, and then you mistakenly believe that it's the food that caused those problems when it wasn't. It was the fact that you were just too high-strung.

- 8.45** Taking a few deep breaths can really go a long way towards chilling you out, making sure that you are extracting—that your body is primed, that your nervous system is primed for nutrient assimilation. Because when you're on the go, when you're stressed out, when your body is having a fight or flight response, the last thing that you're thinking about, from a biological standpoint—the last thing that your body wants is to be eating. It's expecting that there's a threat imminent. So, you want to make sure to send the signal to your body that everything's good, that it's time to eat. Taking a few deep breaths, nose breathing, in particular, can go a long way towards helping.
- 8.46** You also want to make sure that you are chewing slowly. There are many reasons for this, but as I mentioned, we rely on oral bacteria for this conversion of nitrates. But there are probably many other synergistic aspects to that relationship that we have with our food and our oral bacteria. We have a sense of some of them, but you generally want to make sure that you're chewing slowly, allowing those stomach acids and digestive juices to prime.
- 8.47** Also, certain of these detoxifying compounds that I talked a bit about, like in cruciferous vegetables, are actually created when we chew. Sulfurophane, for example, is created when we chew raw cruciferous vegetables. It's not actually found in the plant. It's created when we damage the plant cell walls. There are many other examples: garlic. When you chew garlic, you are creating a compound that's a powerful antimicrobial compound called allicin in raw garlic. So, you really want to make sure that you are chewing slowly.
- 8.48** I'm talking to myself simultaneously here because I can eat fast sometimes, especially when I'm hungry. But I know that I always feel better when I slow down. Also, from a weight regulation standpoint, it takes some time for your brain to register that you've got food in the tank. The more slowly you can chew, the more effortless energy regulation is going to be. There are many, many benefits to just slowing down when you eat.
- 8.49** If you have digestive issues, there's not a ton of evidence on this, but anecdotally, I found digestive enzymes to help. I think we talked a little bit yesterday about supplemental HCL. Many people struggle with low levels of stomach acid. Betaine HCL is a fairly easy-to-find supplement that can help replace some of those acids. I find that I digest very well when I have them. If I am eating on the go, this is a way to—sometimes, I have no choice. It's a way to just buy yourself a little bit of insurance. People who are on acid-blocking medications also struggle with digestion. So, you can definitely—some of these digestive supplements on the market are worth checking into if you struggle from digestive problems.

- 8.50** It's always good to touch on supplements. As you probably have gathered, I'm not a big believer in one-size-fits-all. Everybody's going to be different. But, in general, one of my favorite supplements to touch on is magnesium. It's very hard to get magnesium. The best sources of magnesium are dark leafy greens, almonds; dark chocolate is another good source of magnesium. But, in general, most people, 50 to 60 % of the population doesn't consume adequate magnesium. I don't know where the other 50% of the population is getting their magnesium from, to be honest, but it's a very important mineral. Also, there aren't good blood tests to ascertain magnesium status because most of our magnesium is stored in our bones. We just don't have it in circulation. It's not something that you can see with the blood test. I like to supplement with magnesium. Glycinate is the form that I take. I take it every day. It's also good for migraine prevention among a whole host of other things.
- 8.51** Vitamin D also can be worth supplementing with. Vitamin D, as I alluded to, is a hormone. It regulates gene expression for about 5 % of the human genome. Magnesium, interestingly, helps to activate vitamin D. So, I would probably take magnesium before I would take vitamin D if you have a low vitamin D status to see if that helps to normalize your vitamin D levels. But, in general, vitamin D is great, especially for people with autoimmunity. There's actually a decent amount of evidence on vitamin D in the immune system to help pump the brakes on an overactive immune system. So, that is, generally, thought to be a very, very good supplement. It's hard to get too much of it, but between 2,000 and 5,000 international units a day is a good starting place.
- 8.52** Fish oil seems to be most beneficial for people who don't regularly consume fish, but, in general, it's a great source of omega-3 fatty acids. I often get asked where it sits within krill oil and algae oil, and most of the evidence supporting omega-3 supplementation circles around fish oil. And so, that's usually the supplement that I would choose. You want to make sure that you're buying high-quality fish oil. I would not take fish oil from a drug store or supermarket. You want to make sure that you're buying it from a reputable manufacturer because the fats are very delicate. So, you want to make sure that you're buying it from a manufacturer that goes to adequate lengths to ensure that the oils are of full potency and not going to be oxidized by the time the bottle reaches your hands. One thing that you could look out for in fish oil supplements is called IFOS, International Fish Oil Standards. They certify fish oils. It's an independent lab. They test for purity. So, I'm a huge fan of fish oil. If you eat a lot of fish, you probably don't need to take it, but I generally take it as a form of insurance. I eat fish a few times a month, once or twice a week, maybe. But yeah, I like fish oil.
- 8.53** Creatine, as I mentioned, will make you gain a little bit of weight. It draws water into the muscles, so it doesn't make you look puffy. It should make you look better because it draws water into the musculature. It helps you lift more weight. It's generally considered very safe. Check with your physician if you have a medical issue to see

whether or not creatine is right for you. But, generally, very safe, very effective, and very beneficial to brain health and mental health. There's a lot of evidence now coming out supporting that. But it's a great way to lift more weight, build muscle for men and for women. It's also very cheap. There's a lot of different types of creatine now on the market. Creatine monohydrate is the form that is most evidence-based, and it's the cheapest. So, again, that is creatine monohydrate.

8.54 Then protein. I'm a huge fan of protein supplements. I love whey protein. I drink a protein shake or two every day. It 's a great way to keep your protein up. It feels a little indulgent. You get a whey protein shake, it's sweetened with stevia, a little bit of vanilla; you can make a milkshake with it. You could also blend it with some frozen berries, and you get an ice cream consistency. It's wonderful. So, I'm a huge fan of protein. You don't have to be a fitness junkie to reap benefits from a protein shake. Generally, people that, even incrementally, as I think I alluded to, increase their protein intake, you'll see a change in your body composition because it's so satiating. More protein means you're probably going to consume, in the free living environment, less carbs and fat. It directly supports your musculature. It's a great supplementation choice.

8.55 Thank you guys so much.